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ETSIT ESCUELA TÉCNICA SUPERIOR DE INGENIEROS DE TELECOMUNICACIÓN UPM

MASTER OF SCIENCE IN NEUROTECHNOLOGY
MASTER´S THESIS

"DESIGN, DEVELOPMENT AND EVALUATION OF AN EXPLAINABLE ARTIFICIAL INTELLIGENCE PIPELINE FOR PREDICTING CYBERSICKNESS ESCALATION USING NATIVE VIRTUAL REALITY"

DANIEL LLOPIS CONEJO

2026

MASTER OF SCIENCE IN NEUROTECHNOLOGY

MASTER´S THESIS

Title: **“Design, development and evaluation of an explainable artificial intelligence pipeline for predicting cybersickness escalation using native virtual reality”**

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RESUMEN

La realidad virtual (VR) ha experimentado un crecimiento significativo en ámbitos como la formación, la simulación, la rehabilitación clínica, el entretenimiento y la investigación científica. Sin embargo, uno de los principales obstáculos para su adopción generalizada sigue siendo el cybersickness, un síndrome caracterizado por síntomas como náuseas, mareo, desorientación, fatiga visual y malestar general. Estos síntomas pueden afectar negativamente a la experiencia del usuario y limitar la duración y eficacia de las sesiones de realidad virtual.

El objetivo de este Trabajo Fin de Máster fue desarrollar y evaluar un sistema basado en inteligencia artificial capaz de anticipar eventos de escalada de cybersickness utilizando señales nativas de realidad virtual, incluyendo variables de seguimiento ocular (eye-tracking) y movimiento de cabeza (head-tracking). Para ello, se empleó una base de datos obtenida durante experiencias inmersivas de realidad virtual que incluía medidas oculomotoras, pupilares, de comportamiento visual y de movimiento, junto con evaluaciones temporales del nivel de malestar mediante la Fast Motion Sickness Scale (FMS).

Se diseñó una metodología de predicción temprana basada en ventanas temporales deslizantes. A partir de las señales originales se extrajeron características estadísticas y dinámicas capaces de describir la evolución temporal del comportamiento fisiológico y conductual de los usuarios. Posteriormente, se construyó un conjunto de datos supervisado cuyo objetivo consistía en predecir si se produciría una escalada de cybersickness en los segundos posteriores a cada ventana de observación.

Se evaluaron diferentes enfoques de aprendizaje automático y aprendizaje profundo, incluyendo Logistic Regression, Random Forest, XGBoost, HistGradientBoosting, Support Vector Machines, CatBoost, Multi-Layer Perceptron (MLP), redes convolucionales unidimensionales (1D-CNN) y redes Long Short-Term Memory (LSTM). Asimismo, se analizaron distintas estrategias de tratamiento del desbalanceo de clases, selección de características y explicabilidad mediante SHAP.

Los resultados mostraron que los modelos basados en aprendizaje automático tradicional alcanzaron un rendimiento competitivo para la detección temprana de eventos de escalada. Sin embargo, los modelos de aprendizaje profundo capaces de explotar la estructura temporal de las señales obtuvieron los mejores resultados globales. La 1D-CNN alcanzó la mayor sensibilidad para detectar eventos de escalada, mientras que la LSTM obtuvo los mejores valores de F1-score, PR-AUC y Matthews Correlation Coefficient. Además, el análisis de interpretabilidad permitió identificar patrones neurofisiológicos consistentes asociados al aumento progresivo del riesgo de cybersickness. Las señales relacionadas con el comportamiento de la mirada, la posición ocular, la dinámica pupilar y determinados patrones de movimiento mostraron una contribución especialmente relevante para la predicción.

Finalmente, se demostró la viabilidad de generar advertencias tempranas antes de la aparición de incrementos observables en la severidad de los síntomas, lo que abre la puerta al desarrollo de sistemas adaptativos capaces de modificar dinámicamente la experiencia de realidad virtual para reducir el malestar del usuario. En conjunto, este trabajo contribuye al campo de la neurotecnología aplicada a la realidad virtual al demostrar que las señales nativas disponibles en los visores modernos contienen información suficiente para anticipar cambios fisiológicos y conductuales asociados al cybersickness.

SUMMARY

Virtual Reality (VR) has experienced significant growth in fields such as training, simulation, clinical rehabilitation, entertainment, and scientific research. However, one of the main barriers to its widespread adoption remains cybersickness, a syndrome characterized by symptoms such as nausea, dizziness, disorientation, visual fatigue, and general discomfort. These symptoms can negatively affect user experience and limit the duration and effectiveness of VR sessions.

The objective of this Master's Thesis was to develop and evaluate an artificial intelligence-based system capable of anticipating cybersickness escalation events using native virtual reality signals, including eye-tracking and head-motion measurements. To achieve this, a dataset collected during immersive VR experiences was employed, containing oculomotor, pupillary, visual-behaviour, and movement-related variables together with temporal assessments of symptom severity through the Fast Motion Sickness Scale (FMS).

An early prediction methodology based on sliding temporal windows was designed. Statistical and dynamic features were extracted from the original signals to characterize the temporal evolution of users' physiological and behavioural responses. A supervised learning dataset was then constructed to predict whether a cybersickness escalation event would occur within the seconds following each observation window.

Several machine learning and deep learning approaches were evaluated, including Logistic Regression, Random Forest, XGBoost, HistGradientBoosting, Support Vector Machines, CatBoost, Multi-Layer Perceptron (MLP), one-dimensional Convolutional Neural Networks (1D-CNN), and Long Short-Term Memory (LSTM) networks. Different strategies for class imbalance handling, feature selection, and explainability through SHAP were also explored.

The results demonstrated that traditional machine learning models achieved competitive performance for the early detection of escalation events. However, deep learning models capable of exploiting the temporal structure of the signals achieved the strongest overall performance. The 1D-CNN obtained the highest sensitivity for escalation-event detection, whereas the LSTM achieved the best F1-score, PR-AUC, and Matthews Correlation Coefficient. Furthermore, interpretability analyses revealed consistent neurophysiological patterns associated with the progressive increase in cybersickness risk. Features related to gaze behaviour, eye position, pupillary dynamics, and movement patterns showed particularly relevant contributions to the prediction process.

Finally, the proposed framework demonstrated the feasibility of generating early warnings before observable increases in symptom severity occurred, opening the door to adaptive systems capable of dynamically modifying the VR experience to mitigate user discomfort. Overall, this work contributes to the field of neurotechnology applied to virtual reality by demonstrating that native signals available in modern VR headsets contain sufficient information to anticipate physiological and behavioural changes associated with cybersickness.

PALABRAS CLAVE

Cibermareo, Realidad Virtual, Seguimiento Ocular, Inteligencia Artificial, Aprendizaje Automático, Neurotecnología, Predicción Temprana, Interacción Humano-Computador, Biomarcadores Oculomotores, Sistemas Adaptativos

KEYWORDS

Cybersickness, Virtual Reality, Eye Tracking, Artificial Intelligence, Machine Learning, Neurotechnology, Early Warning Systems, Human-Computer Interaction, Oculomotor Biomarkers, Adaptive Systems

TABLE OF CONTENTS

TABLE OF FIGURES

1. INTRODUCTION

Virtual Reality (VR) has emerged as a transformative technology with applications spanning healthcare, education, military training, and entertainment. Its ability to create highly immersive and interactive environments has positioned VR as a key enabler of next-generation human–computer interaction paradigms. However, despite major advances in display systems, rendering pipelines, and interaction techniques, the widespread adoption of VR remains hindered by a critical limitation: cybersickness [1], [2].

Cybersickness refers to a set of adverse physiological and perceptual symptoms experienced during or after exposure to immersive virtual environments, including nausea, dizziness, eye strain, headaches, and disorientation [2]. In the VR literature, this phenomenon is commonly explained through the sensory conflict account, according to which discomfort arises when visual motion cues are inconsistent with vestibular and proprioceptive information [2], [3]. In addition, several technological and interaction-related factors have been associated with cybersickness, including hardware and software characteristics, navigation design, and broader user-experience conditions in head-mounted display systems [2], [4].

Traditionally, cybersickness has been evaluated using subjective questionnaires, most notably the Simulator Sickness Questionnaire (SSQ) [5]. In studies that require repeated measurements during exposure, the Fast Motion Sickness Scale (FMS) has also been used as a rapid instrument for tracking symptom intensity over time [6]. Although these tools are well established and remain valuable in experimental research, they still provide user-reported measurements at discrete moments and therefore offer limited temporal granularity when compared with the continuous dynamics of VR sensor data [5], [6]. This restricts their usefulness for the development of adaptive systems capable of responding to cybersickness risk as it unfolds.

To address these limitations, recent research has increasingly explored objective measurement approaches based on physiological and behavioral signals. These include variables such as breathing, blinking, and other physiological responses, as well as signals directly captured by VR head-mounted displays, including eye-tracking and head-motion data [1], [7]. Among these alternatives, integrated head-mounted displays (HMD) sensors are especially promising because they enable continuous, non-invasive, and scalable data collection without requiring additional hardware. Prior work has shown that users' gaze behavior can provide significant information for predicting cybersickness during HMD-based VR experiences, reinforcing the value of native headset signals for this task [1], [8], [9].

Building on this paradigm, machine learning (ML) and deep learning (DL) techniques have been increasingly applied to model the relationships between high-dimensional sensor data and cybersickness outcomes. Recent reviews show that these methods have become a major research direction for the automatic detection and prediction of cybersickness from physiological and behavioral signals [8]. However, much of the literature has focused on identifying current sickness states or severity levels rather than anticipating symptom escalation before users explicitly report discomfort. Moreover, many of these models remain difficult to interpret, which limits transparency and hinders their practical adoption in real-world VR systems [1], [8].

In response to these challenges, a growing body of work has begun to incorporate Explainable Artificial Intelligence (XAI) techniques into cybersickness modeling. Explainable approaches aim to clarify which variables contribute most strongly to a model's prediction, thereby improving transparency and supporting a deeper understanding of the mechanisms associated with cybersickness [10]. In this context, interpretability is not merely a desirable complement to predictive accuracy; it is a necessary component for building trustworthy systems that can inform researchers, designers, and end users about why a given cybersickness-related decision has been made.

A notable example of this approach is the RelaxVR framework, which integrates machine learning, XAI, and a dialogue-based interface supported by large language models to create an interactive system capable of predicting, explaining, and mitigating cybersickness in real time [1]. RelaxVR leverages multimodal data collected from VR headsets, including eye-tracking and head-tracking signals, to train predictive models. These models are complemented by explainability techniques that identify the dominant factors contributing to cybersickness, enabling the selection of mitigation strategies from a predefined reduction library [1]. This framework is particularly relevant to the present thesis because it also introduces the MazeSick dataset, which contains high-frequency time-series sensor data collected during immersive VR maze navigation tasks [1].

Despite these advances, an important research gap remains: the early prediction of cybersickness escalation events based solely on signals available from standard VR headsets. While prior work has demonstrated the feasibility of detecting cybersickness severity from native and physiological signals, fewer studies have focused on anticipating its progression before users explicitly report symptoms [1], [8]. Early prediction is especially valuable because it enables proactive intervention, which may prevent discomfort rather than merely reacting once it has already become substantial. In parallel, there is still a need for more systematic analysis of pre-event temporal signatures that precede cybersickness escalation.

In this context, the present thesis aims to investigate whether time-series data from VR-integrated sensors, such as gaze behavior, head movement, and locomotion-related dynamics, can be used to anticipate cybersickness escalation events. The work is grounded in the analysis of the MazeSick dataset introduced in RelaxVR [1]. By designing a pipeline that includes data preprocessing, temporal feature extraction, machine learning modeling, and explainability analysis, this research seeks to uncover the patterns associated with cybersickness onset and escalation.

More specifically, this thesis explores the hypothesis that distinct temporal signatures exist prior to cybersickness escalation, and that these signatures can be captured through sliding-window feature engineering and modeled using supervised learning techniques. In addition, the integration of XAI methods such as SHAP will enable the identification of the most influential features driving model predictions, providing insights into the physiological and behavioral mechanisms associated with cybersickness [10]. Ultimately, the goal is to evaluate the feasibility of an early warning system capable of predicting cybersickness risk in real time, thereby contributing to the development of more adaptive and user-centered VR systems.

In summary, this work is situated at the intersection of virtual reality, time-series machine learning, and explainable artificial intelligence. By moving beyond static detection toward anticipatory and interpretable modeling, it seeks to address one of the most persistent challenges in immersive technologies and to contribute to the design of safer and more comfortable VR experiences.

1.1. OBJECTIVES

1.1.1. GENERAL OBJECTIVE

The primary objective of this thesis is to investigate the feasibility of **anticipating cybersickness escalation events** using machine learning models trained on time-series data obtained exclusively from integrated sensors in virtual reality head-mounted displays (HMDs), such as gaze behavior and head movement.

This work aims to move beyond traditional post hoc detection approaches by developing a predictive framework capable of identifying **pre-event physiological and behavioral patterns**, enabling early warning mechanisms for cybersickness in immersive environments. In addition, the integration of explainable artificial intelligence (XAI) techniques seeks to provide transparency and interpretability, facilitating a deeper understanding of the factors contributing to cybersickness progression.

1.1.2. SPECIFIC OBJECTIVES

To achieve the general objective, the following specific objectives are defined:

DATASET ANALYSIS AND UNDERSTANDING

Exploring and analyzing the **MazeSick dataset** within the RelaxVR framework, focusing on the structure, temporal resolution, and characteristics of multimodal VR sensor signals (e.g., gaze, head motion, locomotion dynamics).

This includes performing exploratory data analysis (EDA) to identify distributions, correlations, noise patterns, and potential anomalies in the data.

DATA PREPROCESSING AND TEMPORAL STRUCTURING

Designing and implementing a robust preprocessing pipeline that transforms raw high-frequency VR signals into structured time-series data suitable for machine learning.

This stage includes:

- Temporal indexing and alignment of signals
- Handling missing values and noise (e.g., smoothing/interpolation)
- Feature normalization and scaling
- Outlier detection and filtering

FEATURE ENGINEERING USING SLIDING WINDOWS

Developing a **temporal feature extraction framework** based on sliding windows over the time-series data, enabling the capture of short-term dynamics preceding cybersickness events.

Extracted features will include:

- Statistical descriptors (mean, variance, median)
- Dynamic features (velocity, acceleration, derivatives)
- Variability measures (interquartile range, zero-crossings)
- Behavioral features derived from gaze and head movement patterns

This step is critical to convert raw signals into meaningful predictive inputs.

PREDICTIVE MODELING OF CYBERSICKNESS ESCALATION

Designing, training, and evaluation of machine learning and deep learning models capable of predicting cybersickness escalation within a defined forecasting horizon.

The models considered include:

- Logistic Regression
- Random Forest
- Histogram Gradient Boosting
- CatBoost
- XGBoost
- SVM
- MLP
- 1D-CNN
- LSTM

Evaluation will be performed using time-aware validation strategies to prevent leakage and ensure realistic performance assessment.

EXPLAINABILITY AND MODEL INTERPRETATION

Applying **Explainable Artificial Intelligence (XAI)** techniques, such as SHAP, to interpret model predictions and identify the most influential features contributing to cybersickness escalation.

This objective aims to:

- Improve model transparency
- Enhance trust in predictive systems
- Provide insights into the physiological and behavioral mechanisms underlying cybersickness

PRE-EVENT PATTERN ANALYSIS

Investigating the existence of **pre-event signatures**, i.e., temporal patterns in sensor data that precede cybersickness escalation events.

This involves aligning predictions with ground truth labels and analyzing whether model outputs increase prior to symptom onset, validating the hypothesis of anticipatory detection.

EVALUATION OF EARLY WARNING CAPABILITY

Assessing the feasibility of the proposed system as an **early warning mechanism** for cybersickness in VR environments.

This includes evaluating:

- Predictive performance (accuracy, recall, F1-score, etc.)
- Temporal anticipation capability
- Practical applicability in real-time VR systems

1.1.3. EXPECTED CONTRIBUTIONS

This thesis is expected to contribute to the field in the following ways:

- Demonstrating the viability of predicting cybersickness escalation using only **native VR sensor data**, without external physiological devices
- Providing a **time-series machine learning pipeline** tailored to VR environments
- Advancing the use of **XAI in immersive technologies** for interpretable prediction
- Identifying **early physiological and behavioral indicators** of cybersickness
- Laying the groundwork for **adaptive, real-time VR systems** capable of preventing discomfort before it occurs

1.2. STATE OF THE ART

1.2.1. CYBERSICKNESS IN VIRTUAL REALITY: DEFINITION, IMPACT, AND EXPLANATORY THEORIES

Cybersickness remains one of the main barriers to the large-scale adoption of immersive virtual reality (VR). Although VR technologies have advanced substantially and are now used in domains such as healthcare, education, training, simulation, and entertainment, many users still experience discomfort during exposure, including nausea, dizziness, oculomotor strain, disorientation, and headache [1], [2]. This problem has direct consequences for usability, task performance, perceived comfort, sense of presence, and willingness to continue using VR systems [1], [11], [12].

Several theoretical frameworks have been proposed to explain cybersickness. Among them, sensory conflict theory remains one of the most influential accounts, arguing that cybersickness arises when visual motion information is inconsistent with vestibular and proprioceptive signals [2], [3]. In immersive VR, this conflict often appears when users visually perceive self-motion while their body remains stationary, or when virtual camera motion, head motion, and physical movement are not fully aligned. Other complementary perspectives, such as postural instability accounts, have also contributed to the understanding of the phenomenon, but the literature broadly agrees that cybersickness is a multifactorial process shaped by the interaction between technological characteristics, task design, navigation strategy, and individual susceptibility [1], [2], [4].

This multifactorial nature is particularly relevant from a modeling perspective. Cybersickness does not emerge from a single isolated cause; rather, it is influenced by factors such as display latency, field of view, visual flow, locomotion method, exposure time, and inter-individual variability [1], [2], [4]. Consequently, cybersickness is well suited to data-driven approaches capable of modeling nonlinear and time-dependent relationships between behavioral or physiological signals and symptom progression.

1.2.2. TRADITIONAL SUBJECTIVE ASSESSMENT OF CYBERSICKNESS

Historically, cybersickness has been assessed primarily through subjective questionnaires. The most widely used instrument is the Simulator Sickness Questionnaire (SSQ), proposed by Kennedy et al., which became the standard tool for quantifying simulator- and VR-related sickness symptoms through the subscales of nausea, oculomotor discomfort, and disorientation [5]. The SSQ remains highly relevant because of its robustness, interpretability, and widespread use in experimental research. However, it is fundamentally a retrospective measure, typically administered before and after exposure, which limits its usefulness for fine-grained temporal analysis or real-time adaptation [5].

To address the need for repeated assessments during exposure, shorter instruments were later introduced. A particularly influential example is the Fast Motion Sickness Scale (FMS), validated by Keshavarz and Hecht as an efficient verbal rating procedure for tracking motion sickness intensity over time [6]. Compared with longer questionnaires, the FMS is better suited to repeated in-experience measurements and therefore has become especially useful in studies where cybersickness must be represented as a dynamic temporal variable rather than as a single final outcome [6].

Even so, subjective measures have intrinsic limitations. They depend on user self-report, can interrupt immersion, and generally provide sparse labels when compared with the high-frequency sensor streams generated during VR interaction [1], [6], [8]. For this reason, recent research has increasingly shifted toward objective or semi-objective sensing approaches that can capture cybersickness continuously and with greater temporal resolution.

1.2.3. OBJECTIVE MEASUREMENT THROUGH PHYSIOLOGICAL AND BEHAVIORAL SIGNALS

One of the clearest trends in the literature is the move from questionnaire-based assessment toward the use of objective physiological and behavioral signals. Dennison et al. showed that physiological responses such as breathing, blinking, and other bodily signals can predict cybersickness-related outcomes, supporting the idea that cybersickness can be measured beyond self-report alone [7]. Likewise, Qu et al. demonstrated the feasibility of real-time cybersickness detection using bio-physiological signals, including electrodermal and cardiovascular variables [13].

Despite their value, approaches based on external physiological sensors present important practical limitations. They often require additional hardware, more complex acquisition protocols, and lower ecological validity for consumer-grade VR environments [7], [13]. For systems intended to operate in native VR conditions, signals already available from commercial head-mounted displays are especially attractive. In this regard, eye-tracking and head-tracking data have become prominent sources of information because they are continuous, non-invasive, and increasingly integrated into modern HMDs [1], [4].

The literature supports the relevance of gaze behavior as an indicator of cybersickness. Chang et al. found that eye-movement patterns in HMD-based VR were significantly associated with cybersickness levels, suggesting that gaze-related variables contain valuable information about the user's cybersickness state during VR exposure [9]. In parallel, head kinematics have also shown predictive value. Salehi et al. reported that head movement patterns, including features derived from motion signals and their temporal derivatives, constitute a promising source of information for cybersickness detection [14]. More broadly, the systematic review by Yang et al. concluded that physiological and behavioral data are central to current machine learning approaches for cybersickness research, although methodological heterogeneity remains considerable across studies [8].

These findings are especially relevant for work focused on native VR signals. This justifies the focus on variables that can be obtained directly from the headset and controller ecosystem, such as pupil-related measures, gaze geometry, eye openness, head rotation, and locomotion-related dynamics. This also aligns with the broader trend toward practical, scalable, and sensor-light systems for cybersickness prediction [1], [8], [9], [14].

1.2.4. MACHINE LEARNING FOR CYBERSICKNESS DETECTION AND PREDICTION

Once objective sensing became feasible, machine learning naturally emerged as a promising strategy for cybersickness modeling. Cybersickness develops through complex and potentially nonlinear interactions among multiple time-varying signals, and these dependencies are difficult to capture through simple handcrafted rules. The systematic review by Yang et al. shows that machine learning and deep learning have become major methodological directions for modeling cybersickness from physiological, behavioral, and multimodal data [8].

A large part of the literature in this area has focused on detection or severity estimation rather than on true anticipation. For example, machine learning pipelines have been trained to classify cybersickness states from physiological signals [7], [13], while more recent work has used built-in VR sensor streams and multimodal architectures to estimate cybersickness with strong performance [1]. RelaxVR is particularly relevant in this context because it introduces not only the MazeSick dataset but also a full predictive pipeline based on eye-tracking and head-tracking data collected during immersive maze navigation [1]. This work is especially important because it demonstrates that native VR signals can be sufficiently informative for high-performing cybersickness prediction and because it brings the field closer to deployable real-time systems.

However, much of the existing literature still frames the task as current-state classification rather than future escalation forecasting. In other words, many models aim to determine whether the user is currently experiencing cybersickness or belongs to a specific severity class, but not whether discomfort is likely to intensify within a forthcoming time horizon. This distinction is important for intervention design. Reactive systems act only after discomfort has become evident, whereas proactive systems seek to intervene before cybersickness escalates. From this perspective, the forecasting formulation remains comparatively underexplored and represents a relevant research gap [1], [8].

From a methodological point of view, the literature also supports the use of temporal feature extraction strategies when working with high-frequency VR streams. In particular, studies based on head and gaze dynamics show that short-time temporal descriptors can capture patterns associated with cybersickness more effectively than raw signals alone [9], [14]. This makes window-based feature engineering especially appropriate when the objective is to analyze pre-event signatures and build early-warning models.

1.2.5. CLASSICAL INTERPRETABLE AND STRONG BASELINE MODELS FOR TIME-WINDOWED VR DATA

In forecasting problems based on engineered temporal features, several classical machine learning models remain highly competitive. Logistic Regression is a particularly valuable baseline because of its interpretability, robustness, and compatibility with binary outcomes through thresholding and class-weighting strategies. In research centered on explainability and early warning, such properties are especially attractive because they facilitate the analysis of how individual features contribute to the final prediction [15].

Tree-based ensembles are equally relevant. Random Forests remain attractive because they are robust to nonlinear relationships, feature interactions, and moderate noise, while generally requiring less preprocessing than linear models [16]. For the type of tabular data generated after temporal windowing and feature extraction, this family of methods often provides a strong balance between predictive accuracy and manageable interpretability.

This is consistent with current practice in sensor-based tabular learning: before moving toward more complex deep architectures, it is methodologically sound to establish strong baselines using interpretable linear models and ensemble methods. In the context of a thesis focused on explainability and early prediction, this choice is not merely pragmatic but scientifically justified, since it allows performance comparisons while preserving analytical transparency [8], [15], [16].

1.2.6. EXPLAINABLE ARTIFICIAL INTELLIGENCE IN CYBERSICKNESS RESEARCH

One of the main limitations of many cybersickness prediction systems is that they behave as black boxes. High predictive performance is important, but in comfort-critical and user-centered applications, understanding why a model produces a given prediction is also essential. This is the rationale behind the growing incorporation of Explainable Artificial Intelligence (XAI) into cybersickness research [1], [17].

A key contribution in this area is SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee as a unified framework for additive feature attribution [10]. SHAP provides both local and global explanations by estimating how much each feature contributes to a prediction while preserving desirable theoretical properties such as local accuracy and consistency [10]. Because it can be applied to tree ensembles and other predictive models, SHAP has become one of the most widely adopted post hoc explainability methods in applied machine learning.

In cybersickness research, explainability has already shown clear value. Kundu et al., in TruVR, proposed explainable machine learning models for cybersickness detection and prediction and showed that the explanatory layer helped identify the variables most strongly associated with cybersickness across different datasets [17]. RelaxVR extends this line of work by integrating XAI into a real-time VR framework and using explanations not only to interpret predictions but also to guide mitigation decisions [1]. This is a significant conceptual shift, since explanation is no longer treated solely as a diagnostic tool for researchers, but also as an operational component that can support adaptive intervention during immersion.

For work focused on pre-event analysis, XAI is especially useful because it helps answer not only whether cybersickness escalation is predictable, but also which temporal patterns tend to precede it. In this way, explanation supports both model transparency and physiological interpretation, making SHAP particularly suitable for pipelines based on engineered features extracted from gaze, pupil, head, and motion variables [10].

1.2.7. MITIGATION STRATEGIES AND THE MOVE TOWARD ADAPTIVE VR SYSTEMS

Alongside cybersickness prediction research, another major branch of the literature has focused on cybersickness mitigation. Among software-based strategies, one of the most studied is dynamic field-of-view restriction. Fernandes and Feiner showed that subtle dynamic field-of-view modification can reduce VR sickness by limiting peripheral motion during user movement [18]. More recently, Wu and Rosenberg proposed an adaptive field-of-view restriction technique based on real-time optical-flow assessment, reinforcing the idea that mitigation should respond dynamically to the current state of the user and the scene rather than rely on a single fixed rule [19].

The literature also indicates that locomotion design strongly affects cybersickness and presence. Mayor et al. compared different interaction and locomotion methods in immersive environments and found meaningful differences in cybersickness, presence, and usability across techniques [12]. Likewise, Kim et al. showed that body position and locomotion method significantly shape both presence and cybersickness when navigating a virtual environment [11]. These findings are highly relevant because they imply that locomotion-related variables should not be treated as peripheral context, but as central explanatory factors in both prediction and mitigation.

Taken together, this body of work suggests a progression in the field: from static post-exposure assessment, to real-time detection, and finally to adaptive systems capable of monitoring users and responding during immersion [1], [18], [19]. Yet the transition from detection to anticipatory control remains incomplete. Many mitigation systems are activated only once cybersickness is already present or once a classifier identifies a problematic current state. Forecasting escalation before symptoms intensify remains comparatively underexplored.

1.2.8. RESEARCH GAP AND POSITIONING OF THE PRESENT THESIS

The literature supports four consolidated conclusions. First, cybersickness is a central obstacle to VR adoption and is associated with measurable changes in users' physiological responses and behavioral patterns [1], [7], [8]. Second, native HMD signals such as eye-tracking and head-tracking data contain useful predictive information [1], [9], [14]. Third, machine learning constitutes an effective framework for modeling cybersickness, especially when working with multimodal and time-dependent data [1], [8]. Fourth, explainability is becoming increasingly important both for scientific understanding and for operational trust in adaptive VR systems [1], [10], [17].

At the same time, an important gap remains open. Much of the existing work is centered on the detection of present sickness or on the classification of already observed severity, whereas fewer studies address the forecasting of imminent cybersickness escalation using a structured early-warning formulation. This gap is especially relevant for real-world VR applications, where the most useful system is not the one that confirms the user is already uncomfortable, but the one that anticipates rising risk and enables intervention before symptoms intensify.

This is precisely where the present thesis is positioned. In line with recent advances, but moving one step further, the work focuses on forecasting cybersickness escalation from native VR headset signals using a time-series pipeline based on temporal windows, engineered statistical and dynamic features, classical and ensemble machine learning baselines, and post hoc explainability through SHAP. From the perspective of the current state of the art, this positioning is well justified: it is technically grounded in the literature, practically relevant for deployable VR systems, and scientifically valuable because it addresses the still underdeveloped problem of pre-event signature analysis and anticipatory prediction [1], [8], [10].

2. DEVELOPMENT

The methodology followed in this work is designed to address the problem of anticipating cybersickness escalation through a structured and explainable machine learning pipeline based on time-series data obtained from native virtual reality (VR) sensors.

To achieve this objective, a multi-stage approach was implemented, integrating data preprocessing, temporal feature engineering, predictive modeling, and explainability analysis. The methodology is grounded in the characteristics of the MazeSick dataset [1], which consists of high-frequency multimodal signals such as gaze behavior, head motion, and locomotion dynamics, collected during immersive VR navigation tasks .

First, raw sensor data were processed to ensure temporal consistency, signal alignment, and data quality. As described in Section 2.2.3, an exploratory data analysis was performed to identify missing values, invalid measurements, duplicated samples, and potential signal artifacts. Subsequently, the preprocessing procedures detailed in Section 2.3 were applied to transform the raw recordings into a structured time-series dataset suitable for machine learning analysis. Given the high temporal resolution of VR data, particular attention was paid to preserving meaningful temporal patterns while reducing noise and artifacts.

Next, a sliding-window strategy was applied to segment the continuous signals into fixed-length temporal windows, as described in Sections 2.4.1 and 2.4.2. This approach enabled the extraction of short-term dynamics preceding potential cybersickness events. Within each window, a set of engineered features was computed, including statistical descriptors, dynamic measures, and behavioral indicators derived from gaze and head movement patterns. The complete feature extraction process is detailed in Section 2.5. This transformation is critical, as it converts raw sensor streams into informative representations that can be effectively used by machine learning models.

Subsequently, supervised learning models were trained to predict cybersickness escalation within a predefined forecasting horizon. The forecasting problem formulation is presented in Section 2.4.2, while the final forecasting dataset generation process is described in Section 2.4.3. Several machine learning and deep learning approaches were evaluated, including Logistic Regression, Random Forest, XGBoost, HistGradientBoosting, CatBoost, Support Vector Machines, Multi-Layer Perceptrons, one-dimensional Convolutional Neural Networks, and Long Short-Term Memory networks. The complete model development methodology is presented in Section 2.6. Time-aware and participant-aware validation strategies were employed to ensure realistic model evaluation conditions and to avoid information leakage.

Finally, explainability techniques based on SHAP (SHapley Additive exPlanations) were applied to interpret model predictions, as described in Section 3.11. This step allowed the identification of the most influential features contributing to cybersickness escalation, providing insights into the underlying physiological and behavioral mechanisms. In addition, it supported the analysis of pre-event temporal patterns, enabling the evaluation of whether the models captured meaningful anticipatory signals prior to symptom onset.

Overall, the methodology summarized in Figure 1 aims not only to achieve accurate prediction of cybersickness escalation, but also to provide interpretable and actionable insights that can contribute to the development of adaptive and user-centered VR systems.

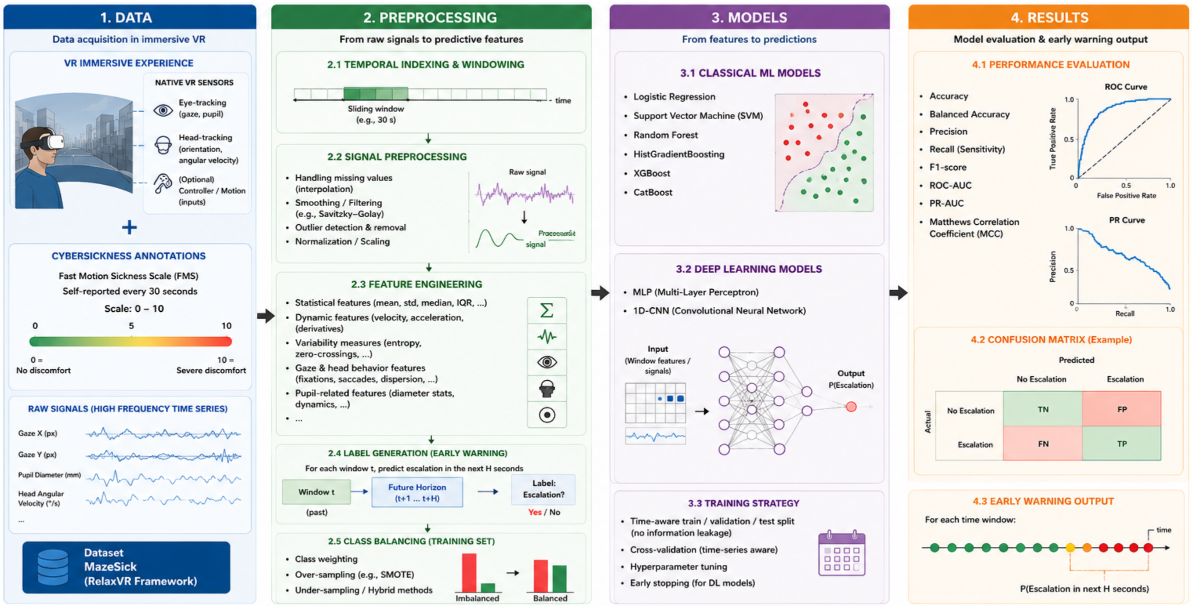


Figure 1. General overview of the proposed methodology, from VR sensor acquisition and preprocessing to machine learning modeling, explainability analysis, and cybersickness escalation forecasting [20].

2.1. NEUROPHYSIOLOGICAL BASIS OF CYBERSICKNESS

Cybersickness is considered a multisensory phenomenon resulting from conflicts between visual, vestibular, and proprioceptive information during exposure to immersive virtual environments. According to sensory conflict theory, discomfort emerges when the motion perceived through the visual system is inconsistent with signals provided by the vestibular system and body position sensors. This mismatch triggers a cascade of physiological and perceptual responses that may progressively evolve into symptoms such as nausea, dizziness, disorientation, visual fatigue, and oculomotor discomfort [2], [3].

From a neurophysiological perspective, eye-tracking signals provide a particularly valuable source of information for studying cybersickness because they reflect multiple processes involved in visual perception and sensorimotor integration. Variables such as gaze direction, pupil diameter, convergence distance, and eye openness are closely linked to visual attention, oculomotor control, cognitive workload, and autonomic nervous system activity.

Several studies have reported that cybersickness is associated with alterations in gaze stability, eye-movement behaviour, blinking activity, and other physiological responses observable during VR exposure [7], [9], [13]. For example, increased visual instability and atypical gaze patterns may emerge as users attempt to resolve conflicts between expected and perceived motion [9]. Similarly, physiological indicators such as blinking behaviour and pupillary responses have been linked to increased cognitive load and stress during immersive experiences [7], [13].

Head-motion signals also play a relevant role in cybersickness research. Head orientation and movement directly influence the visual information received by the user and interact with vestibular processing mechanisms. Abrupt or inconsistent head-motion patterns may contribute to visuovestibular conflicts and have been identified as potential indicators of cybersickness risk.

Figure 2 illustrates the sensory conflict theory, one of the most widely accepted explanations of cybersickness. The figure is adapted from the model originally proposed by Reason and Brand [3], which explains how discrepancies between expected and actual sensory information are processed by neural comparison mechanisms, ultimately leading to motion-sickness-related symptoms.

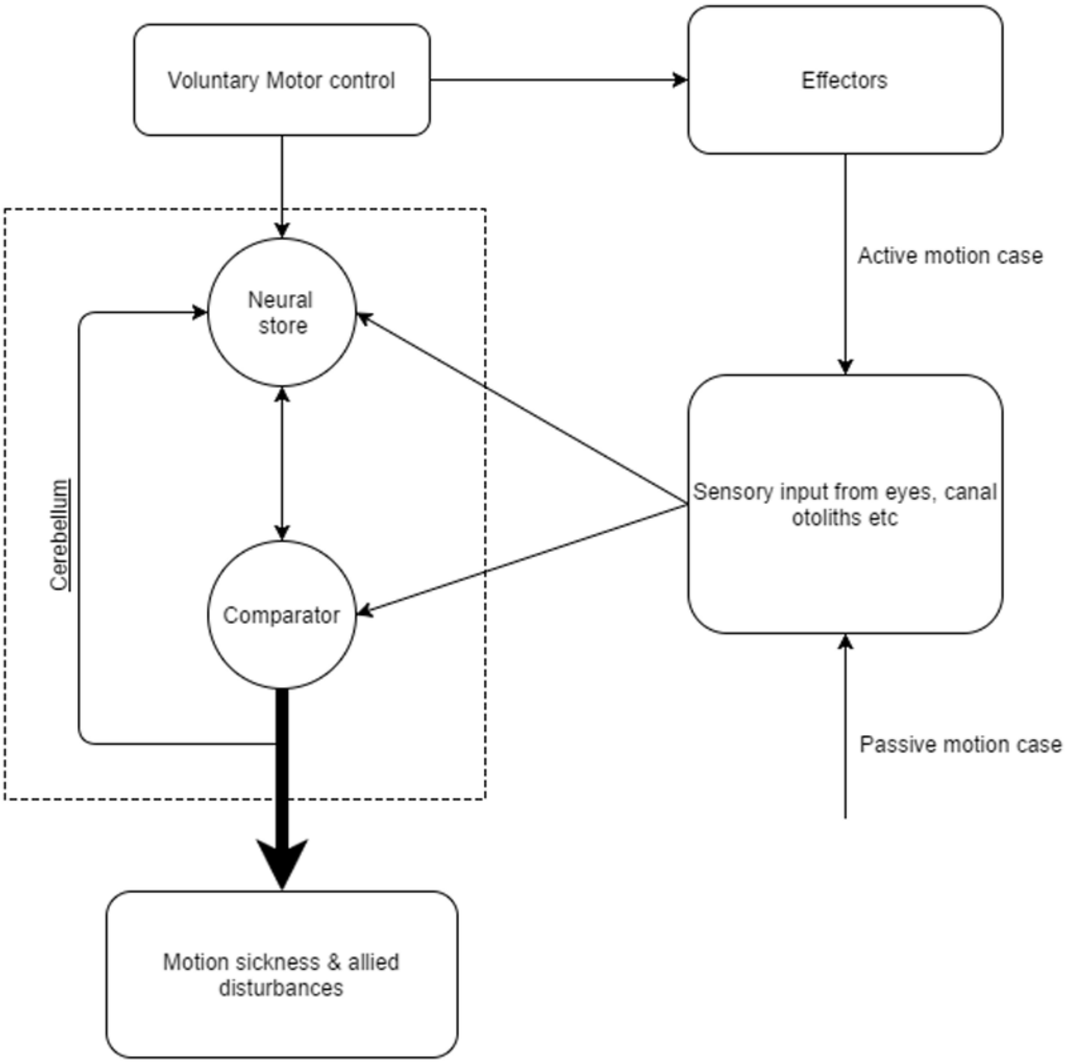


Figure 2. Sensory conflict model explaining the neurophysiological mechanisms underlying motion sickness and cybersickness (adapted from Reason & Brand [3]).

Figure 3 connects these neurophysiological mechanisms with the predictive framework adopted in this thesis. The figure illustrates how sensory conflict may trigger autonomic and perceptual responses that manifest as cybersickness symptoms. These processes can be indirectly observed through VR-native signals, such as pupil diameter, eye openness, gaze behaviour, convergence distance, and head-motion dynamics, which constitute the input variables used for machine learning-based prediction and forecasting.

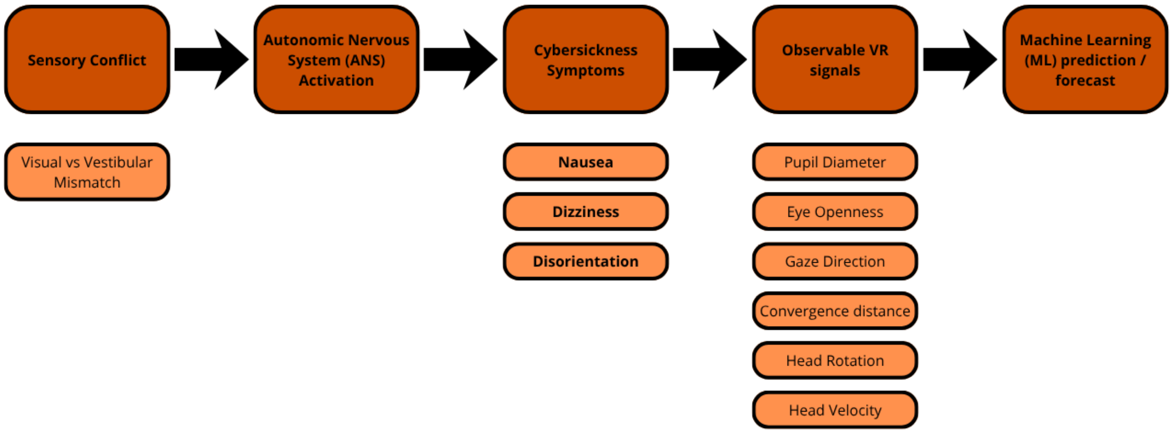


Figure 3. Conceptual relationship between sensory conflict, cybersickness symptoms, observable VR signals, and machine-learning-based prediction.

The neurophysiological relevance of these variables provides the rationale for focusing on eye-tracking and head-motion signals throughout the present work. Since these measurements are available directly from modern VR head-mounted displays, they constitute a practical and scalable source of information for developing objective cybersickness prediction systems.

2.2. DATA LOADING & EXPLORATORY DATA ANALYSIS (EDA)

2.2.1. DATASET ORIGIN AND EXPERIMENTAL PROTOCOL

The dataset [1] employed in this work originates from the MazeSick dataset introduced by Kundu et al. as part of the RelaxVR framework for cybersickness detection and mitigation in virtual reality. The dataset was collected through a controlled user study involving 60 participants (39 male and 21 female, aged between 18 and 38 years) using an HTC Vive Pro Eye headset equipped with integrated eye-tracking capabilities.

Participants were immersed in a custom-designed dungeon-style maze environment developed in Unity. The experimental task consisted of navigating through a series of randomly generated mazes using smooth locomotion while searching for and collecting virtual objectives. The use of smooth locomotion was intentional, as it is known to induce stronger cybersickness responses than teleportation-based navigation, making it suitable for studying cybersickness development under realistic VR conditions.

Before the experiment, participants completed a short training session and underwent eye-tracking calibration. The main VR session lasted up to 12 minutes, during which eye-tracking and head-tracking signals were continuously recorded. In parallel, participants verbally reported their perceived discomfort using the Fast Motion Sickness Scale (FMS), ranging from 0 (no discomfort) to 10 (severe discomfort), every 30 seconds following an automated audio prompt, providing a compromise between temporal resolution and participant burden during immersion. This procedure enabled the collection of synchronized objective sensor measurements and subjective cybersickness assessments throughout the VR exposure.

The resulting dataset contains multimodal time-series recordings combining eye-tracking variables (e.g., pupil diameter, gaze direction, gaze origin, convergence distance, and eye openness) together with head-motion information derived from head position and orientation signals. These measurements constitute the basis for the predictive analyses developed in this thesis.

Figure 4 summarizes the complete data acquisition pipeline, from participant recruitment and calibration to sensor acquisition and cybersickness annotation. Figure 5 shows the virtual maze environment used during the experiment.

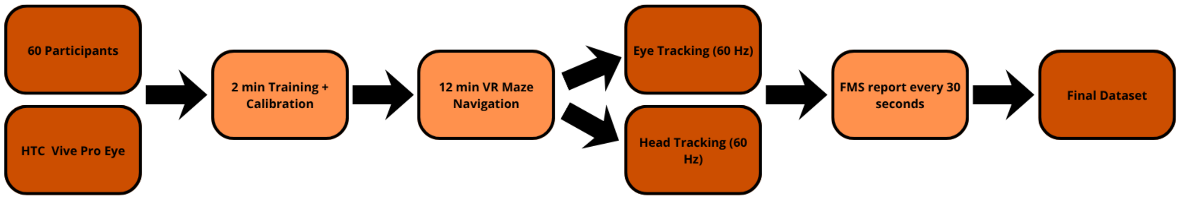


Figure 4. Overview of the MazeSick data acquisition protocol, including participant preparation, VR exposure, multimodal sensor acquisition, and FMS-based cybersickness annotation.

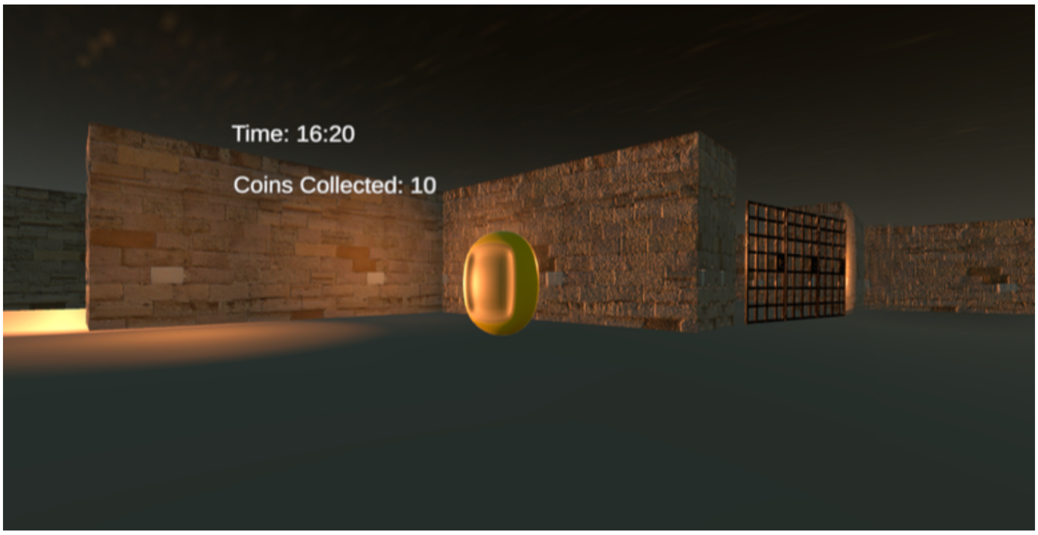


Figure 5. First-person view of the virtual maze environment used during the MazeSick data collection protocol (adapted from Kundu et al. [1]).

2.2.2. DATASET STRUCTURE AND VARIABLE CHARACTERIZATION

Following the experimental protocol described above and summarized in Figure 4 and Figure 5, an initial structural inspection was performed to determine the dimensionality and composition of the dataset. The dataset contained 76,500 samples and 42 original variables. All variables were numerical, with most features stored as floating-point values, whereas the target variable, FMS, was stored as an integer value. This confirmed that the dataset was directly suitable for numerical analysis and machine learning modelling after appropriate preprocessing.

The variables were organized into functional groups according to the type of neurotechnological information represented. Basic eye-related variables included pupil diameter, eye openness, pupil position in the sensor plane, and convergence distance. Gaze-related variables were separated into local-space and world-space representations, allowing both headset-relative and environment-relative visual behavior to be considered. Normalized eye variables represented standardized gaze origin and gaze direction information for both eyes. Finally, head-motion variables included quaternion rotations, Euler angles, and velocity. This grouping was introduced to facilitate subsequent interpretation, since eye dynamics, gaze orientation, and head movement are directly related to oculomotor control, visuovestibular integration, and sensorimotor adaptation during VR exposure.

To better understand the composition of the recorded signals, Table 1 presents the distribution of variables across the main sensor categories used throughout this work. As expected, eye-tracking measurements constitute the majority of the available features, reflecting the central role of gaze behaviour and oculomotor activity in the study of cybersickness.

Table 1. Nº features by category

Total Numerical Features	**41**
Eye Basic	9
Gaze Local	6
Gaze World	6
Normalized Eye	12
Head	8

2.2.3. DATA QUALITY ASSESSMENT

A preliminary data quality analysis was then conducted. No missing values were detected in the dataset, which avoided the need for imputation at this stage. This is particularly important in time-series modelling, since imputation may introduce artificial temporal patterns if not carefully controlled. In contrast, duplicated samples were identified, although their proportion was very small in comparison to the full dataset. Constant and near-constant variables were also examined to detect features with limited informational value. In addition, descriptive statistics revealed that several variables contained values equal to -1. These values were considered potentially invalid acquisition values, especially in eye-tracking-related measurements such as pupil diameter or eye openness, where negative values are not physiologically meaningful. Consequently, these observations were flagged for later preprocessing rather than being immediately removed during the exploratory phase.

2.2.4. ANALYSIS OF THE TARGET VARIABLE (FMS)

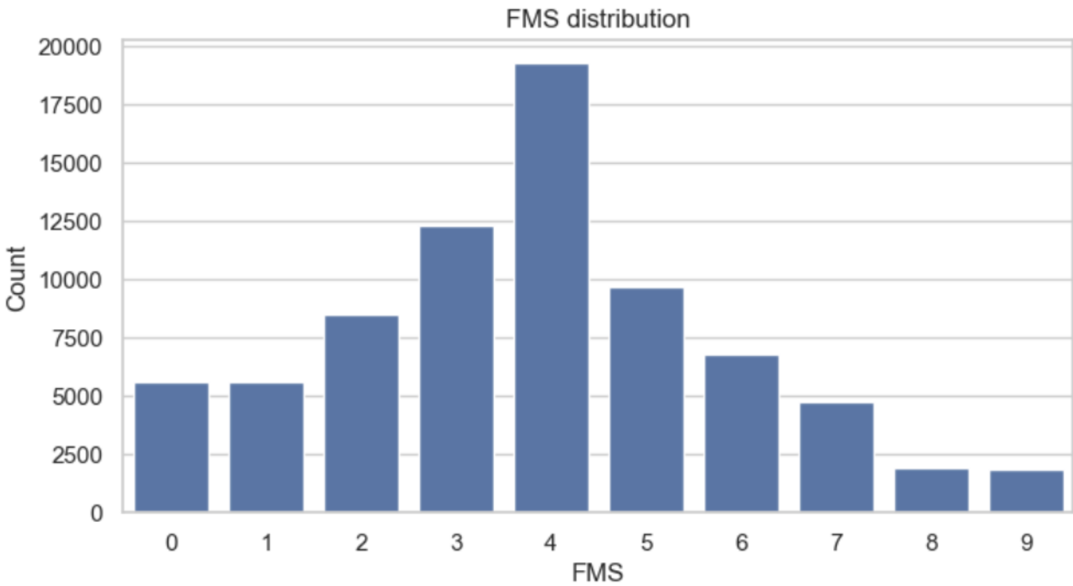


Figure 6. Distribution of FMS values across the dataset.

The target variable selected for the analysis was the Fast Motion Sickness Scale (FMS), which represents the subjective intensity of cybersickness reported by the participant. The distribution of FMS values is shown in Figure 6. All severity levels from 0 to 9 were present, indicating that the dataset covered the full range of reported cybersickness intensity. However, the distribution was not uniform. Most samples were concentrated around intermediate values, especially FMS levels 3, 4, and 5, whereas extreme high values such as 8 and 9 were less frequent. From a modelling perspective, this indicates a moderate class imbalance that must be considered in later stages, particularly because the objective of the thesis is not only to model common discomfort states, but also to anticipate escalation toward higher cybersickness levels.

From a neurotechnology perspective, this distribution is relevant because cybersickness is not represented as a binary phenomenon but as a gradual perceptual and physiological process. Intermediate FMS values may correspond to early or moderate stages of sensory conflict, in which the nervous system is already processing mismatched visual, vestibular, and proprioceptive information but severe symptoms have not yet fully developed [2], [3]. Therefore, the presence of many intermediate samples is useful for studying the transition between comfortable and uncomfortable states, which is central to the early-warning objective of this thesis.

2.2.5. TEMPORAL STRUCTURE OF CYBERSICKNESS REPORTS

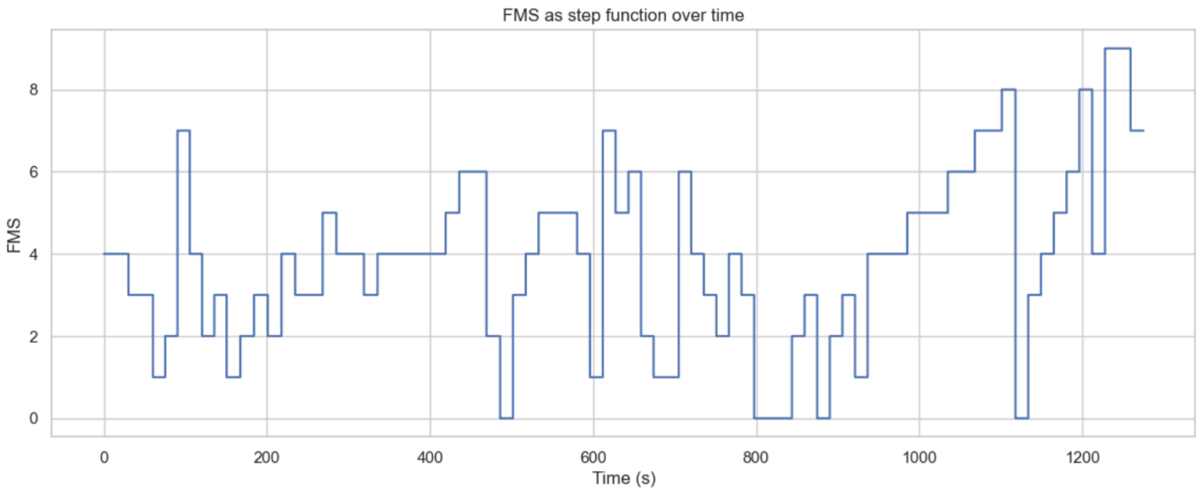


Figure 7. Distribution of FMS values across time.

The temporal evolution of FMS was then analyzed over the sample index and over time. Since the FMS value was maintained until a new report was available, the signal appeared as a step function rather than as a continuous measurement. This behavior is shown in Figure 7. The step-like structure reflects the subjective reporting mechanism: cybersickness ratings are updated discretely, whereas gaze, pupil, and head-motion signals are recorded continuously at 60 Hz. This mismatch between label frequency and sensor frequency is a key methodological aspect of the study, because the predictive model must learn from continuous physiological and behavioral dynamics while the target variable changes only at specific reporting points.

To account for the repeated structure of the dataset, participant-level temporal variables were created. First, participant_id was generated by dividing the global dataframe index into 60 equally sized participant segments. This variable identifies which participant each sample belongs to. Then, participant_sample_idx was created to represent the temporal order of each sample within each participant recording. Based on this index, participant_time_sec was computed by dividing participant_sample_idx by the sampling frequency of 60 Hz. This variable represents the elapsed time in seconds within each participant's VR exposure. In parallel, two global variables, sample_idx and time_sec, were also created to preserve compatibility with previous exploratory visualizations. These global variables describe the position of each sample in the complete concatenated dataset, but participant-level time variables are more appropriate for analyses that must avoid mixing temporal information across participants.

After constructing the temporal index, changes in FMS were explicitly identified. A binary variable named fms_change was created within each participant sequence. This variable takes the value 1 when the current FMS value differs from the previous sample of the same participant, and 0 otherwise. Therefore, it marks the moments at which a new cybersickness rating block begins. Based on the cumulative sum of fms_change, a second variable named fms_block_id was created. This variable assigns a block identifier to each continuous segment in which the FMS value remains constant. Since block identifiers are repeated across participants, a global_fms_block_id was also created by combining participant_id and fms_block_id. This ensured that each FMS block could be uniquely identified across the complete dataset.

The block-level representation was summarized by computing, for each participant and FMS block, the FMS value, the starting sample, the ending sample, the number of samples, the duration in seconds, and the start and end times. This transformation was important because it converted the original sample-level dataset into a temporal description of subjective cybersickness episodes. In other words, each block represented a period during which the participant reported a stable level of discomfort.

The distribution of FMS block durations is shown in Figure 8. Most blocks lasted between approximately 1 and 21 seconds, with a concentration around short-to-intermediate durations. This confirms that the subjective label does not change at every sensor sample, but remains stable over short temporal periods. The existence of these blocks supports the use of temporal windows in later stages of the pipeline, because cybersickness escalation can be formulated as a prediction problem based on the sensor dynamics observed before a future change in reported discomfort.

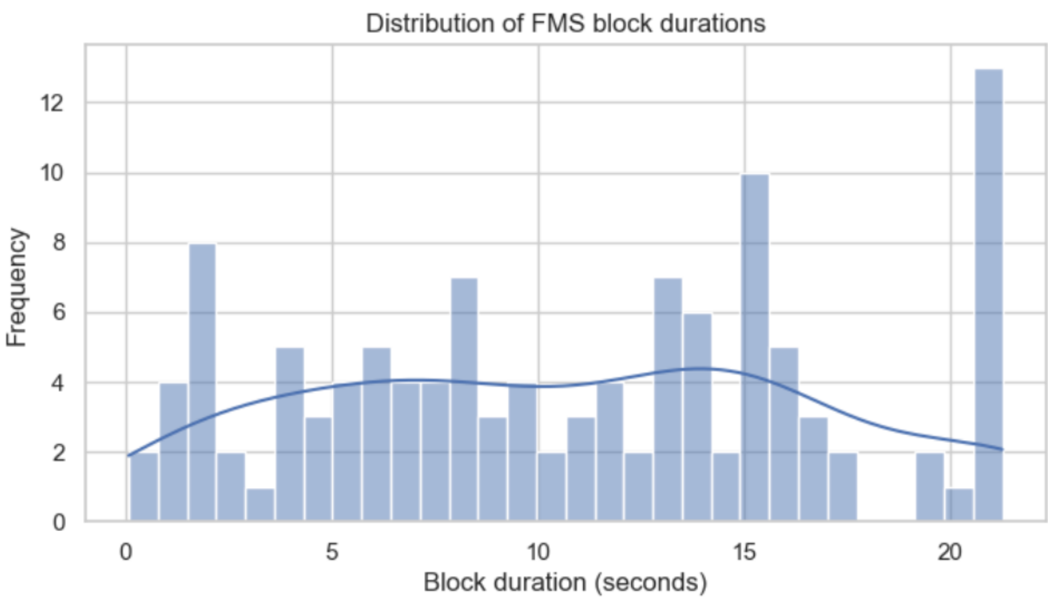


Figure 8. Distribution of FMS block durations.

The transition matrix between consecutive FMS blocks is shown in Figure 9. The matrix was computed by comparing the FMS value of each block with the FMS value of the following block. The strongest values were located along the diagonal and near-diagonal positions, indicating that many consecutive blocks either maintained the same FMS level or transitioned to nearby levels. This suggests that cybersickness evolves progressively rather than through completely random changes. For example, transitions around FMS levels 3, 4, 5, and 6 were particularly frequent, which is consistent with the central concentration observed in the FMS distribution.

This result is important for the objective of the thesis. If cybersickness progression tends to follow gradual trajectories, then early prediction becomes more plausible, because the sensor signals preceding an escalation may contain detectable pre-event patterns. From a neurophysiological point of view, this gradual evolution is consistent with the idea that cybersickness emerges from the progressive accumulation of sensory mismatch, oculomotor strain, and postural or vestibular instability rather than from an instantaneous event [2], [3]. Therefore, the transition analysis supports the decision to model cybersickness escalation as a time-dependent forecasting problem instead of treating each sample as an independent classification instance.

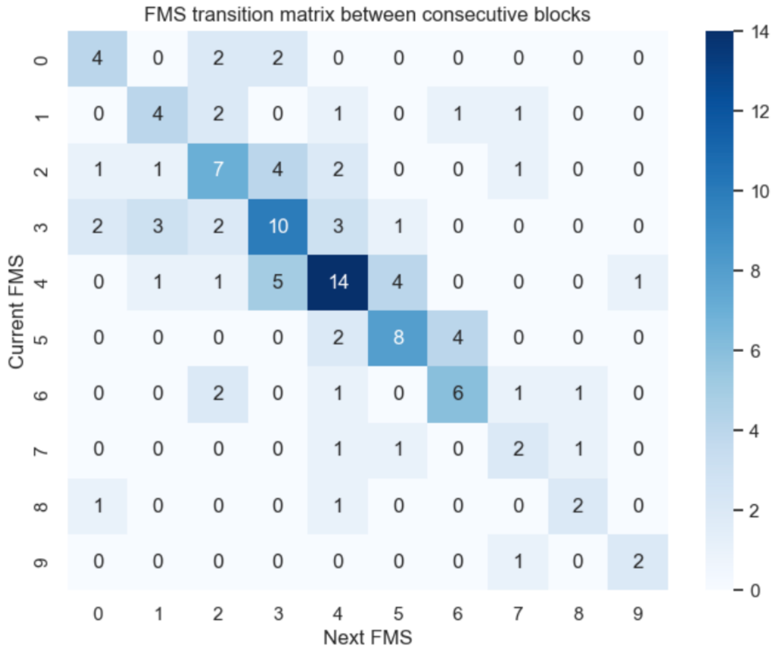


Figure 9. FMS transition matrix between consecutive blocks.

2.2.6. CORRELATION AND REDUNDANCY ANALYSIS

A correlation analysis was then performed to obtain an initial estimate of the linear association between each numerical feature and the FMS score. Temporal helper variables such as sample_idx, time_sec, fms_change, and fms_block_id were excluded from this analysis to avoid introducing artificial correlations derived from indexing or label construction. The most positively and negatively correlated features are summarized in Figure 10.

The strongest positive correlations with FMS were moderate rather than high. These included participant_id, NrmLeftEyeOriginY, HeadEulX, GazeOriginLclSpc_Y, NrmRightEyeOriginY, convergence distance, and right pupil diameter. Negative correlations were observed for variables such as gaze direction components, eye openness, and HeadEulZ. The presence of only moderate correlations suggests that cybersickness cannot be explained by a single isolated signal. Instead, it is likely associated with multivariate interactions between gaze behavior, ocular dynamics, head orientation, and locomotion-related variables.

This finding is particularly relevant from a neurotechnology perspective. Cybersickness is understood as a multisensory phenomenon involving visual, vestibular, proprioceptive, and oculomotor systems [2], [3]. Therefore, it is expected that no individual sensor variable fully captures the underlying state of discomfort. Rather, the predictive information may emerge from temporal combinations of variables, such as changes in gaze stabilization, head orientation, convergence behavior, eye openness, or movement dynamics. This justifies the later use of feature engineering and machine learning models capable of capturing nonlinear and multivariate relationships.

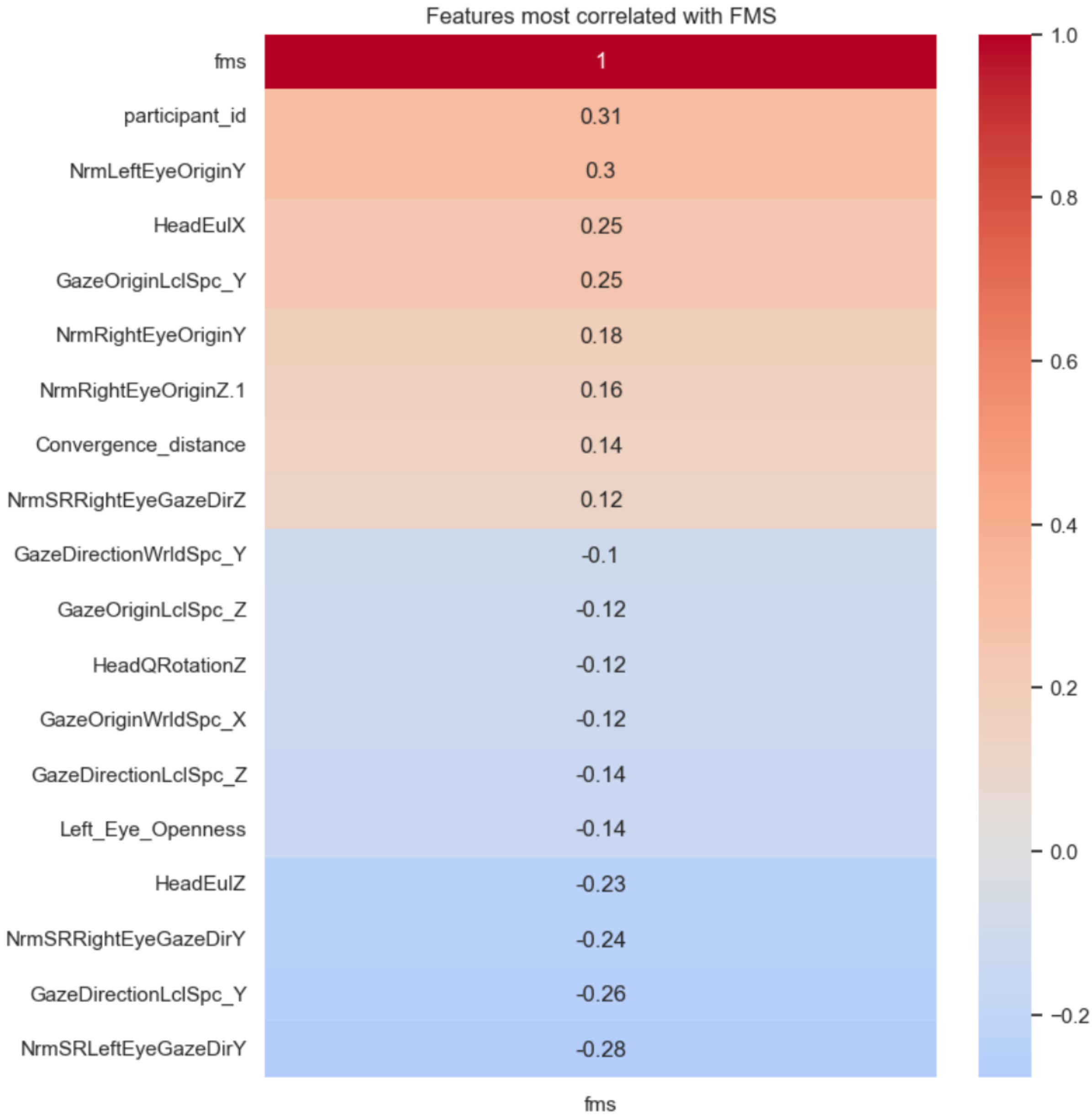


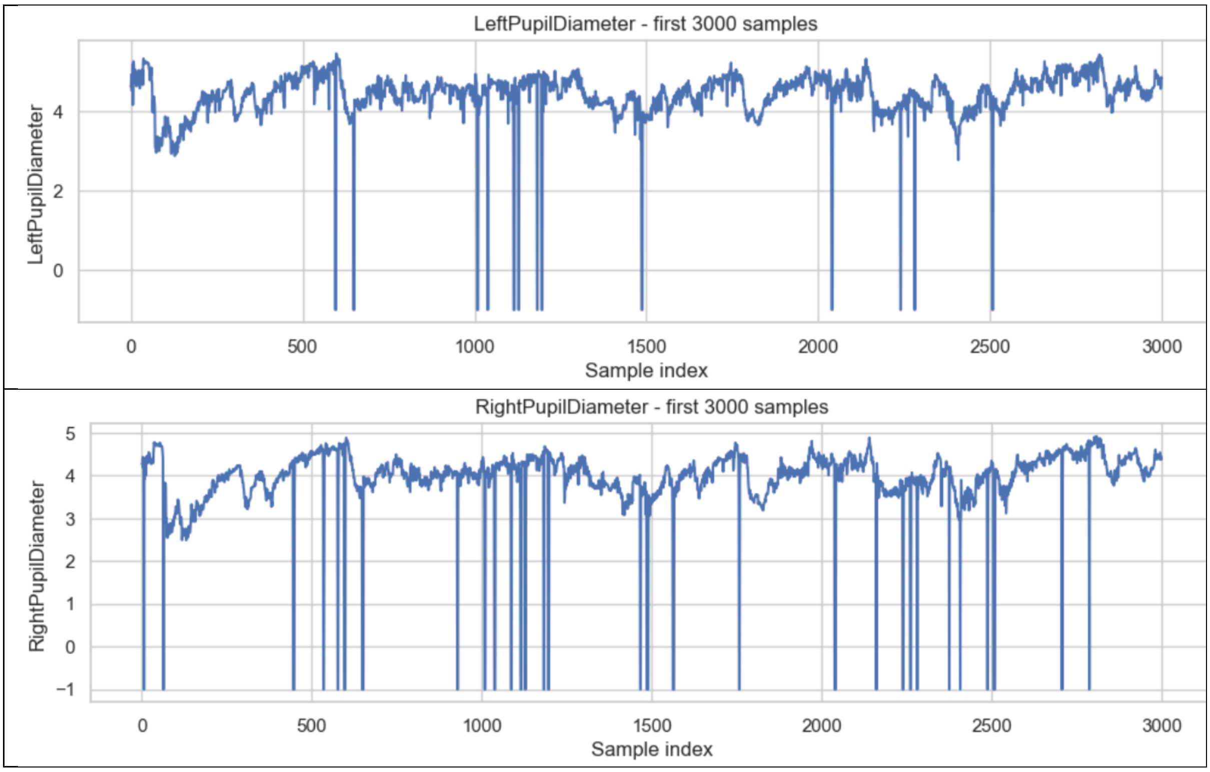
Figure 10. Features most correlated with FMS.

Feature-to-feature correlations were also analyzed to identify redundant variables. Pairs with absolute correlation values above 0.95 were considered highly redundant. These highly correlated pairs mainly involved variables representing closely related eye or gaze measurements, such as pupil diameter, pupil position, and normalized gaze direction components. This redundancy is expected in VR datasets because the same underlying physiological or spatial behavior can be encoded through several coordinate systems or sensor-derived representations. However, it also indicates that feature selection or regularization may be necessary in later modelling stages to reduce multicollinearity and improve model stability.

2.2.7. RAW SIGNAL INSPECTION

Finally, raw signal visualizations in Figure 11 were generated for selected variables during the first 3,000 samples. These plots included pupil diameter, convergence distance, head Euler angles, and velocity. The visual inspection revealed several important characteristics. First, pupil diameter signals showed abrupt drops to negative values, supporting the previous hypothesis that -1 values correspond to acquisition artifacts or invalid eye-tracking readings. Second, convergence distance displayed occasional sharp peaks, suggesting transient tracking instability or momentary changes in binocular convergence estimation. Third, head-angle variables showed discontinuities and abrupt jumps, especially in Euler-angle representations, which may reflect angular wrapping effects or rapid changes in head orientation. These observations confirmed that preprocessing was required before extracting predictive features.

From a neurotechnological perspective, these artifacts must be handled carefully. Pupil diameter, eye openness, and convergence distance are meaningful only when the eye-tracking system reliably captures ocular behavior. Invalid values may correspond to blinks, partial eyelid occlusions, gaze loss, or headset tracking errors. Similarly, head orientation signals are essential for studying visuovestibular interaction, but discontinuities in angular representations may distort derivative-based features such as velocity or acceleration if they are not treated appropriately. Therefore, the exploratory analysis provided the basis for the preprocessing stage, where invalid values, artifacts, and redundant representations were addressed before constructing the final machine learning dataset.



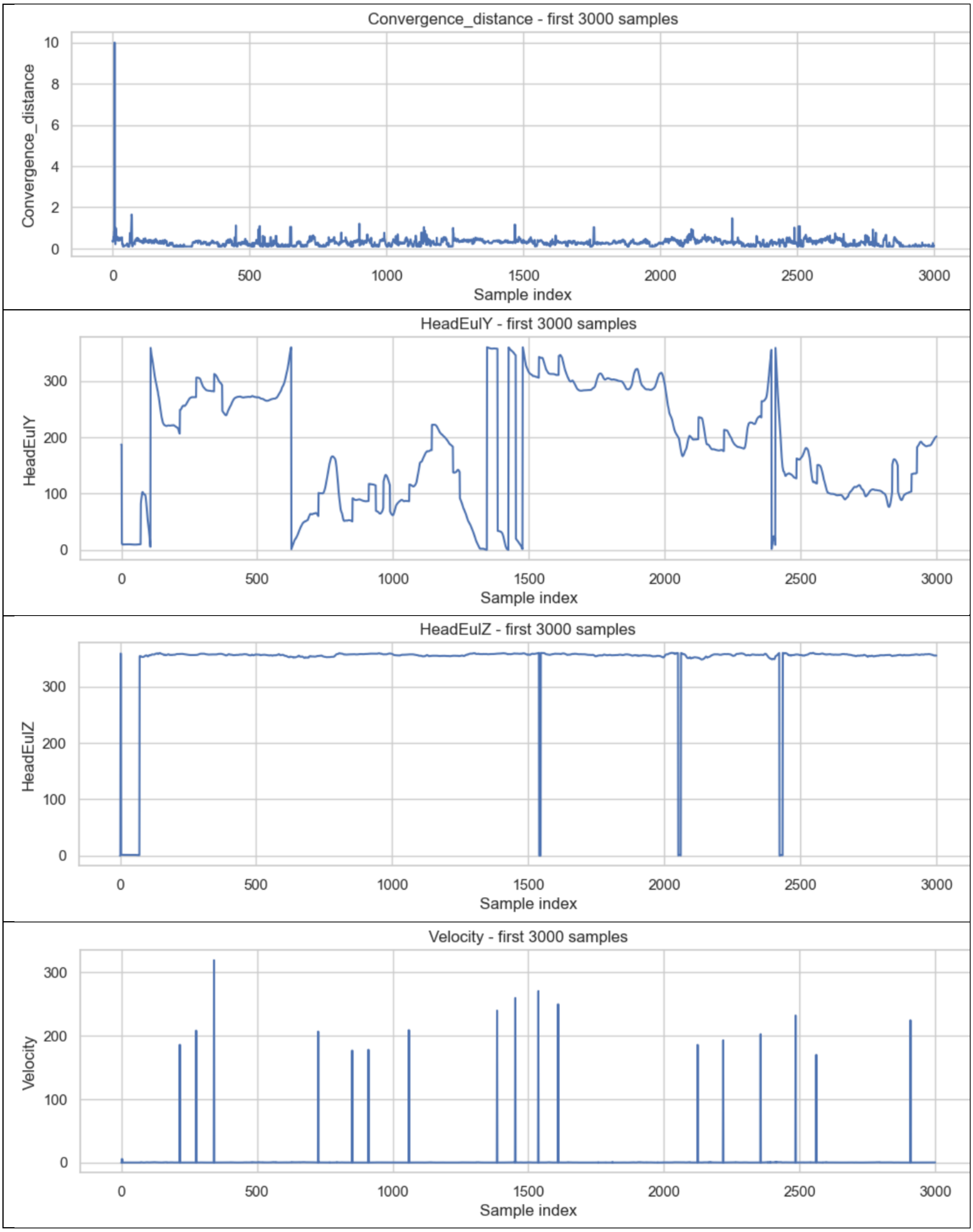


Figure 11. Raw signal inspection for selected eye-tracking and head-motion variables.

2.2.8. EXPLORATORY NEUROPHYSIOLOGICAL FEATURES

The original dataset contained a large number of low-level eye-tracking, gaze, and head-motion variables. While these measurements provide detailed information about participant behavior, some of them are difficult to interpret directly from a physiological or neurotechnological perspective. For this reason, a set of derived exploratory features was created to obtain more compact and interpretable descriptors of ocular and head-motion behavior.

The objective of this step was not to generate the final features used for modelling, but rather to explore whether higher-level representations of eye activity, gaze behavior, and head movement exhibited meaningful relationships with cybersickness severity.

Four derived variables were generated.

The first variable, **pupil_asymmetry**, was computed as the difference between the left and right pupil diameters:

$$pupil\_asymmetry = LeftPupilDiameter - RightPupilDiameter$$

This variable quantifies the degree of asymmetry between both pupils. Under normal conditions, pupil diameters are expected to remain relatively similar between eyes. Therefore, large deviations may indicate measurement artifacts, tracking instability, or unusual oculomotor responses.

The second variable, **eye_openness_mean**, was calculated as the average openness of both eyes:

$$eye\_openness\_mean = \frac{Left\_Eye\_Openness + Right\_Eye\_Openness}{2}$$

This feature provides a global measure of eyelid opening and may reflect blinking behavior, fatigue, visual discomfort, or temporary eye-tracking loss. From a neurophysiological perspective, changes in eye openness may be associated with oculomotor strain and visual fatigue during prolonged virtual reality exposure.

The third variable, **head_rotation_mag**, was computed as the Euclidean magnitude of the three Euler-angle components describing head orientation:

$$head\_rotation\_mag = \sqrt{HeadEulX^2 + HeadEulY^2 + HeadEulZ^2}$$

This variable summarizes the overall magnitude of head orientation in a single scalar descriptor. Since head movement directly influences vestibular stimulation and visual scene perception, it may be associated with sensory conflict mechanisms involved in cybersickness development.

The fourth variable, **gaze_direction_world_mag**, was obtained as the magnitude of the world-space gaze direction vector:

$$gaze\_direction\_world\_mag = \sqrt{GazeDirectionWrldSpc\_X^2 + GazeDirectionWrldSpc\_Y^2 + GazeDirectionWrldSpc\_Z^2}$$

Because gaze direction vectors are expected to be normalized, this variable mainly serves as a quality-control indicator rather than as a direct physiological measurement.

Figure 12 shows the distributions of the four derived variables across the different FMS levels.

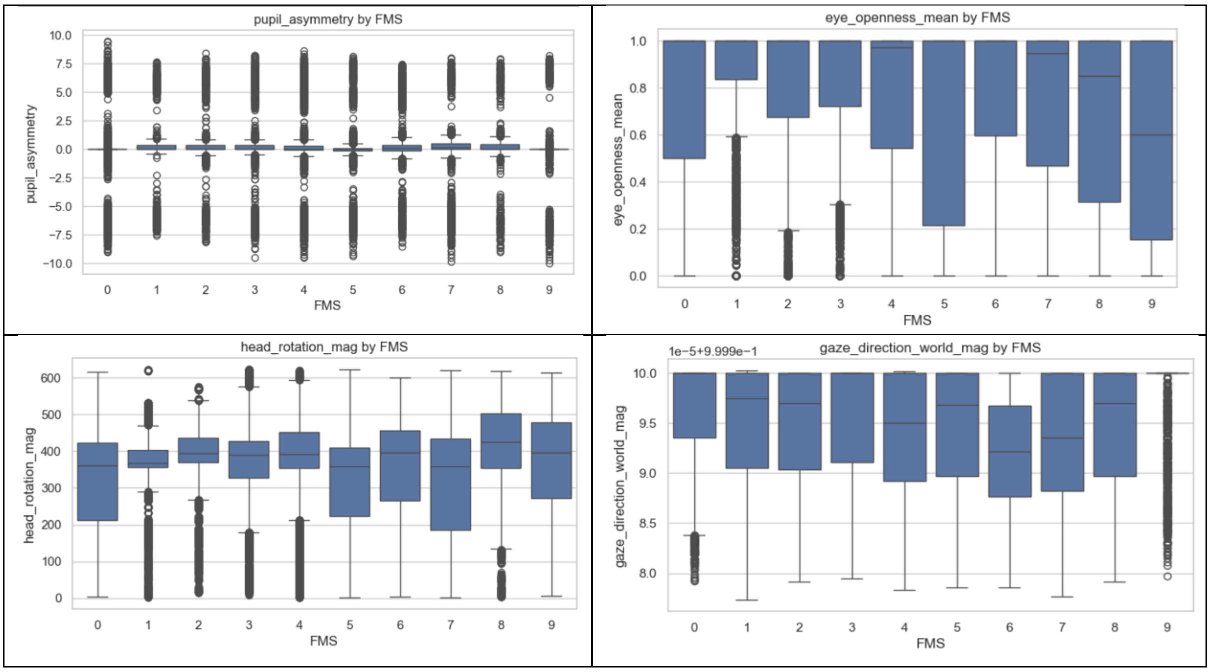


Figure 12. Distribution of derived exploratory features across FMS levels. The figure shows the distributions of pupil asymmetry, mean eye openness, head rotation magnitude, and gaze direction magnitude for each Fast Motion Sickness Scale (FMS) level.

CONCLUSIONS

The distributions reveal several relevant patterns. The **pupil asymmetry** variable remained centered around zero for most FMS levels, indicating that the average pupil diameter was generally similar in both eyes. However, a considerable number of extreme positive and negative values was observed. Since strong physiological anisocoria is not expected in healthy participants under these experimental conditions, these extreme values likely correspond to eye-tracking inaccuracies, partial eye occlusions, blinks, or temporary loss of pupil detection.

The **eye_openness_mean** distributions exhibited substantial variability across FMS levels. Although most observations remained close to complete eye opening, several samples showed values approaching zero. These observations may correspond to blinking events, partial eyelid closure, or tracking failures. Interestingly, higher FMS levels appeared to present greater variability in some cases, suggesting that ocular behavior may become less stable as discomfort increases.

The **head_rotation_mag** variable displayed the clearest differences between FMS levels. While a perfectly monotonic relationship was not observed, higher FMS values generally exhibited larger variability and elevated median values. This observation is consistent with the hypothesis that head-motion behavior may contribute to cybersickness development. From a neurotechnological perspective, head rotations modify vestibular stimulation while simultaneously altering the visual information received through the headset. When the resulting sensory information does not match the predictions generated by the central nervous system, sensory conflict may emerge, contributing to the appearance of cybersickness symptoms [2], [3].

Finally, **gaze_direction_world_mag** remained nearly constant across all FMS levels, as expected for a normalized direction vector. This confirms the internal consistency of the gaze-direction measurements and suggests that the magnitude itself is unlikely to provide discriminative information regarding cybersickness severity.

To complement the previous analysis, the average value of both original and derived variables was computed separately for each FMS level. This aggregation provides a higher-level view of how physiological and behavioral measurements evolve as cybersickness severity changes.

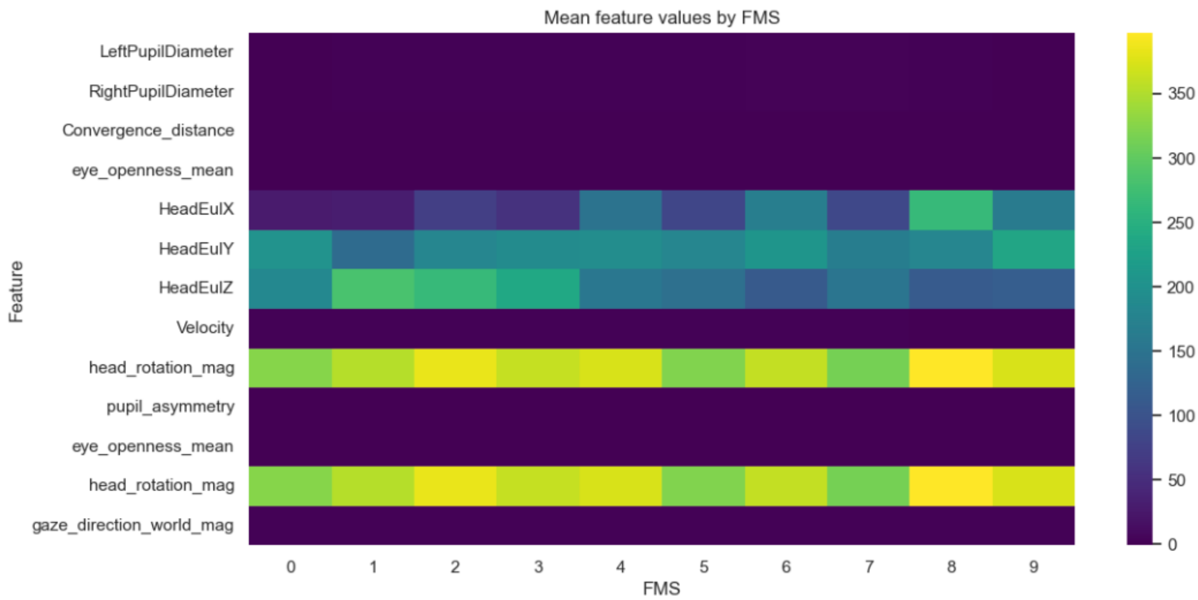


Figure 13. Mean feature values grouped by FMS level. Rows correspond to physiological and behavioral variables, whereas columns represent the different FMS scores

The heatmap (Figure 13) highlights that head-related variables exhibit the largest variation across FMS levels. In particular, **HeadEulX**, **HeadEulY**, **HeadEulZ**, and **head_rotation_mag** show noticeable changes in their average values between different cybersickness levels.

In contrast, variables such as pupil diameter, pupil asymmetry, gaze-direction magnitude, and velocity remain relatively stable when averaged across all samples belonging to the same FMS category.

This result suggests that head-orientation behavior may contain information associated with cybersickness severity. However, it is important to note that the observed differences are relatively moderate. Therefore, the relationship between physiological signals and cybersickness is unlikely to be explained by a single variable. Instead, it is expected that meaningful predictive information emerges from combinations of multiple signals and their temporal evolution.

This observation is consistent with current neurophysiological theories of cybersickness, which describe the phenomenon as the result of complex interactions between visual, vestibular, proprioceptive, and oculomotor systems rather than as a response driven by a single physiological mechanism [2], [3].

2.2.9. PARTICIPANT-LEVEL TEMPORAL DYNAMICS

The temporal behavior of cybersickness was further examined at the participant level by analyzing the number of FMS transitions observed during each recording session.

For each participant, the number of times that the reported FMS value changed was computed. The resulting distribution is shown in Figure 14.

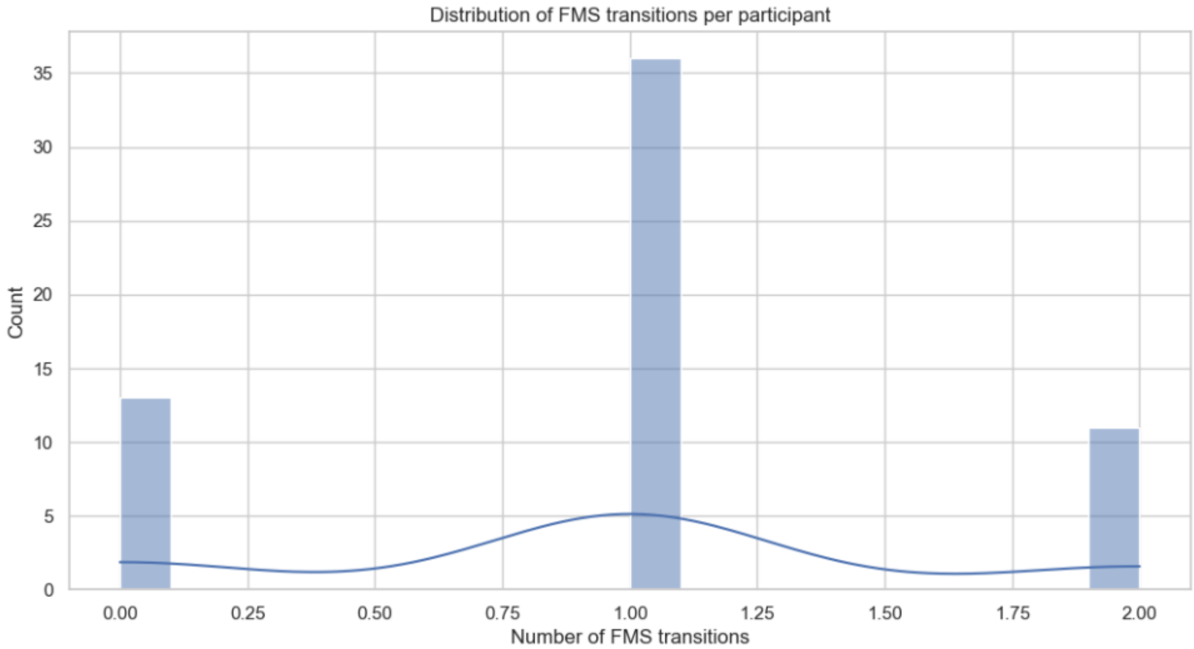


Figure 14. Distribution of the number of FMS transitions per participant.

The distribution reveals that most participants experienced only one FMS transition throughout the experiment, while a smaller number of participants presented either no transitions or two transitions.

This finding confirms that subjective cybersickness ratings evolve considerably more slowly than the physiological signals recorded at 60 Hz. Consequently, consecutive samples cannot be considered independent observations from the perspective of the target variable.

The relatively low number of transitions also supports the use of temporal modelling approaches. Since FMS values remain stable during extended periods and change only occasionally, predictive models must rely on patterns emerging in the physiological signals before a transition occurs rather than on instantaneous measurements alone.

This observation is particularly relevant for the subsequent feature-engineering stage, where temporal windows will be constructed to capture the physiological dynamics preceding changes in perceived discomfort.

2.3. DETECTION OF CYBERSICKNESS ESCALATION EVENTS

After characterizing the overall distribution of FMS values and their temporal structure, a dedicated analysis was conducted to identify cybersickness escalation events. These events constitute the central phenomenon investigated in this thesis, since the final objective is not only to estimate the current discomfort level but also to anticipate future increases in cybersickness severity.

To detect escalation events, the difference between consecutive FMS values was computed independently for each participant. This operation generated a new variable named **fms_diff**, defined as the difference between the current FMS value and the previous FMS value within the same participant sequence:

$$fms\_diff_t = FMS_t - FMS_{t-1}$$

Positive values indicate that the participant reported a higher cybersickness level than in the previous observation period, whereas negative values correspond to reductions in perceived discomfort.

Based on this variable, a binary indicator named **is_escalation** was created. This variable takes the value 1 whenever the FMS score increases with respect to the previous observation and 0 otherwise:

$$is\_escalation = \begin{cases} 1 & \text{if } fms\_diff > 0 \\ 0 & \text{otherwise} \end{cases}$$

The introduction of this variable is particularly important because it transforms the original subjective rating signal into a set of discrete events representing worsening cybersickness states. From a machine learning perspective, these events define the future occurrences that the predictive system will attempt to anticipate using physiological and behavioral measurements recorded beforehand.

Only positive transitions were considered escalation events because the primary objective of the study is to identify the physiological patterns that precede increasing discomfort. In practical virtual reality applications, anticipating worsening symptoms is considerably more relevant than detecting symptom recovery, as it enables adaptive interventions before severe cybersickness develops.

The detected escalation events were then extracted together with their corresponding participant identifier, occurrence time, current FMS value, and escalation magnitude.

Figure 15 shows the temporal distribution of all detected escalation events across participants.

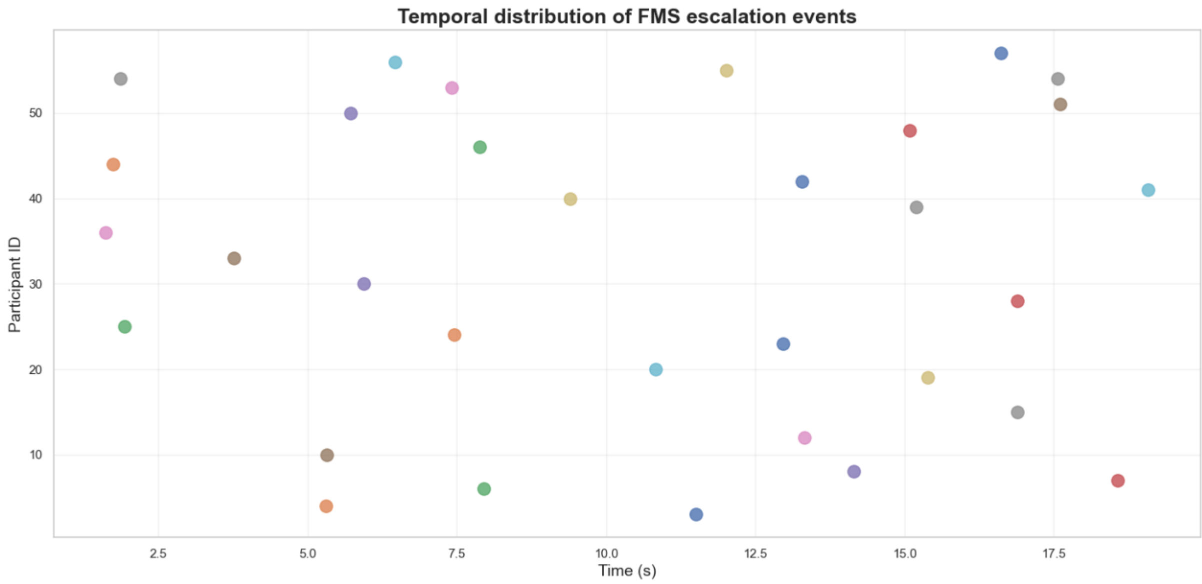


Figure 15. Temporal distribution of detected FMS escalation events. Each point represents a positive transition in the Fast Motion Sickness Scale (FMS), indicating an increase in reported cybersickness severity.

The figure reveals that escalation events are distributed throughout the entire duration of the virtual reality exposure rather than being concentrated at a specific time interval. Escalations were observed both during the early stages of exposure and near the end of the recording sessions.

This observation suggests that cybersickness progression cannot be explained solely as a function of exposure duration. Although prolonged exposure is generally associated with increased discomfort, individual susceptibility and moment-to-moment interactions between visual, vestibular, and proprioceptive information appear to play a substantial role in determining when symptom escalation occurs.

From a neurophysiological perspective, this result is consistent with sensory conflict theories of cybersickness, which propose that discomfort emerges when discrepancies accumulate between incoming sensory information and the predictions generated by the central nervous system [2], [3]. Since the magnitude and timing of these conflicts may vary considerably between individuals, escalation events are expected to occur at different moments even under similar experimental conditions.

The observed dispersion of escalation events across participants further highlights the strong inter-individual variability characteristic of cybersickness. Some participants experienced escalations relatively early, whereas others maintained stable discomfort levels for most of the experiment before exhibiting a sudden increase. This variability reinforces the importance of participant-specific physiological responses and suggests that personalized adaptation mechanisms may influence cybersickness progression.

To obtain a broader view of the temporal dynamics of cybersickness, a participant-level heatmap was generated using the complete FMS trajectories.

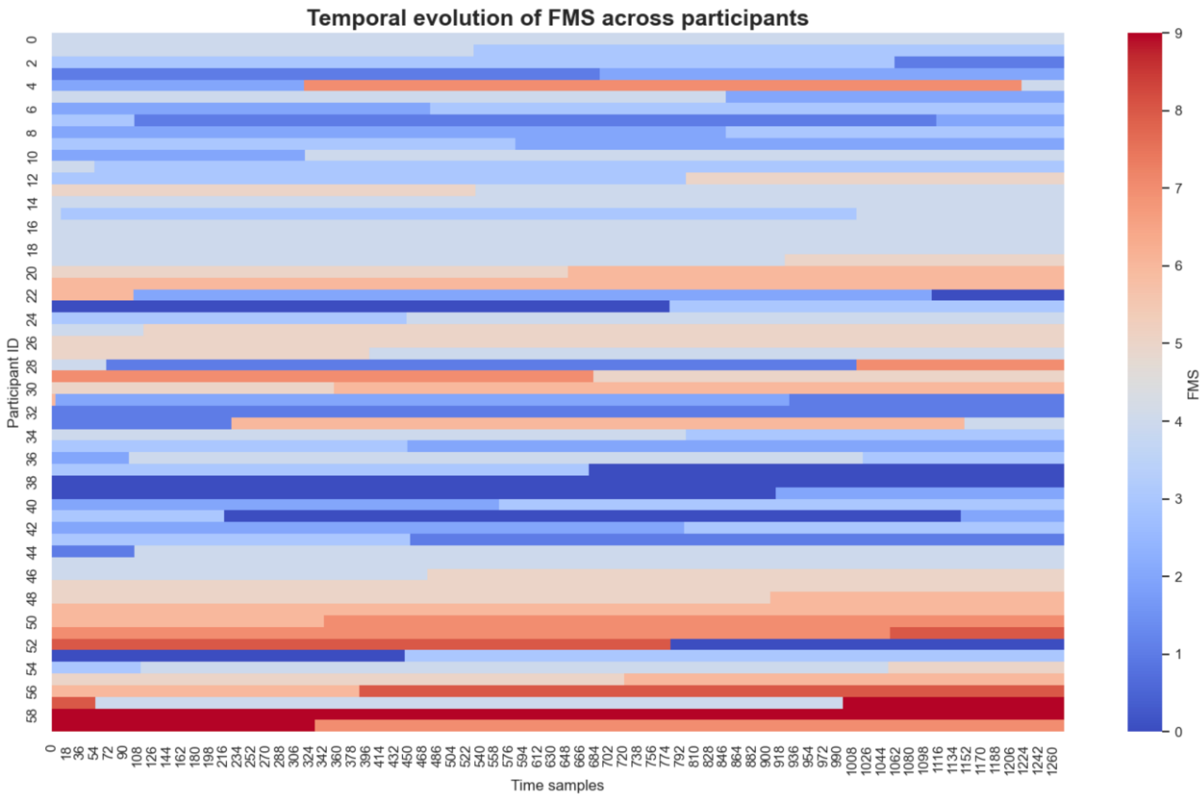


Figure 16. Temporal evolution of FMS values across participants. Rows correspond to participants, columns represent time samples, and colors indicate the reported FMS level.

The heatmap (Figure 16) provides a compact representation of the evolution of subjective discomfort throughout the experiment. Several relevant patterns can be observed.

First, large regions of constant color are present for many participants, confirming the block-like structure identified during the previous exploratory analysis. This indicates that subjective cybersickness ratings remain stable during extended periods and change only occasionally.

Second, substantial variability exists between participants. Some individuals remained at relatively low FMS levels throughout the entire session, whereas others reported moderate or severe discomfort for prolonged periods. Several participants also exhibited clear transitions between different FMS states, visible as abrupt color changes along the temporal axis.

Third, the heatmap reveals that high FMS values are not restricted to a small subset of participants. Instead, elevated discomfort levels appear in multiple individuals, although with different temporal trajectories. This finding supports the assumption that cybersickness progression follows participant-specific patterns rather than a single common trajectory.

From a neurotechnology perspective, these observations are highly relevant. The heatmap suggests that the neural and behavioral mechanisms underlying cybersickness do not evolve uniformly across participants. Differences in visuovestibular integration, sensory reweighting processes, oculomotor adaptation, and susceptibility to motion-induced discomfort may contribute to the heterogeneous patterns observed in the data [2], [3].

Importantly, the heatmap also confirms that the dataset contains both stable and transitional periods. This combination is particularly valuable for predictive modelling because it allows the extraction of physiological patterns immediately preceding escalation events while simultaneously providing examples of periods in which no escalation occurs.

Overall, the escalation-event analysis provides two key insights. First, increases in cybersickness severity occur throughout the entire duration of the VR experience rather than at a single predictable moment. Second, substantial participant-to-participant variability exists in both the timing and magnitude of symptom progression. These findings justify the subsequent formulation of the problem as a temporal prediction task, in which physiological and behavioral signals are used to anticipate future increases in cybersickness severity before they become subjectively noticeable.

2.4. FORECASTING WINDOW CONSTRUCTION AND TEMPORAL FEATURE EXTRACTION

The objective of this thesis is not to estimate the current cybersickness level but rather to anticipate future increases in discomfort before they become subjectively noticeable. Consequently, the original sample-level dataset was transformed into a forecasting dataset based on temporal windows.

This transformation allows the predictive models to learn physiological and behavioral patterns that precede cybersickness escalation events rather than relying on information available only after the increase has already occurred.

2.4.1. FEATURE SELECTION AND LEAKAGE PREVENTION

Before constructing the forecasting windows, several variables were excluded from the candidate feature set.

The excluded variables included the target variable itself (*fms*), participant identifiers, temporal indexing variables, and variables derived directly from the target signal:

- participant_id
- participant_sample_idx
- participant_time_sec
- sample_idx
- time_sec
- fms_change
- fms_block_id
- global_fms_block_id
- fms_diff
- is_escalation

These variables were intentionally removed because they either contained explicit information about participant identity or were generated directly from the target labels. Including them would introduce information leakage and artificially inflate model performance.

In addition, automatically renamed duplicated columns generated during dataset import were identified through the suffixes *.1* and *.2*. These duplicated representations were removed to avoid redundancy and prevent multiple versions of the same signal from influencing the learning process.

A final verification was then performed to ensure that the selected features did not contain missing values or infinite values that could compromise the subsequent feature extraction process.

2.4.2. FORECASTING PROBLEM DEFINITION

The prediction problem was formulated as a binary forecasting task.

A window of historical physiological and behavioral data was used to predict whether a cybersickness escalation event would occur in the immediate future.

Three temporal parameters were defined:

- Window duration: 3 seconds
- Window displacement (step): 0.5 seconds
- Forecast horizon: 2 seconds

Given the acquisition frequency of 60 Hz, these parameters correspond to:

$$WINDOW\_SIZE = 180\ samples$$

$$STEP\_SIZE = 30\ samples$$

$$FORECAST\_HORIZON = 120\ samples$$

The use of a three-second observation window was intended to capture short-term physiological dynamics immediately preceding possible symptom escalation. At the same time, the two-second forecasting horizon was selected to provide a realistic anticipation interval while maintaining a sufficiently close temporal relationship between physiological behavior and future cybersickness progression.

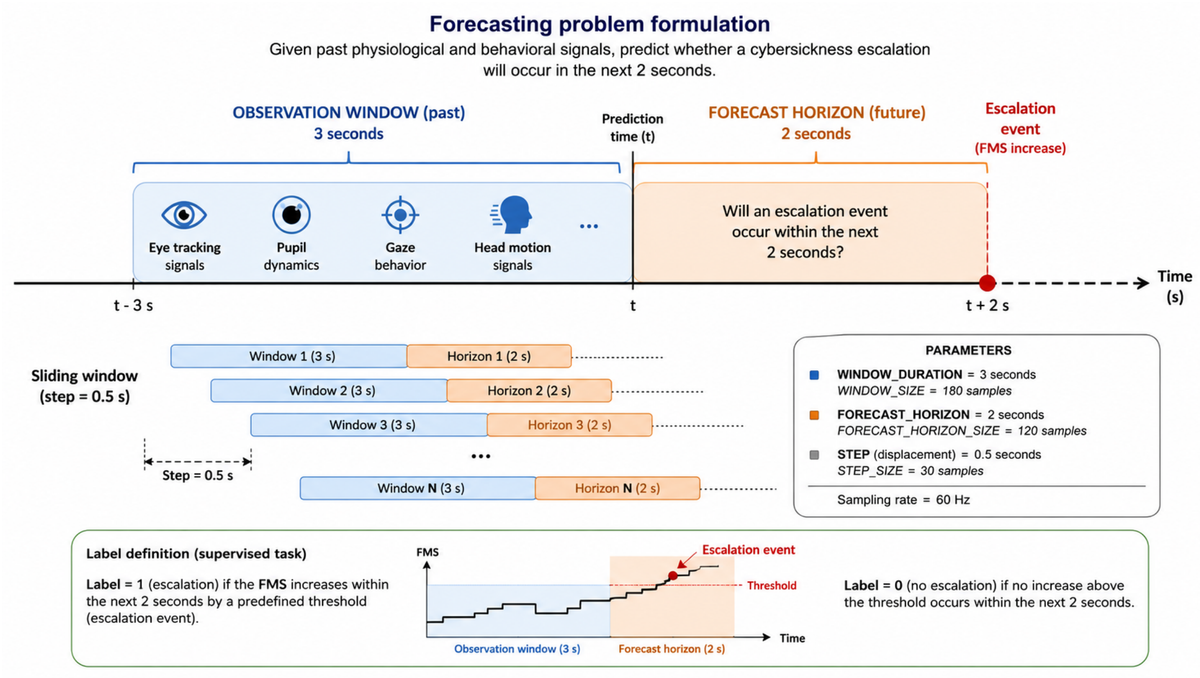


Figure 17. Forecasting formulation used in this work. Physiological and behavioural signals recorded during a 3-second observation window are used to predict whether a cybersickness escalation event will occur within the subsequent 2-second forecasting horizon.

From a neurotechnological perspective, this formulation presented in Figure 17 is particularly relevant because physiological responses associated with cybersickness often emerge before the participant consciously reports increased discomfort. Changes in gaze behavior, oculomotor control, pupil dynamics, convergence patterns, and head movement may therefore act as early indicators of an impending escalation event.

2.4.3. TEMPORAL FEATURE EXTRACTION

For each forecasting window, a set of statistical and temporal descriptors was computed for every physiological signal.

Given a signal x contained within a window, the following features were extracted:

- Mean
- Standard deviation
- Minimum
- Maximum
- Median
- Range
- Temporal slope

The mean provides information about the central tendency of the signal, whereas the standard deviation quantifies short-term variability. Minimum and maximum values capture extreme observations, while the median offers a robust measure of central tendency that is less sensitive to outliers.

The range was computed as:

$$Range = Max - Min$$

and provides a simple estimate of signal amplitude.

Finally, a temporal slope was calculated using linear regression across the samples contained within the window. This feature summarizes the overall direction of change occurring during the observation period.

The inclusion of temporal slopes is particularly relevant in the context of cybersickness prediction because physiological alterations associated with increasing discomfort may appear as progressive trends rather than abrupt instantaneous changes.

As a result of this procedure, each original signal was transformed into a set of descriptive temporal features capable of characterizing both its magnitude and its short-term evolution.

2.4.4. WINDOW LABELING STRATEGY

Windows were generated independently for each participant in order to preserve temporal consistency and avoid mixing observations belonging to different individuals.

For each window, the corresponding future segment was defined as the interval immediately following the observation window and extending over the forecasting horizon.

A positive label was assigned whenever at least one escalation event occurred inside this future segment:

$$y = \begin{cases} 1 & \text{if at least one future escalation exists} \\ 0 & \text{otherwise} \end{cases}$$

Consequently, positive windows do not contain the escalation itself. Instead, they contain only the physiological information available before the escalation occurs.

This distinction is essential because it ensures that the models learn predictive patterns rather than simply detecting changes that have already happened.

2.4.5. DATASET SUMMARY

The resulting forecasting dataset contained:

Table 2. Summary of the forecasting dataset

METRIC	VALUE
Nº windows	1.980
Nº features	308
Positive windows	110
Negative windows	1.870
Positive ratio	5.56%

The resulting class distribution shown in Table 2 reveals a substantial imbalance between positive and negative windows.

This imbalance is expected because cybersickness escalation events are relatively infrequent compared with the total duration of stable physiological activity. From a practical perspective, this situation closely resembles real-world monitoring scenarios, where discomfort escalation events represent rare but highly relevant occurrences.

The imbalance identified at this stage motivated the later use of dedicated class-balancing strategies during model training.

Figure 18 illustrates the temporal distribution of positive forecasting windows.

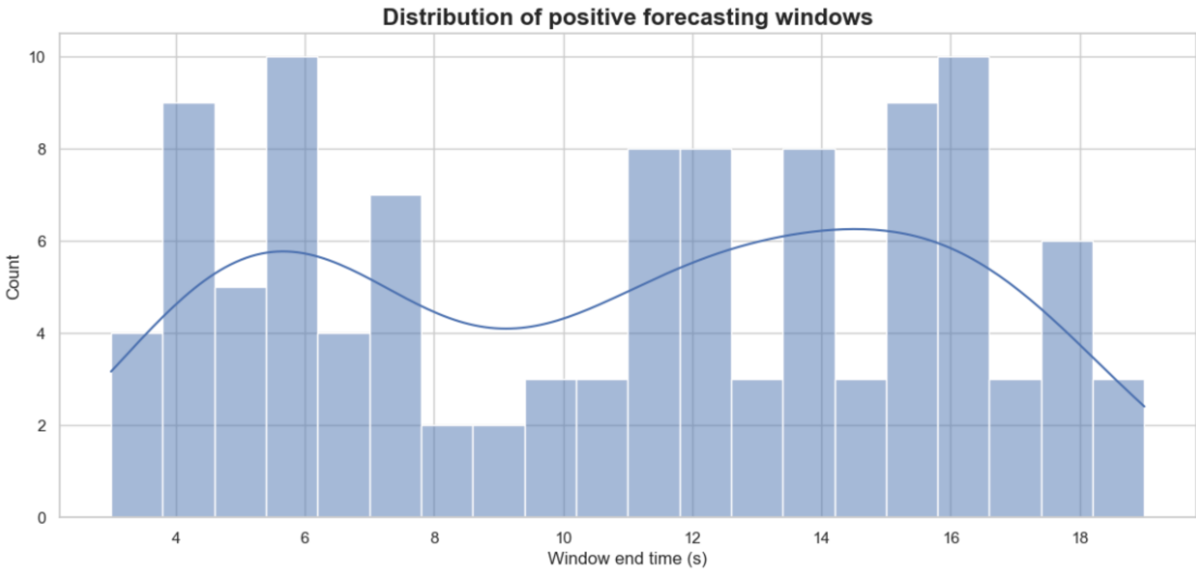


Figure 18. Distribution of positive forecasting windows. The figure shows the ending time of windows that were labeled as positive because an escalation event occurred within the subsequent forecasting horizon.

The distribution indicates that positive forecasting windows are present throughout the entire experiment duration. No single temporal region dominates the occurrence of positive samples.

This observation is consistent with the previous escalation-event analysis and further supports the assumption that cybersickness progression cannot be explained solely by exposure duration. Instead, physiological predictors must be extracted directly from participant behavior and sensor measurements.

To visually verify the forecasting strategy, an illustrative participant example was generated.

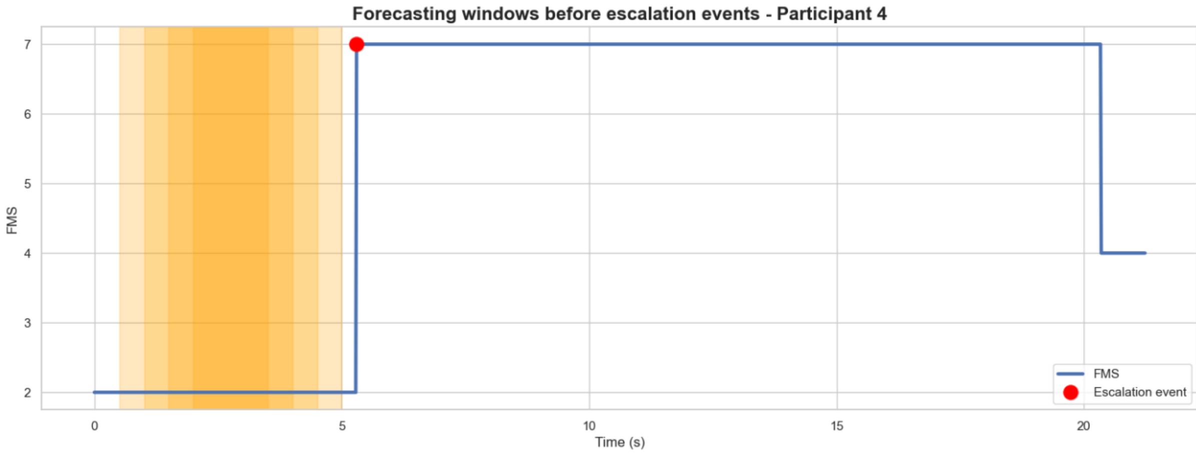


Figure 19. Example of forecasting windows preceding an escalation event. The orange regions represent positive forecasting windows, while the red marker indicates the subsequent escalation event.

Figure 19 demonstrates the intended prediction mechanism. The highlighted windows occur entirely before the increase in FMS score and therefore contain no information about the future escalation itself.

The escalation event occurs after the end of the forecasting windows, ensuring that only information available prior to symptom worsening is used for prediction.

This design closely resembles the intended deployment scenario of a real-time cybersickness monitoring system. At any given instant, the system only has access to past and present physiological measurements and must estimate whether discomfort is likely to increase in the near future.

From a neurotechnology perspective, this formulation transforms the problem from symptom detection into symptom anticipation. Rather than identifying discomfort after it has already become subjectively noticeable, the model aims to recognize early physiological signatures of sensory conflict, oculomotor instability, and visuovestibular mismatch before participants explicitly report increased cybersickness.

2.4.6. DATA LEAKAGE VERIFICATION

A final verification step was performed to ensure that the generated labels were consistent with the forecasting formulation and that no future information was inadvertently included in the observation windows.

A random subset of forecasting windows was inspected by comparing the assigned labels with the actual presence of escalation events inside the corresponding future forecasting horizon.

The consistency check confirmed that the generated labels matched the future escalation events with a very high agreement rate, indicating that the forecasting windows were correctly constructed and that no evidence of information leakage was detected.

This validation is particularly important because data leakage is one of the most common sources of artificially inflated performance in temporal prediction problems. By ensuring strict temporal separation between observation windows and future labels, the resulting forecasting dataset more accurately reflects the conditions that would be encountered in a real-time cybersickness prediction system.

2.5. PRE-EVENT SIGNATURE ANALYSIS

Before training the predictive models, an exploratory analysis was conducted to investigate whether physiological signals exhibited characteristic temporal patterns prior to cybersickness escalation events.

The objective of this analysis was to determine whether individual physiological variables showed observable differences between windows immediately preceding an escalation event and windows corresponding to stable periods without future symptom worsening.

As an illustrative example, the variable **HeadEulX** was selected. This signal represents one of the Euler-angle components describing head orientation and therefore provides information regarding participant head posture and movement throughout the virtual reality experience.

For each forecasting window, the complete temporal trajectory of the signal was extracted. Windows labeled as positive (i.e., windows followed by an escalation event within the forecasting horizon) were grouped together and compared against windows labeled as negative.

To facilitate comparison, all windows were aligned onto a common temporal axis spanning the three-second observation period. The mean trajectory and standard deviation were then computed separately for positive and negative windows.

Figure 20 presents the resulting average trajectories.

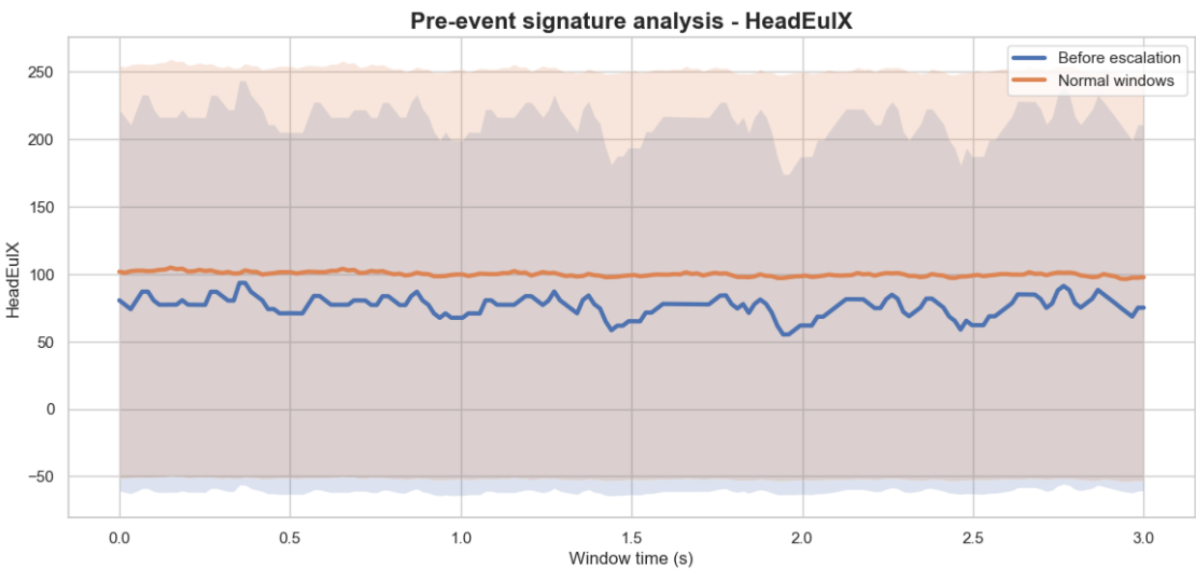


Figure 20. Average temporal trajectory of HeadEulX for windows preceding cybersickness escalation events and windows not followed by future escalation. Shaded regions represent one standard deviation around the mean.

The figure shows that the average trajectories of positive and negative windows remain relatively similar throughout the observation period. No abrupt changes or clearly separable patterns can be observed in the mean signal values immediately preceding escalation events.

This result suggests that HeadEulX alone does not provide sufficient discriminative information to reliably distinguish between pre-escalation and non-escalation periods. The absence of a strong separation is not unexpected, given the multifactorial nature of cybersickness and the complexity of the physiological mechanisms involved.

However, an important difference emerges when considering signal variability rather than mean values. The windows preceding escalation events exhibit noticeably wider variability bands, indicating greater dispersion of head-orientation measurements during the observation period.

In contrast, windows not followed by escalation events display more stable trajectories with comparatively reduced variability.

From a neurotechnological perspective, this observation may indicate that periods preceding symptom worsening are associated with increased instability in head-motion behavior. Such instability could reflect compensatory movements, altered postural control strategies, or changes in visuomotor behavior occurring as participants progressively experience sensory conflict.

According to prevailing theories of cybersickness, discomfort emerges when discrepancies arise between visual, vestibular, and proprioceptive information [2], [3]. Increased variability in head-orientation signals may therefore represent an indirect manifestation of these adaptive processes, as participants unconsciously modify their movement patterns in response to growing sensory mismatch.

Nevertheless, the substantial overlap between both distributions indicates that this signal alone is unlikely to constitute a reliable biomarker of impending cybersickness escalation.

This finding is scientifically relevant because it suggests that cybersickness prediction cannot be reduced to the analysis of a single physiological variable. Instead, meaningful predictive information is expected to emerge from the combination of multiple physiological, ocular, gaze-related, and head-motion features analyzed simultaneously.

Consequently, the subsequent modelling stage focuses on multivariate representations derived from the complete feature set rather than relying on individual signals in isolation.

2.6. ARTIFICIAL INTELLIGENCE TECHNIQUES

After constructing the forecasting dataset, several artificial intelligence models were implemented to predict whether a cybersickness escalation event would occur within the predefined future horizon. At this stage, the objective was to define the modelling strategy, preprocessing transformations, validation design, and model architectures. The quantitative performance of the models is presented later in the Results section.

The prediction task was formulated as a time series binary classification problem. Each temporal window was assigned a label equal to 1 if an FMS escalation occurred within the following forecasting horizon, and 0 otherwise. Therefore, the models were trained to distinguish between windows preceding symptom worsening and windows corresponding to non-escalation periods.

A participant-aware validation strategy was used throughout the modelling process. The dataset was split using StratifiedGroupKFold, where the groups corresponded to participant identifiers. This ensured that windows from the same participant were not simultaneously present in the training and test sets. This decision was essential to avoid participant leakage, since consecutive windows from the same subject may share physiological, behavioral, and temporal characteristics. In neurotechnology applications, this type of leakage is particularly problematic because a model may otherwise learn participant-specific traits rather than generalizable markers of cybersickness escalation.

An outer participant-based split was first used to separate training and test data. Then, an inner StratifiedGroupKFold cross-validation scheme was applied during hyperparameter optimization. This inner validation preserved both the class distribution and the participant grouping structure. Hyperparameter tuning was performed using GridSearchCV, and the optimization metric was average precision. This metric was selected because the forecasting dataset was highly imbalanced, with escalation windows representing only a small proportion of the total number of windows. In this context, average precision is more informative than accuracy because it focuses on the model's ability to correctly identify the minority positive class across different decision thresholds. This pipeline is summarized in Figure 21.

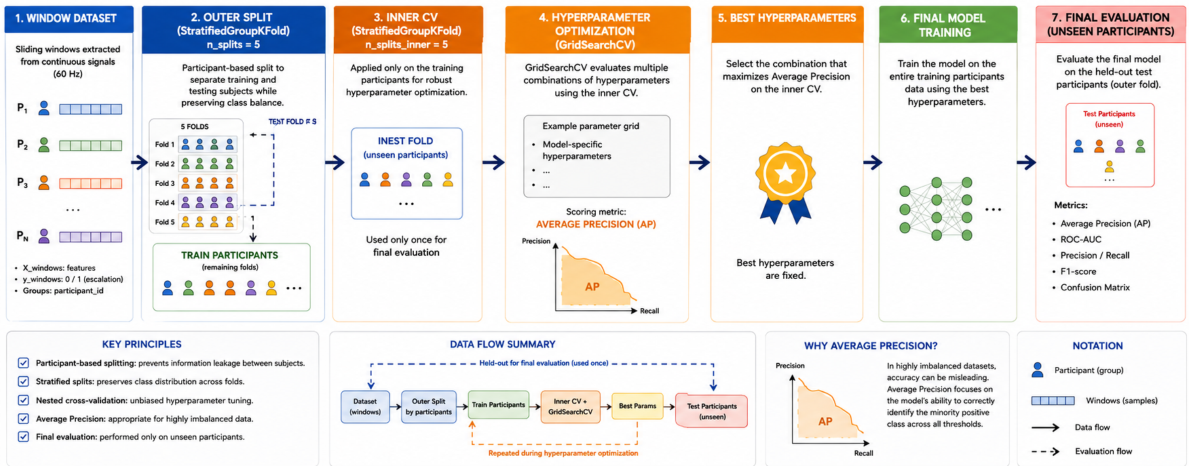


Figure 21. Overview of the nested participant-aware validation strategy used [20].

Several families of models were evaluated: Logistic Regression, Random Forest, XGBoost, Histogram Gradient Boosting, Support Vector Machine, CatBoost, Multilayer Perceptron, and a one-dimensional Convolutional Neural Network. These models were selected to cover a broad methodological spectrum, ranging from interpretable linear baselines to nonlinear tree ensembles and neural architectures.

For each model family, different variants were implemented. A baseline version was first trained using the complete engineered feature set. Then, imbalance-aware versions were developed using class weighting and oversampling techniques to address the strong class imbalance present in the forecasting dataset. Oversampling increases the representation of minority-class samples in the training data, helping the models learn patterns associated with cybersickness escalation events more effectively. In particular, the Synthetic Minority Over-sampling Technique (SMOTE) was explored as a data-level balancing strategy. Finally, SHAP-based feature selection was investigated by retaining the 20 most relevant features identified from model explanations. This strategy allowed the comparison of models trained with all available features against models trained with a compact and more interpretable subset of features.

2.6.1. COMMON PREPROCESSING STRATEGY

A median imputation step was included in all classical machine learning pipelines. Although the exploratory analysis indicated that no missing values were initially detected, imputation was retained as a safety mechanism to ensure robustness during cross-validation and model fitting. Median imputation was selected because it is less sensitive to extreme values than mean imputation.

Feature scaling was applied to models that are sensitive to feature magnitude, including Logistic Regression, Support Vector Machine, Multilayer Perceptron, and the 1D-CNN. StandardScaler was used to transform each feature to zero mean and unit variance. This transformation was necessary because models based on distances, gradients, or linear coefficients can be strongly affected by variables measured on different numerical scales.

Tree-based models, including Random Forest, XGBoost, HistGradientBoosting, and CatBoost, were not standardized because decision-tree algorithms are generally insensitive to monotonic transformations of the input variables. These models split the feature space according to thresholds and therefore do not require all variables to be expressed on the same scale.

To address class imbalance, two strategies were considered. The first strategy consisted of using class weights, which assign greater importance to the minority class during training. The second strategy used SMOTE, a synthetic oversampling method that generates artificial minority-class samples by interpolating between existing minority examples [21]. SMOTE was applied only inside the training folds through imbalanced-learn pipelines, ensuring that synthetic samples were never created using test data.

Feature selection was implemented in some model variants using SelectKBest with the ANOVA F-test. This method ranks features according to their univariate association with the target label and retains only the top k variables. Although this approach does not capture nonlinear interactions, it provides a simple and computationally efficient dimensionality reduction step.

2.6.2. LOGISTIC REGRESSION

Logistic Regression was used as the main interpretable linear baseline. This model estimates the probability of cybersickness escalation as a logistic transformation of a weighted sum of input features. Its inclusion was important because it provides a transparent reference model against which more complex nonlinear methods can be compared [15].

Three Logistic Regression variants were implemented. The baseline pipeline consisted of median imputation, standard scaling, and Logistic Regression. The regularization strength was controlled through the hyperparameter C, and both unweighted and class-balanced configurations were evaluated.

A second variant incorporated feature selection and SMOTE. In this case, the pipeline included median imputation, standard scaling, SelectKBest feature selection, SMOTE oversampling, and Logistic Regression. The number of selected features, the number of SMOTE neighbors, the regularization strength, and the class-weighting strategy were optimized during cross-validation.

A third variant used SHAP-based feature selection. First, the best baseline Logistic Regression model was fitted using the training data. Then, SHAP values were computed using a linear SHAP explainer. The mean absolute SHAP value of each feature was used as a global relevance score, and the 20 most influential features were retained [10]. A new Logistic Regression model was then trained using only these SHAP-selected features.

This SHAP-based variant was included to evaluate whether a compact interpretable subset of features could preserve relevant predictive information while improving model simplicity.

2.6.3. RANDOM FOREST

Random Forest was implemented as a nonlinear ensemble baseline. This model combines multiple decision trees trained on bootstrapped samples of the data and aggregates their predictions [16]. Its main advantage is its ability to capture nonlinear relationships and feature interactions without requiring strong assumptions about the data distribution.

The Random Forest baseline pipeline included median imputation followed by the classifier. Hyperparameters included the number of trees, maximum tree depth, minimum number of samples required to split a node, minimum number of samples per leaf, and class-weighting strategy.

A second Random Forest variant incorporated SMOTE before model training. Since Random Forest models can already handle complex nonlinear patterns, SMOTE was used only as an imbalance-reduction mechanism.

A third variant used SHAP-based feature selection. TreeSHAP was applied to the best baseline Random Forest model to estimate feature contributions for the escalation class. The 20 features with the highest mean absolute SHAP values were selected, and a new Random Forest model was trained using only this reduced feature set.

2.6.4. XGBOOST

XGBoost was included as a gradient-boosted decision-tree model. Unlike Random Forest, which builds independent trees, XGBoost constructs trees sequentially, where each new tree attempts to correct the errors of the previous ensemble [22]. This makes it especially suitable for structured tabular datasets with nonlinear interactions.

The XGBoost baseline pipeline consisted of median imputation and an XGBClassifier configured for binary logistic classification. The loss was evaluated using log-loss, and hyperparameters included the number of estimators, tree depth, learning rate, subsampling ratio, column subsampling ratio, and scale_pos_weight.

The scale_pos_weight parameter was included to compensate for class imbalance by increasing the importance of positive escalation windows during training.

A second XGBoost variant used SMOTE instead of scale_pos_weight. In this version, synthetic positive samples were generated within the training folds, and scale_pos_weight was fixed to 1 to avoid combining two imbalance-correction strategies simultaneously.

A third XGBoost variant used SHAP-based feature selection. TreeSHAP values were computed from the best baseline XGBoost model, and the 20 most influential features were retained. A new XGBoost model was then trained using only those selected features.

2.6.5. HISTOGRAM GRADIENT BOOSTING

Histogram Gradient Boosting was also evaluated as an additional boosting-based tree ensemble. This model follows the gradient boosting principle but uses histogram-based binning of continuous variables to improve computational efficiency [23].

The baseline pipeline consisted of median imputation followed by HistGradientBoostingClassifier. The hyperparameter grid included the learning rate, maximum number of boosting iterations, maximum number of leaf nodes, maximum depth, L2 regularization strength, and class-weighting strategy.

A second variant included SMOTE before model fitting. A third variant used SHAP-based feature selection. For this model, SHAP values were computed on a random sample of the preprocessed training data to reduce computational cost. The 20 features with the highest mean absolute SHAP values were selected, and a reduced HistGradientBoosting model was trained using only those variables.

2.6.6. SUPPORT VECTOR MACHINE

A Support Vector Machine was included to evaluate margin-based classification. SVMs aim to find a decision boundary that maximizes the margin between classes [24]. Both linear and radial basis function kernels were considered. The linear kernel allows a simpler decision boundary, whereas the RBF kernel enables nonlinear separation in a transformed feature space.

The SVM baseline pipeline included median imputation, standard scaling, and SVC with probability estimation enabled. Hyperparameters included the regularization parameter C, kernel type, gamma value, and class-weighting strategy.

A second SVM variant incorporated feature selection and SMOTE. This pipeline included imputation, scaling, SelectKBest feature selection, SMOTE, and SVC.

A third SVM variant used a SHAP-selected feature subset. Direct SHAP explanation for SVMs generally requires KernelExplainer, which is computationally expensive for high-dimensional datasets. Therefore, the top 20 SHAP features obtained from the Logistic Regression model were used as a proxy feature subset. This decision was explicitly treated as a methodological approximation rather than as a direct SVM-specific explanation.

2.6.7. CATBOOST

CatBoost was included as an additional gradient boosting algorithm. CatBoost is based on ordered boosting and is designed to reduce prediction shift and improve robustness in gradient-boosted decision-tree models [25]. Although CatBoost is especially known for handling categorical variables, it can also be applied effectively to numerical tabular data.

The CatBoost baseline pipeline consisted of median imputation followed by CatBoostClassifier. The model was trained using Logloss as the loss function and AUC as the internal evaluation metric. Hyperparameters included the number of iterations, tree depth, learning rate, L2 leaf regularization, and automatic class weighting.

A second CatBoost variant incorporated SMOTE before the classifier. In this case, synthetic minority samples were generated in the training folds.

A third variant used SHAP-based feature selection. TreeSHAP was applied to a sample of the preprocessed training data, the 20 most influential features were selected, and a reduced CatBoost model was trained on this subset.

2.6.8. MULTILAYER PERCEPTRON

A Multilayer Perceptron was implemented as a fully connected neural-network model. Unlike tree-based methods, the MLP learns nonlinear combinations of features through successive layers of weighted transformations and nonlinear activation functions.

The baseline MLP pipeline included median imputation, standard scaling, and MLPClassifier. Early stopping was enabled to reduce overfitting by monitoring validation performance during training. Several architectures were explored, including one hidden layer with 32 units, one hidden layer with 64 units, and two hidden layers with 64 and 32 units. Both ReLU and tanh activations were evaluated. L2 regularization was controlled through the alpha parameter, and different initial learning rates were tested.

A second MLP variant included SelectKBest feature selection and SMOTE before model training. This combination was used to reduce dimensionality and address the strong imbalance between escalation and non-escalation windows.

A third MLP variant was trained using the SHAP-selected feature subset obtained from the Logistic Regression model. As with the SVM, direct SHAP explanation for the MLP would require a model-agnostic explainer with high computational cost. Therefore, the Logistic Regression SHAP subset was used as a practical interpretable approximation.

2.6.9. ONE-DIMENSIONAL CONVOLUTIONAL NEURAL NETWORK

One-dimensional convolutional neural networks (1D-CNNs) are particularly suitable for time-series classification tasks because they can automatically learn local temporal patterns directly from sequential data without requiring extensive manual feature engineering [26].

For this model, each forecasting window was represented as a sequence rather than as a vector of handcrafted features. Therefore, the input tensor had the shape:

$$N \times T \times F$$

where N represents the number of windows, T corresponds to the number of time steps in each window, and F represents the number of original signal features. Since each window lasted 3 seconds and data were recorded at 60 Hz, each sequence contained 180 time steps.

The same participant-based train-test split used for the classical machine learning models was applied to the CNN dataset. Before training, the three-dimensional sequence data were temporarily reshaped into a two-dimensional matrix in order to apply median imputation and standard scaling. The transformed data were then reshaped back into sequence format.

A participant-aware validation split was also created from the training set using StratifiedGroupKFold. This ensured that the validation data contained participants not present in the CNN training subset.

The CNN architecture consisted of the following layers:

1. Input layer with shape $180 \times F$. First one-dimensional convolutional layer with ReLU activation and same padding.
2. Batch normalization layer.
3. Max-pooling layer.
4. Second one-dimensional convolutional layer with twice the number of filters.
5. Batch normalization layer.
6. Global average pooling layer.
7. Fully connected dense layer with ReLU activation.
8. Dropout layer.
9. Output dense layer with sigmoid activation.

The convolutional layers were included to detect local temporal patterns within the physiological signals. In the context of cybersickness prediction, these patterns may correspond to short-term changes in gaze dynamics, eye openness, pupil behavior, convergence, or head-motion trajectories. Max-pooling reduced the temporal resolution while preserving relevant activations. Global average pooling transformed the temporal feature maps into a compact representation, reducing the number of trainable parameters compared with flattening. Dropout was included as a regularization mechanism to reduce overfitting.

The output layer used a sigmoid activation because the task was binary. The model was trained using binary cross-entropy loss and the Adam optimizer. ROC-AUC and PR-AUC were monitored during training. Since the dataset was imbalanced, class weights were computed from the training labels and applied during model fitting.

A small hyperparameter search was performed over the number of convolutional filters, kernel size, number of dense units, dropout rate, and learning rate. Early stopping was applied by monitoring validation PR-AUC, and the best model weights were restored after training.

2.6.10. LONG SHORT-TERM MEMORY NETWORK (LSTM)

Long Short-Term Memory (LSTM) networks are a specialized type of recurrent neural network designed to model sequential and time-dependent data. Unlike traditional feed-forward architectures, LSTMs incorporate memory cells and gating mechanisms that enable the network to retain relevant information over extended temporal intervals while mitigating the vanishing-gradient problem commonly encountered in standard recurrent neural networks [27].

This capability makes LSTM architectures particularly suitable for forecasting tasks involving physiological and behavioral time-series data, where relevant patterns may emerge progressively over time rather than being contained within isolated observations. In the context of cybersickness prediction, temporal dependencies may exist between gaze behavior, eye-tracking variables, head motion dynamics, and the subsequent progression of discomfort symptoms. Therefore, LSTMs provide a natural framework for evaluating whether sequential modeling can improve the anticipation of cybersickness escalation events.

In this work, the LSTM model was included as an additional deep-learning baseline to complement the previously evaluated 1D-CNN architecture. The objective was not to perform an exhaustive optimization of recurrent architectures, but rather to assess whether an explicitly sequential model could extract useful temporal information beyond that captured by convolutional filters. The network received the same three-second temporal windows used by the 1D-CNN and was trained using binary cross-entropy loss, class weighting, and early stopping based on validation performance.

The evaluated architecture consisted of a single LSTM layer followed by batch normalization, dropout regularization, and fully connected layers for binary classification. Hyperparameter selection was performed through a limited validation-based search considering different numbers of LSTM units, dense-layer dimensions, dropout rates, and learning rates. The final configuration was selected according to the highest validation PR-AUC score.

2.6.11. MODEL EVALUATION FRAMEWORK

Although model performance is not reported in this chapter, a common evaluation framework was implemented for all models. After hyperparameter optimization, each best estimator was applied to the held-out test set. Predicted labels and predicted probabilities were obtained, and several metrics were computed.

The metrics included accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and Matthews correlation coefficient [28] [29]. Because the dataset is highly imbalanced. A model could achieve high accuracy by predicting mostly negative windows. Therefore, Balanced Accuracy, PR-AUC, and MCC provide a more informative assessment of performance under these conditions.

$$\boldsymbol{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Overall proportion of correctly classified samples.

$$\boldsymbol{Balanced\ Accuracy} = \frac{Recall_{positive} + Recall_{negative}}{2}$$

Average recall across all classes, reducing the effect of class imbalance.

$$\boldsymbol{Precision} = \frac{TP}{TP + FP}$$

Proportion of predicted positive cases that are actually positive.

$$\boldsymbol{Recall} = \frac{TP}{TP + FN}$$

Ability of the model to correctly identify positive cases.

$$\boldsymbol{F1} = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Harmonic mean of Precision and Recall, balancing both metrics.

$$\boldsymbol{MCC} = \frac{TP \cdot TN - FP \cdot FN}{} \sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}$$

Balanced measure of classification quality that considers all elements of the confusion matrix and remains reliable under class imbalance.

ROC-AUC measures the model's ability to discriminate between positive and negative classes across all decision thresholds.

PR-AUC summarizes the trade-off between Precision and Recall across different decision thresholds and is particularly informative for imbalanced datasets.

Confusion matrices and classification reports were also generated for each model, but their interpretation is reserved for the Results section.

Finally, all model results were stored in structured dataframes and exported as CSV files. This allowed systematic comparison between model families and variants in the Results chapter while keeping the Development chapter focused on methodological design.

2.6.12. EXPLAINABILITY FRAMEWORK

While predictive performance is essential, understanding the factors driving model decisions is equally important in neurotechnology applications. Since cybersickness emerges from complex interactions between visual, vestibular, and oculomotor processes, interpretable models can provide insights into the physiological mechanisms associated with symptom escalation.

To improve model interpretability, SHAP (SHapley Additive exPlanations) was employed as the primary explainability framework. SHAP is based on concepts from cooperative game theory and estimates the contribution of each feature to an individual prediction by computing Shapley values. These values quantify how much each feature increases or decreases the predicted probability relative to the model's baseline output [10].

In this work, SHAP was used for two complementary purposes. First, global explanations were generated to identify the features that contributed most consistently to cybersickness escalation predictions across the dataset. Second, SHAP values were used as a feature-selection mechanism by retaining the 20 most influential variables for selected model variants. This approach enabled the comparison between models trained on the complete feature set and models trained on a compact subset of highly informative variables.

The explainability analysis was performed using summary plots, feature importance rankings, and participant-level interpretations. These analyses are presented in Chapter 3 and are used to investigate whether the predictive models capture meaningful neurophysiological patterns associated with cybersickness escalation.

2.6.13. REPRODUCIBILITY AND CODE AVAILABILITY

In accordance with open-science and reproducible-research principles, the source code developed for this work has been released in a public GitHub repository.

The repository provides access to the complete experimental pipeline, including preprocessing procedures, forecasting-window generation, feature extraction, machine learning and deep learning implementations, model evaluation, and explainability analyses.

GitHub repository:

https://github.com/danillopis/Forecasting-Cybersickness-Escalation-Using-Eye-Tracking-and-Head-Motion-Signals

This resource is intended to facilitate result verification, methodological transparency, and future extensions of the proposed cybersickness forecasting framework.

3. RESULTS

This chapter presents the experimental results obtained from the artificial intelligence models developed to predict cybersickness escalation events from eye-tracking and head-motion signals. While the previous chapter described the methodological design, preprocessing procedures, feature engineering strategy, and model architectures, the focus of this chapter is on evaluating and interpreting the predictive performance achieved by each modelling approach.

The prediction task consisted of identifying whether an escalation in the Fast Motion Sickness (FMS) score would occur within a predefined forecasting horizon. As discussed previously, this constitutes a highly challenging classification problem due to several factors. First, escalation events are relatively infrequent compared with non-escalation periods, leading to a substantially imbalanced dataset. Second, cybersickness is a complex physiological phenomenon that emerges from the interaction of multiple sensory and behavioral processes rather than from a single dominant biomarker. Finally, strong inter-individual variability exists across participants, making generalization particularly difficult.

To ensure a fair and realistic evaluation, all models were assessed using participant-independent train-test partitions. Consequently, the reported results reflect the ability of the models to generalize to unseen individuals rather than their capacity to memorize participant-specific patterns. This evaluation strategy is especially important in neurotechnology applications, where models are ultimately expected to operate on new users whose physiological responses were not available during training.

Several complementary performance metrics were considered throughout the evaluation process. In addition to traditional classification measures such as accuracy, precision, recall, and F1-score, particular attention was given to balanced accuracy, area under the precision-recall curve (PR-AUC), area under the receiver operating characteristic curve (ROC-AUC), and Matthews Correlation Coefficient (MCC). These metrics are more informative under severe class imbalance and provide a more reliable assessment of the practical utility of the forecasting models.

The results are first presented separately for each model family, including linear models, tree-based ensemble methods, kernel-based approaches, and neural networks. Different variants of each model are compared, including baseline configurations, imbalance-aware versions incorporating SMOTE or class weighting, and reduced-feature versions based on SHAP-driven feature selection. This analysis makes it possible to evaluate not only predictive performance but also the impact of feature selection and class imbalance mitigation strategies.

After the individual model analyses, a global comparison is conducted to identify the most suitable approaches for cybersickness forecasting. Finally, an extensive Explainable Artificial Intelligence (XAI) analysis is presented. The objective of this final section is to investigate which physiological and behavioral variables contributed most strongly to the prediction process, providing insights into the mechanisms that precede cybersickness escalation and improving the interpretability of the proposed forecasting framework.

3.1. LOGISTIC REGRESSION

Logistic Regression was selected as the first predictive model due to its simplicity, interpretability, and widespread use as a baseline classifier in biomedical and physiological signal analysis. Despite its linear nature, Logistic Regression provides an important reference point against which more sophisticated machine learning and deep learning approaches can be compared.

Three different variants were evaluated:

1. Baseline Logistic Regression.
2. Logistic Regression with SMOTE and feature selection.
3. Logistic Regression trained using the Top-20 SHAP features.

The dataset partitioning strategy resulted in 1,584 training windows and 396 testing windows, with only 90 positive forecasting windows in the training set and 20 positive windows in the testing set. This distribution highlights the severe class imbalance characterizing the forecasting problem.

3.1.1. BASELINE LOGISTIC REGRESSION

The baseline model consisted of median imputation, feature standardization, and Logistic Regression with hyperparameter optimization through GridSearchCV. The best configuration is available in the accompanying GitHub repository.

The confusion matrix shown in Figure 22 revealed that the model classified every test instance as belonging to the majority class (non-escalation). Consequently, all escalation events were missed. Although this behaviour produced a seemingly high overall accuracy, the model failed to provide useful predictive capability for the target class.

This phenomenon is common in highly imbalanced datasets, where a classifier can achieve high accuracy simply by predicting the dominant class. The balanced accuracy remained at 0.50, equivalent to random performance, while recall and F1-score for escalation detection were equal to zero.

Interestingly, the ROC-AUC value remained above random expectation. This indicates that the model was able to assign slightly different probabilities to positive and negative windows, but the default decision threshold was insufficient to convert these probability estimates into effective positive predictions.

Overall, the baseline results demonstrate that a standard linear classifier is unable to adequately address the class imbalance inherent to cybersickness escalation forecasting.

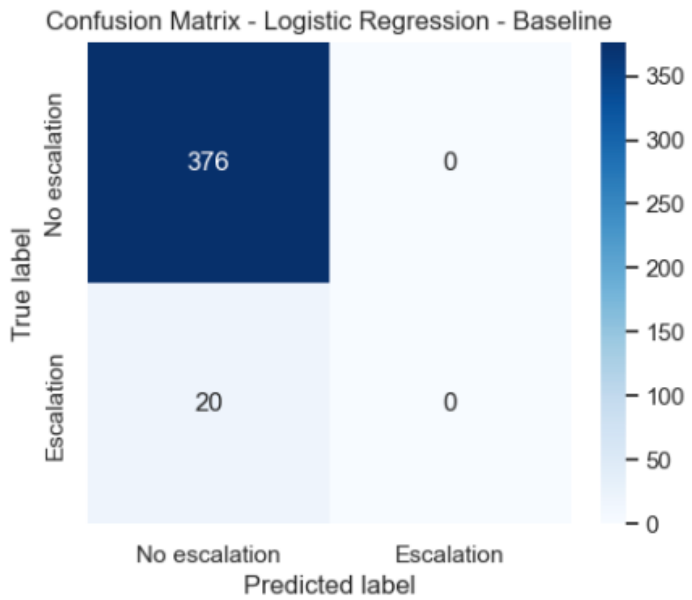


Figure 22. Confusion matrix obtained by the baseline Logistic Regression model.

3.1.2. LOGISTIC REGRESSION WITH SMOTE AND FEATURE SELECTION

To mitigate the severe imbalance between escalation and non-escalation windows, a second variant incorporated two additional mechanisms:

- Univariate feature selection using SelectKBest and ANOVA F-statistics.
- Synthetic Minority Oversampling Technique (SMOTE).

The feature selection process reduced the original feature space to the most informative variables, while SMOTE generated synthetic minority-class examples within the training folds. Hyperparameter optimization selected a subset of 20 features and a SMOTE neighbourhood size of three samples.

Compared with the baseline model, this configuration substantially changed the classifier behaviour, as shown in Figure 23. Rather than predicting exclusively the majority class, the model became capable of identifying a meaningful proportion of escalation events. The recall increased considerably, indicating that a larger fraction of future cybersickness escalations could be detected before they occurred.

This improvement, however, came at the cost of a substantial increase in false positives. The confusion matrix shows that many normal windows were incorrectly classified as impending escalation events. Such behaviour is expected when optimizing for rare-event detection, since increasing sensitivity often requires accepting a reduction in specificity.

Although the overall accuracy decreased relative to the baseline model, metrics designed for imbalanced classification improved. Balanced accuracy, F1-score, and Matthews Correlation Coefficient all increased, indicating a more balanced predictive behaviour. From an application perspective, this trade-off may be acceptable because missing an escalation event can be more problematic than issuing a false warning.

These findings suggest that handling class imbalance is a critical requirement for cybersickness forecasting and that the predictive information contained in the physiological signals can only be exploited effectively after correcting the skewed class distribution.

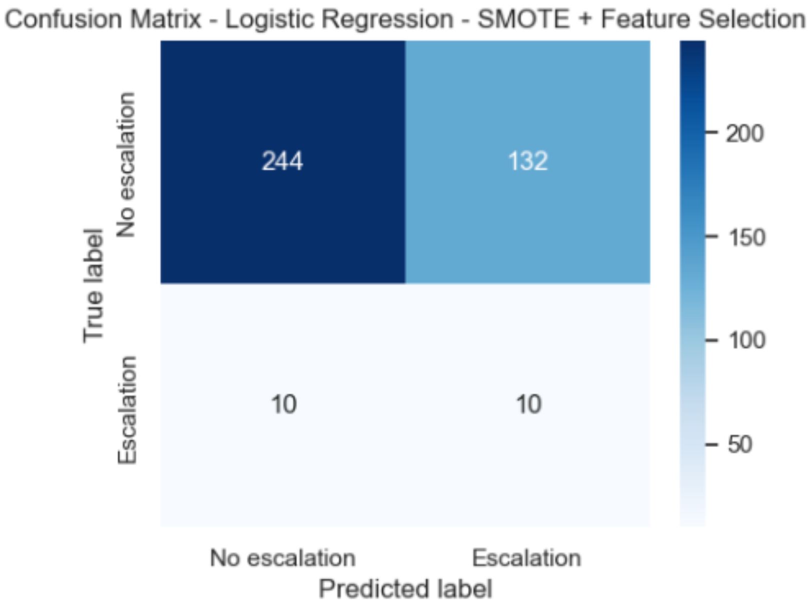


Figure 23. Confusion matrix obtained by Logistic Regression with SMOTE and feature selection.

3.1.3. LOGISTIC REGRESSION WITH SHAP-BASED FEATURE SELECTION

The third variant aimed to investigate whether model performance could be improved by restricting the input space to the twenty most influential variables identified through SHAP analysis of the baseline model.

The selected features were primarily related to gaze behaviour, eye openness, head rotations, head orientation variability, pupil measurements, and convergence dynamics. These variables are physiologically plausible indicators of visual discomfort and sensory conflict, suggesting that the feature selection process successfully identified meaningful cybersickness-related signals.

However, despite using the most influential predictors, the resulting classifier exhibited behaviour similar to the baseline model, as shown in Figure 24. The model again defaulted to predicting only the majority class, failing to identify escalation events in the testing set.

This outcome suggests that the limitation of the baseline Logistic Regression model was not primarily caused by the presence of irrelevant features. Instead, the main challenge appears to stem from the complexity of the class distribution itself. Restricting the model to a smaller subset of variables reduced dimensionality but did not provide sufficient discriminatory power to overcome the imbalance problem.

Consequently, the SHAP-based reduction strategy did not improve predictive performance when used in conjunction with a linear classifier.

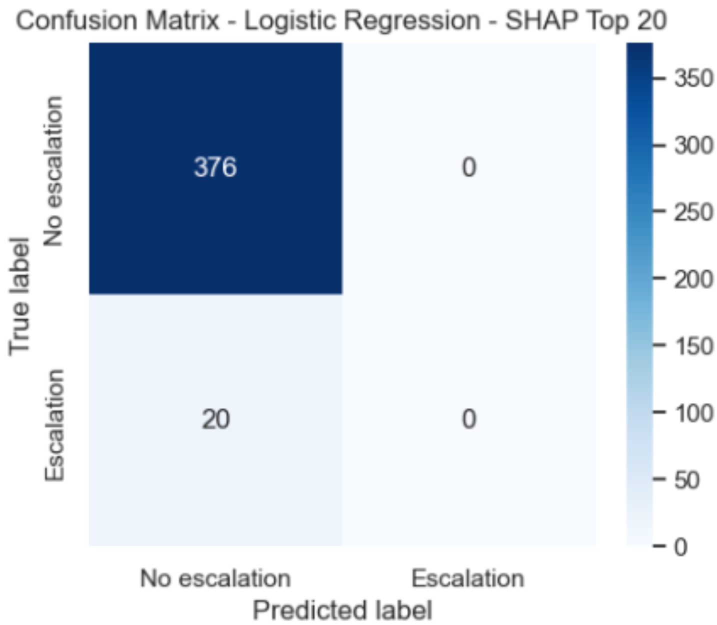


Figure 24. Confusion matrix obtained by Logistic Regression with SHAP.

3.1.4. COMPARATIVE ANALYSIS

Table 3 and Figure 25 present a comparison of the three Logistic Regression variants across the evaluation metrics.

The results reveal a clear pattern. Both the baseline and SHAP-based models exhibited strong majority-class bias, achieving high apparent accuracy but failing to detect escalation events. In contrast, the SMOTE-enhanced version sacrificed overall accuracy in favour of substantially improved sensitivity to future escalations.

These findings indicate that the primary obstacle for Logistic Regression is not feature dimensionality but class imbalance. The improvements obtained through SMOTE demonstrate that meaningful predictive information exists within the extracted physiological features. However, the relatively modest overall performance suggests that the relationship between physiological measurements and future cybersickness escalation is unlikely to be purely linear.

This observation provides initial evidence that more flexible non-linear models may be better suited to capture the complex interactions underlying cybersickness development.

Table 3. Performance comparison of the Logistic Regression variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**LR-baseline**	0.95	0.50	0.00	0.00	0.00	0.70	0.09	0.00
1	**LR – SMOTE + FS**	0.64	0.57	0.07	0.50	0.12	0.69	0.08	0.07
2	**LR - SHAP**	0.95	0.50	0.00	0.00	0.00	0.51	0.06	0.00

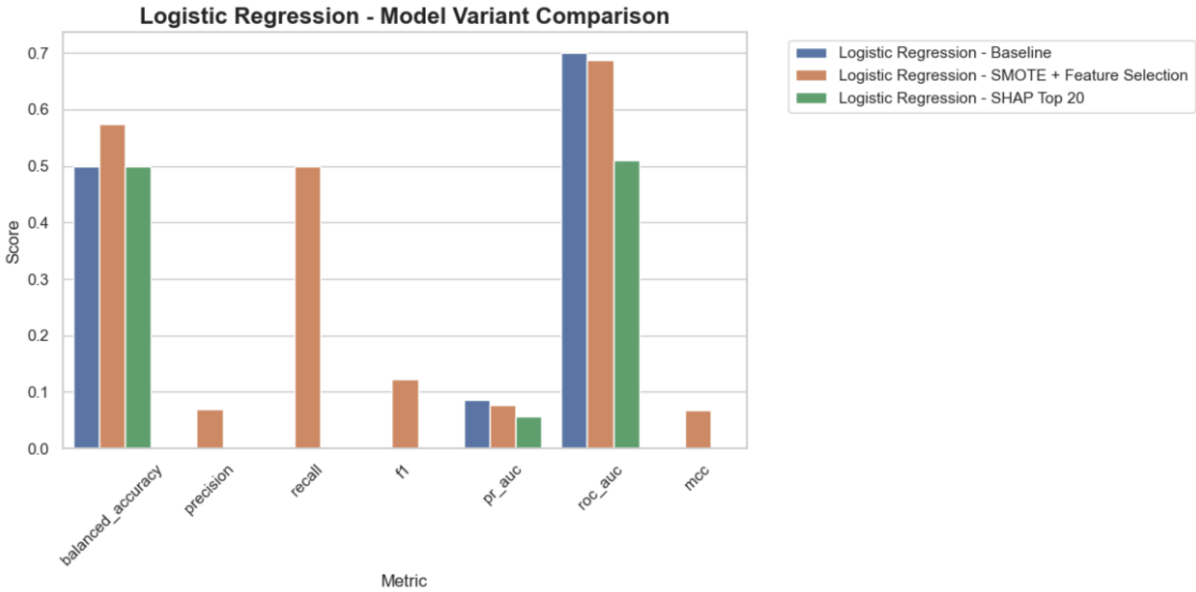


Figure 25. Logistic Regression Model Variant Comparison.

3.2. RANDOM FOREST

Random Forest was selected as the first non-linear ensemble learning method evaluated in this study. Unlike Logistic Regression, Random Forest constructs multiple decision trees and aggregates their predictions through a voting mechanism, allowing the model to capture complex interactions between physiological variables without assuming linear relationships.

The rationale for including Random Forest was twofold. First, cybersickness development is likely driven by non-linear interactions among gaze behaviour, eye movements, convergence metrics, and head motion signals. Second, Random Forests are generally robust to noise and high-dimensional feature spaces, making them particularly suitable for physiological datasets containing hundreds of engineered descriptors.

Three Random Forest configurations were investigated:

1. Baseline Random Forest.
2. Random Forest with SMOTE.
3. Random Forest using the Top-20 SHAP features.

3.2.1. BASELINE RANDOM FOREST

The baseline model consisted of median imputation followed by Random Forest classification.

Despite the increased modelling flexibility compared with Logistic Regression, the resulting classifier exhibited an almost identical behaviour. The confusion matrix (Figure 26) shows that every testing window was assigned to the majority class, leading to a complete failure in detecting escalation events.

As a consequence, recall, precision, and F1-score for the escalation class remained equal to zero. Balanced accuracy was equivalent to random guessing, despite the apparently high overall accuracy obtained from the dominant non-escalation class.

Interestingly, the ROC-AUC score remained above chance level, indicating that the model learned some degree of ranking capability between positive and negative windows. However, these probability estimates were not sufficiently separated to generate positive predictions under the default classification threshold.

This result suggests that simply increasing model complexity does not automatically solve the forecasting problem. Although Random Forest can model non-linear relationships, the extreme imbalance of the dataset appears to dominate the learning process, causing the classifier to favour conservative majority-class predictions.

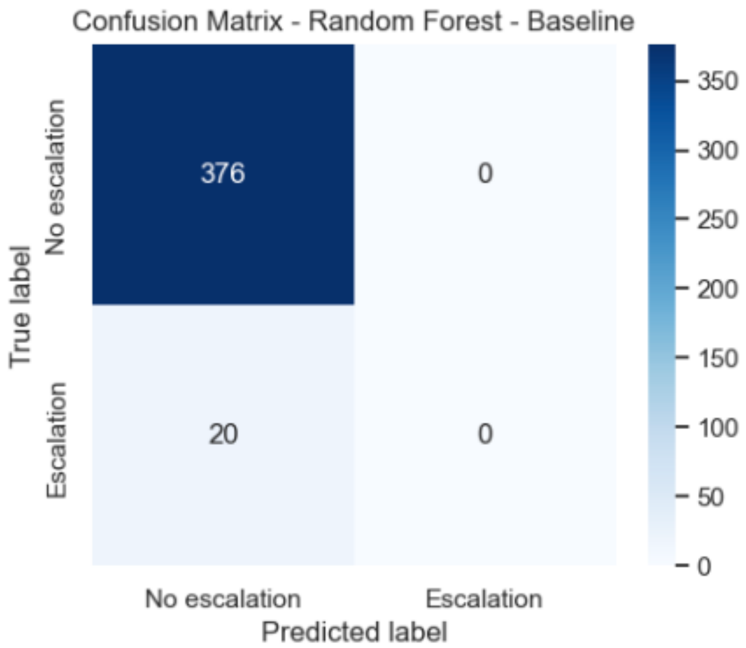


Figure 26. Confusion matrix obtained by the baseline Random Forest model.

3.2.2. RANDOM FOREST WITH SMOTE

To address the imbalance problem, a second Random Forest variant incorporated SMOTE during the training process.

The expectation was that generating synthetic escalation examples would encourage the ensemble to learn decision boundaries associated with future cybersickness events. Hyperparameter optimization selected a model with 500 trees, a maximum depth of 10, and three-neighbour SMOTE oversampling.

Surprisingly, the introduction of synthetic minority samples did not improve escalation detection. Although a small number of non-escalation windows were now classified as escalation events, the model still failed to correctly identify any true escalation case in the testing set.

The confusion matrix (Figure 27) reveals that the increase in positive predictions resulted exclusively in additional false positives rather than true positives. Consequently, recall remained equal to zero while overall accuracy decreased slightly due to the larger number of classification errors.

This outcome indicates that the synthetic samples generated through SMOTE were not sufficient to produce a meaningful separation between escalation and non-escalation windows within the Random Forest decision space. The model became marginally less conservative, but not enough to capture genuine escalation patterns.

From a methodological perspective, these findings illustrate an important limitation of oversampling approaches. While SMOTE can be highly effective in some domains, its success depends on the minority class occupying a coherent region of the feature space. If escalation windows are highly heterogeneous, synthetic interpolation between minority examples may not generate informative samples.

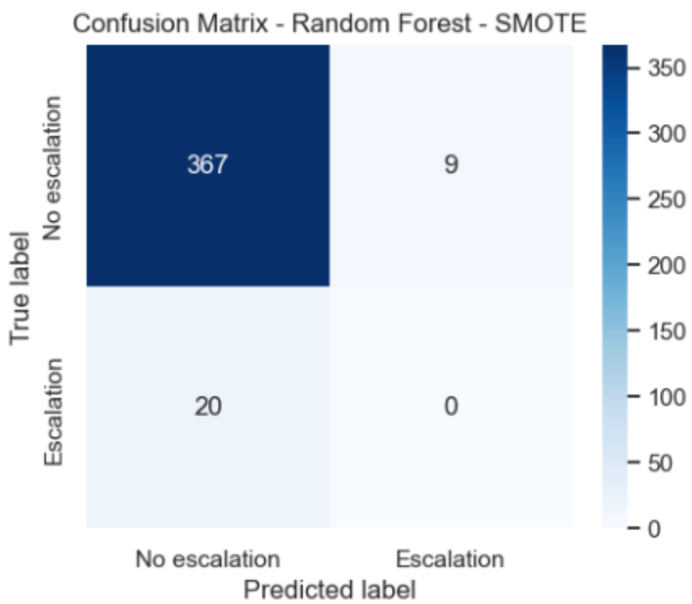


Figure 27. Confusion matrix obtained by the Random Forest with SMOTE.

3.2.3. RANDOM FOREST WITH SHAP-BASED FEATURE SELECTION

The third Random Forest variant investigated whether reducing the feature space to the twenty most influential variables could improve generalization.

SHAP analysis identified a set of variables primarily associated with:

- Eye gaze direction dynamics.
- Head rotation magnitude and variability.

- Head orientation measures.
- Convergence-related visual metrics.
- Pupil asymmetry characteristics.

These variables are physiologically meaningful because they represent mechanisms frequently associated with sensory conflict, visual discomfort, and instability during immersive virtual reality experiences.

The selected feature subset was subsequently used to train a new Random Forest model. Hyperparameter optimization favoured a configuration with 500 trees and balanced class weighting.

In Figure 28 we can see the confusion matrix. Although this model generated more positive predictions than the baseline version, none of these predictions corresponded to actual escalation events. The increase in sensitivity was therefore entirely driven by false positives.

Consequently, the SHAP-based feature reduction strategy did not improve escalation detection. In fact, some performance indicators deteriorated slightly relative to the baseline configuration.

This finding suggests that the inability of Random Forest to identify escalation events was not caused by excessive dimensionality. Instead, the underlying challenge appears to originate from the intrinsic difficulty of distinguishing future escalation windows from normal windows based solely on the available engineered features.

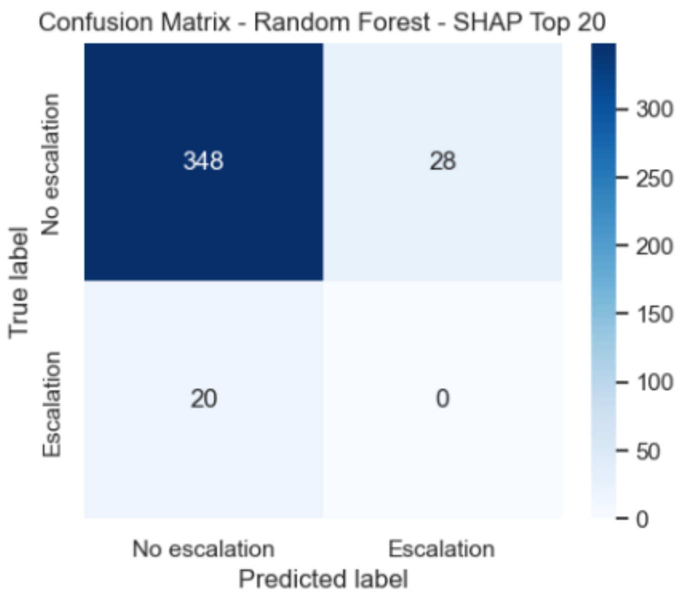


Figure 28. Confusion matrix obtained by the Random Forest SHAP Top-20 model.

3.2.4. COMPARATIVE ANALYSIS

Table 4 and Figure 29 present the comparison of the three Random Forest variants.

Table 4. Performance comparison of the Random Forest variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**RF - baseline**	0.95	0.50	0.00	0.00	0.00	0.64	0.07	0.00
1	**RF – SMOTE**	0.93	0.49	0.00	0.00	0.00	0.64	0.08	-0.03
2	**RF - SHAP**	0.88	0.46	0.00	0.00	0.00	0.63	0.07	-0.06

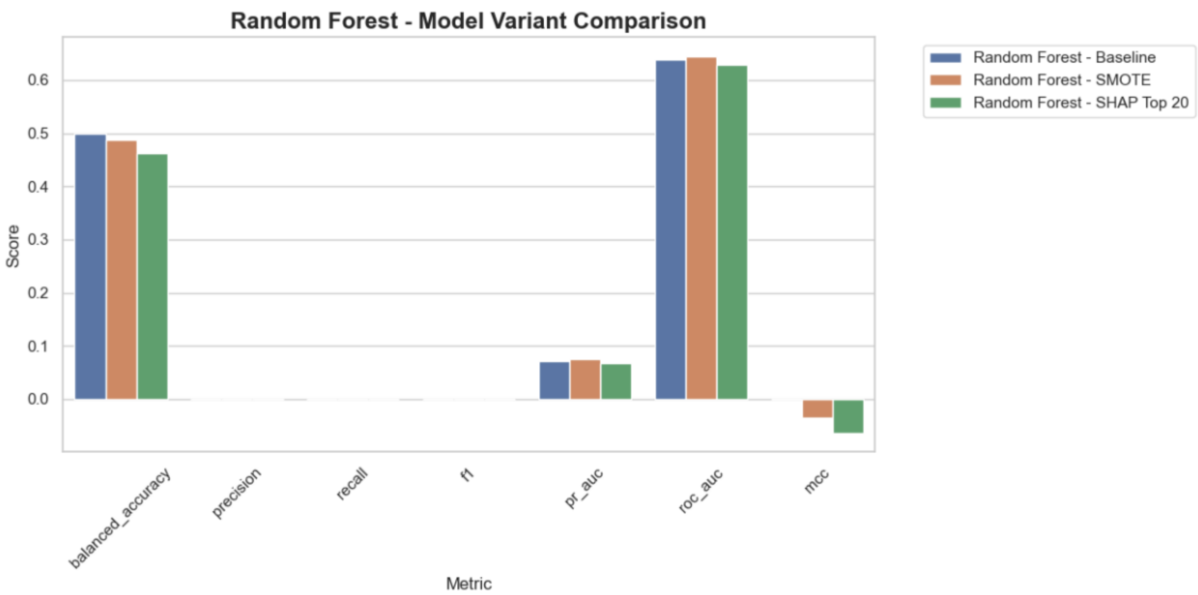


Figure 29. Random Forest Model Variant Comparison.

Several observations emerge from these results.

First, all Random Forest configurations exhibited a strong tendency toward majority-class predictions. Neither the baseline model, nor the SMOTE-enhanced version, nor the SHAP-based variant succeeded in correctly identifying escalation events within the independent test set.

Second, the incorporation of SMOTE and feature selection produced only marginal changes in model behaviour. Unlike the improvements observed in Logistic Regression after imbalance correction, Random Forest appeared largely insensitive to these modifications.

Third, the consistently low values of PR-AUC and Matthews Correlation Coefficient indicate that the ensemble failed to exploit the predictive information present in the minority class. While ROC-AUC values remained moderately above random levels, these scores primarily reflect ranking capability rather than effective event detection.

Overall, the Random Forest results suggest that ensemble tree methods are not particularly well suited to the current forecasting formulation. The model was able to learn general characteristics of normal behaviour but struggled to extract discriminative patterns associated with impending cybersickness escalation.

These findings provide additional evidence that escalation forecasting represents a highly challenging rare-event prediction problem. More sophisticated boosting algorithms and neural architectures may therefore be required to capture the subtle temporal signatures preceding cybersickness escalation.

3.3. XGBOOST

After evaluating linear models and bagging-based ensembles, XGBoost was investigated as a more advanced gradient boosting approach. XGBoost constructs trees sequentially, where each new tree attempts to correct the errors made by previous trees. This iterative optimization process often provides superior predictive performance in tabular datasets, particularly when the underlying relationships between variables are complex and non-linear.

The motivation for including XGBoost in this study was particularly strong. Unlike Random Forest, which builds trees independently, XGBoost focuses explicitly on difficult-to-classify observations. Since escalation events represent rare and potentially subtle physiological transitions, boosting-based methods were expected to be more capable of identifying these minority-class patterns.

Three XGBoost variants were evaluated:

1. Baseline XGBoost.
2. XGBoost with SMOTE.
3. XGBoost using SHAP-selected features.

3.3.1. BASELINE XGBOOST

The baseline XGBoost model represented the first classifier capable of detecting escalation events with meaningful effectiveness.

Hyperparameter optimization selected a relatively conservative boosting configuration, consisting of 100 trees, maximum tree depth of five, a low learning rate of 0.01, and a substantial positive-class weighting factor (`scale_pos_weight = 16.6`). The use of class weighting is particularly important because the forecasting dataset contains a strong imbalance between escalation and non-escalation windows.

The resulting confusion matrix shown in Figure 30 demonstrates a substantial behavioural shift compared with both Logistic Regression and Random Forest. Whereas previous models largely collapsed toward predicting only the majority class, XGBoost successfully identified eight of the twenty escalation windows present in the independent test set.

Specifically:

- True positives: 8
- False negatives: 12

- False positives: 62
- True negatives: 314

This corresponds to a recall of 0.40, meaning that 40% of future escalation events were successfully anticipated. Although precision remained relatively low (0.11), this outcome is expected in rare-event forecasting problems where sensitivity is prioritized over minimizing false alarms.

Several additional metrics reinforce the superiority of this configuration:

- Balanced Accuracy = 0.618
- F1-score = 0.178
- PR-AUC = 0.144
- MCC = 0.135

Importantly, the baseline XGBoost achieved the highest recall observed so far among all evaluated machine learning approaches. From an application perspective, missing an impending cybersickness escalation is generally more costly than generating an additional warning. Consequently, this increase in sensitivity represents a meaningful improvement.

The results suggest that XGBoost was able to identify non-linear physiological signatures associated with future cybersickness escalation that remained inaccessible to both Logistic Regression and Random Forest.

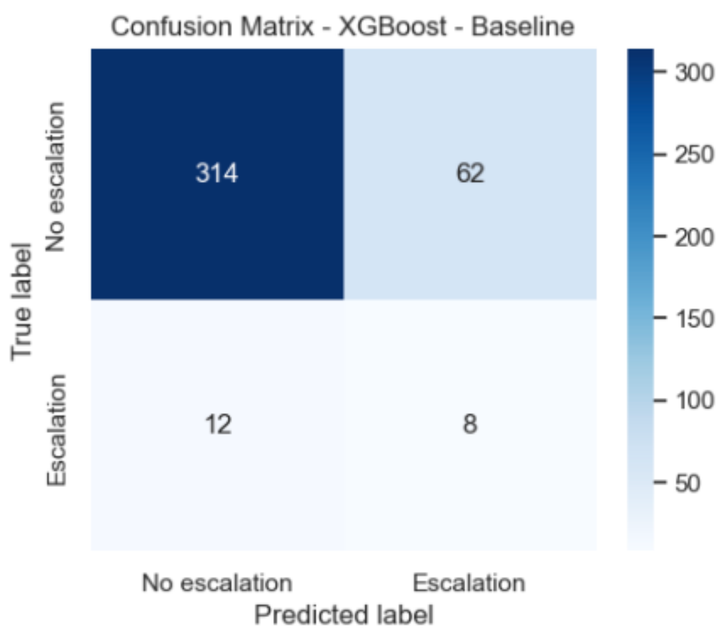


Figure 30. Confusion matrix obtained by the baseline XGBoost model.

3.3.2. XGBOOST WITH SMOTE

A second XGBoost variant incorporated SMOTE oversampling during training.

Given the improvements observed for Logistic Regression after balancing the training distribution, it was hypothesized that synthetic escalation examples could further enhance the ability of XGBoost to learn minority-class patterns.

The optimized configuration selected shallower trees (maximum depth = 3) combined with a higher learning rate (0.05) and SMOTE oversampling using three neighbours.

Contrary to expectations, the introduction of SMOTE led to a deterioration in predictive performance.

The confusion matrix (Figure 31) reveals that the model correctly identified only four escalation events, reducing recall from 0.40 to 0.20. Precision also decreased, indicating that the synthetic samples generated through SMOTE did not improve the discriminative structure of the minority class.

Several explanations may account for this behaviour.

First, XGBoost already incorporates mechanisms specifically designed to address class imbalance through instance weighting. The baseline model explicitly used a large `scale_pos_weight`, which may have provided a more realistic correction than synthetic oversampling.

Second, escalation windows likely exhibit substantial heterogeneity. Under such circumstances, interpolation between minority examples may create synthetic observations that do not correspond to realistic physiological states. Instead of strengthening the minority signal, SMOTE may introduce additional noise into the learning process.

The decline in Balanced Accuracy, F1-score, PR-AUC, and MCC consistently supports this interpretation.

These findings suggest that for this particular forecasting problem, class weighting appears more effective than synthetic oversampling when using gradient boosting models.

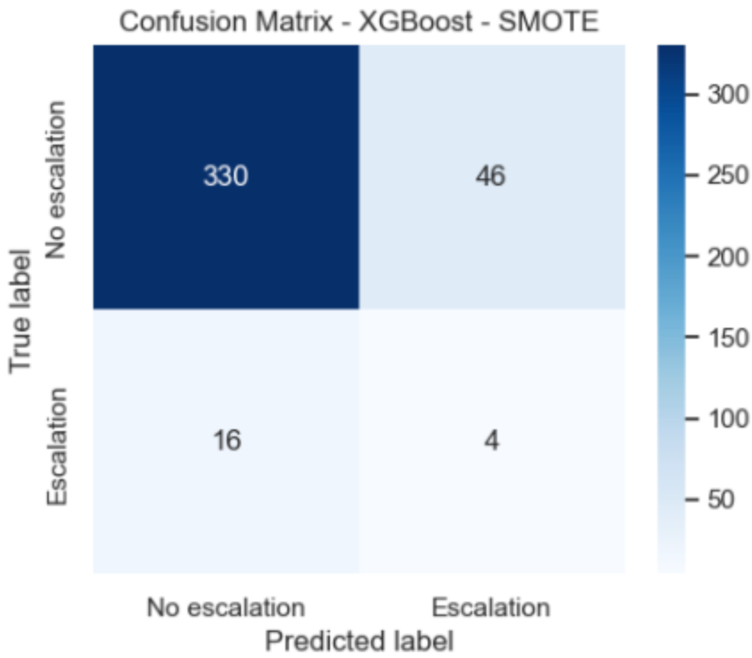


Figure 31. Confusion matrix obtained by the XGBoost-SMOTE model.

3.3.3. XGBOOST WITH SHAP-BASED FEATURE SELECTION

The third XGBoost variant explored whether reducing the feature space to the twenty most informative variables could improve generalization.

SHAP analysis identified a highly interpretable set of physiological descriptors. The most influential variables included:

- Eye origin variability metrics.
- Gaze direction dynamics.
- Head rotation magnitude statistics.
- Head orientation extrema.
- Velocity-related descriptors.
- Pupil diameter variability measures.

These variables are consistent with established theories of cybersickness, which emphasize the importance of visual instability, gaze behaviour, and head movement dynamics during symptom development.

The SHAP-selected features were subsequently used to train a new XGBoost classifier.

The resulting model produced an interesting trade-off.

Compared with the baseline configuration:

- Recall decreased from 0.40 to 0.20.
- Precision increased from 0.11 to 0.20.

- MCC improved from 0.135 to 0.157.

The confusion matrix (Figure 32) indicates that the model became substantially more conservative. Instead of generating a large number of warnings, the SHAP-based version focused on fewer but more reliable escalation predictions.

Specifically:

- True positives: 4
- False positives: 16

compared with

- True positives: 8
- False positives: 62

in the baseline model.

This behaviour is particularly relevant in practical deployment scenarios. A clinical or real-time VR monitoring system may prefer fewer false alarms, even at the cost of reduced sensitivity. Consequently, the SHAP-based variant provides a more interpretable and potentially more actionable decision-making framework.

Another important observation is that reducing the feature space from 308 engineered descriptors to only 20 variables caused only a modest reduction in predictive capability. This suggests that a relatively small subset of physiological signals captures a substantial portion of the information relevant to escalation forecasting.

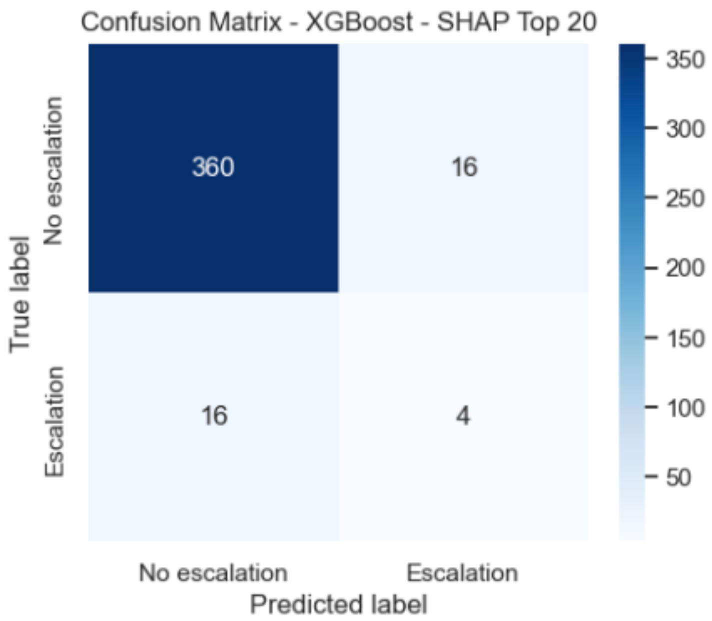


Figure 32. Confusion matrix obtained by the XGBoost SHAP Top-20 model.

3.3.4. COMPARATIVE ANALYSIS

Table 5 and Figure 33 present the comparison between the three XGBoost variants.

Table 5. Performance comparison of the XGBoost variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**XGB – Baseline**	0.81	0.62	0.11	0.40	0.18	0.70	0.14	0.13
1	**XGB – SMOTE**	0.84	0.54	0.08	0.20	0.11	0.64	0.09	0.05
2	**XGB - SHAP**	0.92	0.58	0.20	0.20	0.20	0.67	0.10	0.16

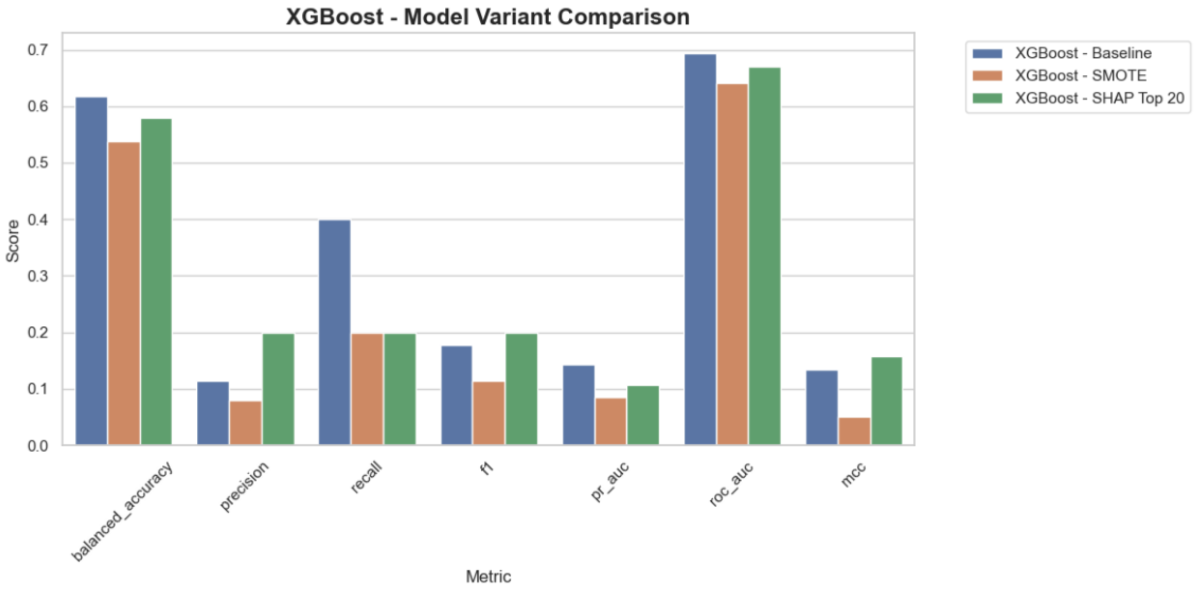


Figure 33. XGBoost Model Variant Comparison.

Several important conclusions emerge.

First, XGBoost clearly outperformed all previously evaluated models. Unlike Logistic Regression and Random Forest, every XGBoost configuration was capable of identifying at least some escalation events.

Second, the baseline model achieved the strongest detection capability, producing the highest recall and PR-AUC among the XGBoost variants. This indicates that retaining the complete feature set and leveraging class weighting provides the richest representation of escalation-related physiological patterns.

Third, the introduction of SMOTE consistently reduced performance. This result reinforces the hypothesis that synthetic minority examples may not adequately capture the complexity of real escalation events.

Finally, SHAP-based feature selection revealed that most predictive information is concentrated within a relatively small subset of variables. Although sensitivity decreased, the increase in precision and MCC demonstrates that dimensionality reduction can improve prediction reliability and model interpretability.

Overall, XGBoost represents the first model family in this study capable of generating practically meaningful escalation forecasts. The results indicate that gradient boosting is particularly well suited to modelling the subtle and highly non-linear physiological signatures that precede cybersickness escalation.

3.3.5. INTERPRETATION OF THE SHAP FEATURE SET

An additional advantage of the SHAP-based XGBoost model is the insight it provides into the physiological mechanisms underlying prediction.

The most influential features were dominated by:

- Head rotation statistics (`head_rotation_mag_*`).
- Head orientation dynamics (`HeadEulX`, `HeadRotationX`).
- Eye origin variability metrics.
- Gaze direction descriptors.
- Velocity-related measures.

This pattern is highly consistent with cybersickness theory. Visual instability, abnormal gaze behaviour, and increased head movement variability are widely recognized as indicators of sensory conflict and discomfort in immersive environments.

The fact that these variables repeatedly emerged among the most influential predictors suggests that the model is learning physiologically plausible representations rather than relying on spurious correlations.

This observation will be further explored in the Explainable Artificial Intelligence (XAI) section, where global and local explanations will be analysed in greater detail.

3.4. HISTGRADIENTBOOSTING

Following the evaluation of XGBoost, HistGradientBoosting (HGB) was investigated as an alternative gradient boosting framework available within the Scikit-learn ecosystem. HistGradientBoosting is a highly optimized implementation of gradient boosting decision trees that discretizes continuous features into histograms before tree construction, significantly reducing computational cost while maintaining strong predictive capabilities.

The motivation for including HistGradientBoosting was twofold. First, it belongs to the same family of boosting algorithms that demonstrated promising results with XGBoost. Second, it provides an efficient mechanism for modelling complex non-linear interactions without requiring the additional dependencies associated with external gradient boosting libraries.

Three variants were evaluated:

1. Baseline HistGradientBoosting.
2. HistGradientBoosting with SMOTE.
3. HistGradientBoosting using SHAP-selected features.

3.4.1. BASELINE HISTGRADIENTBOOSTING

The baseline HistGradientBoosting model employed class balancing and optimized boosting hyperparameters through grid search.

The best configuration consisted of:

- Learning rate = 0.01
- 100 boosting iterations
- 15 maximum leaf nodes
- No depth restriction
- No L2 regularization
- Balanced class weighting

Although the cross-validation performance appeared promising during training, the independent test results revealed substantial difficulties when generalizing to unseen participants.

The confusion matrix (Figure 34) shows:

- True positives: 0
- False negatives: 20
- False positives: 42
- True negatives: 334

Consequently, the model failed to identify any escalation event in the test set.

Performance metrics reflect this limitation:

- Recall = 0.00
- Precision = 0.00
- F1-score = 0.00
- Balanced Accuracy = 0.444
- PR-AUC = 0.067
- MCC = -0.079

Interestingly, despite the complete absence of detected escalation events, the model still produced a relatively large number of false alarms. This indicates that HistGradientBoosting assigned elevated risk scores to certain normal windows but was unable to correctly align those predictions with actual escalation episodes.

From a practical perspective, this behaviour is problematic because it combines the two least desirable outcomes: missed escalation events and unnecessary warnings.

The results suggest that the model learned participant-specific or noise-related patterns rather than robust physiological indicators of future cybersickness escalation.

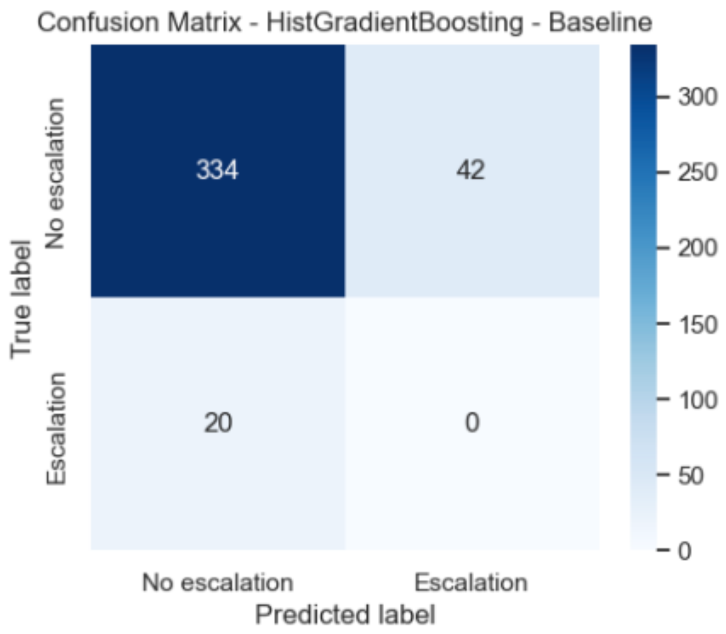


Figure 34. Confusion matrix obtained by the baseline HistGradientBoosting model.

3.4.2. HISTGRADIENTBOOSTING WITH SMOTE

To address class imbalance more explicitly, a second version incorporated SMOTE oversampling during training.

The optimized configuration increased the number of leaf nodes from 15 to 31 while maintaining the same learning rate and number of boosting iterations.

Despite these modifications, the resulting behaviour remained almost unchanged.

The confusion matrix (Figure 35) shows:

- True positives: 0
- False negatives: 20
- False positives: 38
- True negatives: 338

The model again failed to detect any escalation event.

Performance metrics remained nearly identical:

- Recall = 0.00
- Precision = 0.00
- F1-score = 0.00
- Balanced Accuracy = 0.449
- PR-AUC = 0.056
- MCC = -0.075

A slight reduction in false positives was observed compared with the baseline model; however, this improvement was not accompanied by any increase in sensitivity.

This finding is particularly interesting because it contrasts sharply with the behaviour observed in Logistic Regression, where SMOTE partially improved escalation detection. In HistGradientBoosting, synthetic oversampling appears unable to provide meaningful information beyond what the model already captures through its boosting mechanism.

A likely explanation is that the escalation class is not simply underrepresented but also highly heterogeneous. Under such conditions, generating synthetic observations through interpolation may fail to create physiologically meaningful examples.

Consequently, the model remains unable to learn a sufficiently discriminative boundary between escalation and non-escalation states.

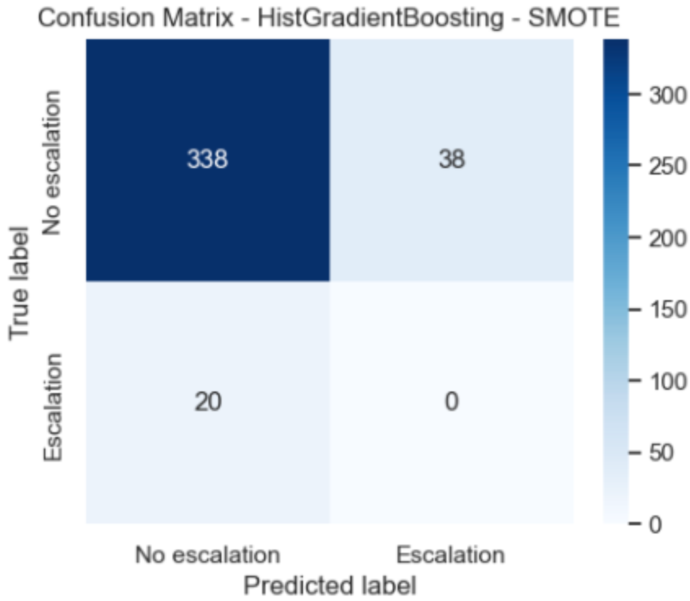


Figure 35. Confusion matrix obtained by the HistGradientBoosting-SMOTE model.

3.4.3. HISTGRADIENTBOOSTING WITH SHAP-BASED FEATURE SELECTION

The third variant explored whether restricting the model to the twenty most influential features could improve generalization.

SHAP analysis identified a feature set dominated by:

- Head rotation extrema and variability metrics.
- Eye origin variability descriptors.
- Gaze direction dynamics.
- Head Euler angle statistics.
- Pupil asymmetry measures.
- Eye-tracking sensor features.

These variables are physiologically plausible and closely resemble those identified previously by XGBoost and Random Forest, suggesting a consistent set of movement and gaze-related biomarkers across multiple model families.

The reduced feature space was subsequently used to train a new HistGradientBoosting model.

Although the resulting classifier remained unable to detect escalation events, several metrics improved slightly.

The confusion matrix (Figure 36) shows:

- True positives: 0
- False negatives: 20
- False positives: 23
- True negatives: 353

Compared with the previous variants, the number of false alarms was reduced substantially.

This behaviour produced:

- Balanced Accuracy = 0.469
- ROC-AUC = 0.595
- PR-AUC = 0.064
- MCC = -0.057

While the model still failed to identify escalation windows, the reduction in false positives indicates that the SHAP-selected features removed a considerable amount of noise from the original feature space.

The model became more conservative and produced cleaner decision boundaries. However, this increased specificity was insufficient to overcome the fundamental challenge of detecting the minority class.

These findings suggest that feature selection alone cannot compensate for the absence of sufficiently informative predictive patterns within the HistGradientBoosting framework.

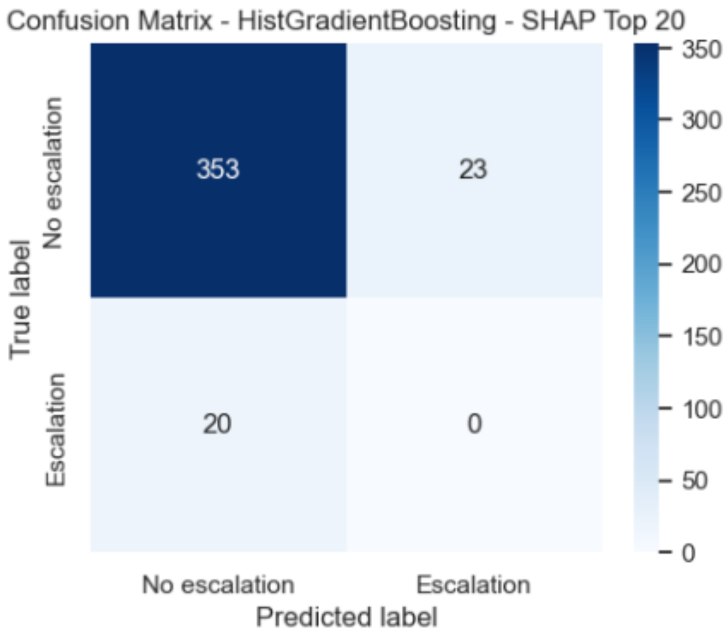


Figure 36. Confusion matrix obtained by the HistGradientBoosting SHAP Top-20 model.

3.4.4. COMPARATIVE ANALYSIS

Table 6 and Figure 37 present a comparison of all HistGradientBoosting variants.

Table 6. Performance comparison of the HistGradientBoosting variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**HGB – baseline**	0.84	0.44	0.00	0.00	0.00	0.58	0.07	-0.08
1	**HGB – SMOTE**	0.85	0.45	0.00	0.00	0.00	0.52	0.06	-0.08
2	**HGB - SHAP**	0.89	0.47	0.00	0.00	0.00	0.59	0.06	-0.06

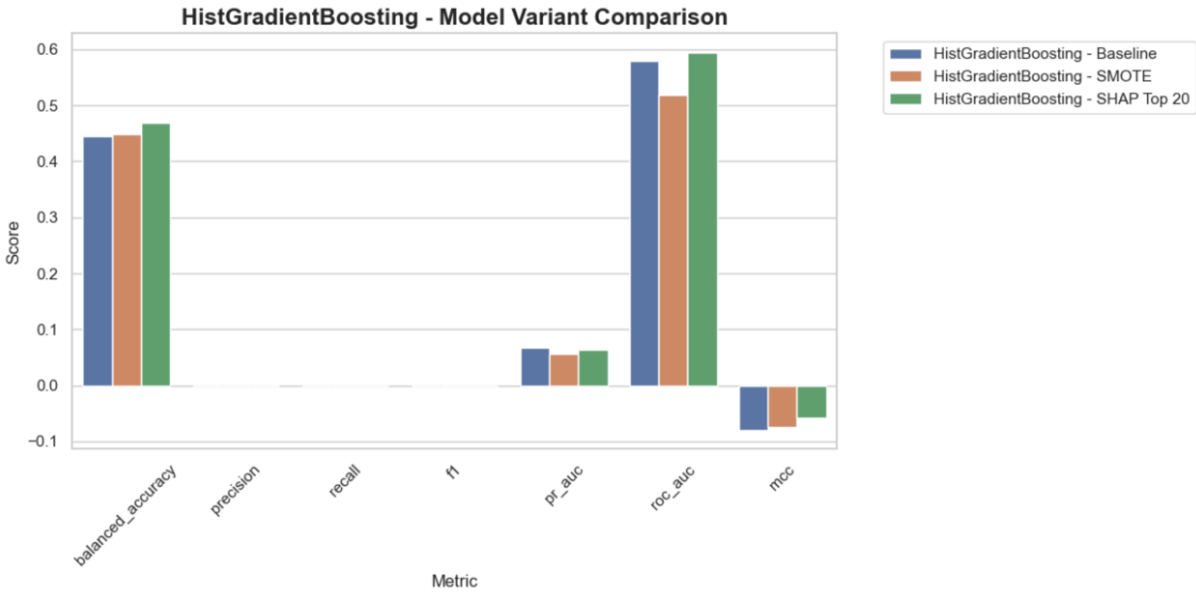


Figure 37. HistGradientBoosting Model Variant Comparison.

Several observations emerge.

First, unlike XGBoost, HistGradientBoosting was unable to detect any escalation event across all evaluated configurations. Recall, precision, and F1-score remained equal to zero regardless of whether oversampling or feature selection was applied.

Second, both SMOTE and SHAP-based dimensionality reduction produced only marginal changes in performance. Although small improvements were observed in Balanced Accuracy and MCC, none of these modifications translated into actual escalation detection.

Third, the consistently negative MCC values indicate that the model's predictions were not meaningfully aligned with the true class labels. This contrasts with XGBoost, where positive MCC values demonstrated genuine predictive capability.

Finally, the comparison between XGBoost and HistGradientBoosting highlights an important methodological finding. Although both algorithms belong to the gradient boosting family, their behaviour differed substantially. XGBoost successfully captured a portion of the escalation signal, whereas HistGradientBoosting failed to generalize beyond the majority class.

This suggests that the additional regularization mechanisms, class weighting strategy, and boosting formulation employed by XGBoost were better suited to the highly imbalanced and noisy nature of the cybersickness forecasting problem.

3.4.5. INTERPRETATION OF THE SHAP FEATURE SET

Despite the disappointing predictive performance, the SHAP analysis remains informative from a physiological perspective.

The most influential features were consistently associated with:

- Head movement dynamics.
- Head rotation magnitude.
- Eye origin variability.
- Gaze direction changes.
- Pupil asymmetry behaviour.

These variables closely match those identified by both Random Forest and XGBoost, reinforcing the hypothesis that cybersickness escalation is preceded by subtle alterations in gaze control and head movement stability.

Therefore, the poor predictive performance of HistGradientBoosting should not be interpreted as evidence that these variables lack relevance. Rather, it suggests that the specific learning mechanism employed by this algorithm was unable to exploit these physiological signatures effectively.

This convergence across multiple SHAP analyses strengthens confidence in the biological plausibility of the identified biomarkers and provides valuable context for the explainability analyses presented later in the study.

3.5. SUPPORT VECTOR MACHINE (SVM)

Support Vector Machines (SVMs) were evaluated as an alternative family of classifiers capable of learning complex decision boundaries in high-dimensional feature spaces. Unlike linear models, SVMs can project the original feature space into higher-dimensional representations through kernel functions, enabling the detection of non-linear relationships that may not be captured by simpler classifiers.

Given the large number of engineered physiological features and the potentially subtle separation between escalation and non-escalation states, SVMs were considered particularly attractive because of their ability to model high-dimensional data while maintaining good generalization properties.

Three variants were investigated:

1. Baseline SVM.
2. SVM with SMOTE and feature selection.
3. SVM with SHAP-selected features.

3.5.1. BASELINE SVM

The baseline configuration employed feature standardization followed by an SVM classifier with probability estimation enabled.

Hyperparameter optimization selected:

- RBF kernel.
- C = 10.
- Gamma = scale.
- No class weighting.

The resulting model exhibited behaviour similar to the baseline Logistic Regression and Random Forest models.

The confusion matrix (Figure 38) shows:

- True positives: 0
- False negatives: 20
- False positives: 0
- True negatives: 376

Thus, the classifier assigned every test sample to the majority class.

Performance metrics confirm this behaviour:

- Recall = 0.00
- Precision = 0.00
- F1-score = 0.00
- Balanced Accuracy = 0.50
- ROC-AUC = 0.532
- PR-AUC = 0.064
- MCC = 0.00

Although the RBF kernel theoretically allows the model to capture highly non-linear patterns, the extreme class imbalance appears to dominate the learning process. As a result, the classifier converged to a trivial solution that maximizes overall accuracy while completely ignoring escalation events.

This outcome highlights an important challenge of cybersickness forecasting: sophisticated classification algorithms do not necessarily outperform simpler approaches when the minority class is scarce and highly heterogeneous.

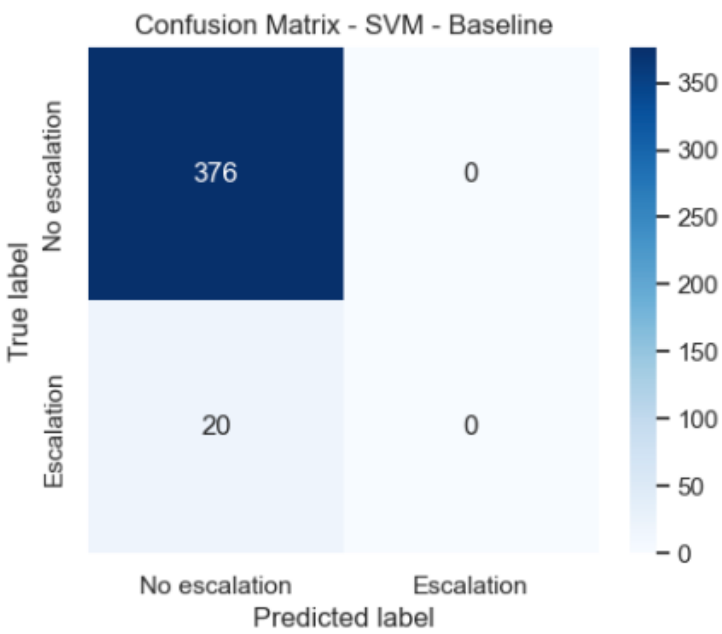


Figure 38. Confusion matrix obtained by the baseline SVM model.

3.5.2. SVM WITH SMOTE AND FEATURE SELECTION

The second configuration incorporated two additional mechanisms specifically designed to address the characteristics of the dataset:

1. Feature selection through SelectKBest.
2. SMOTE oversampling of the escalation class.

The optimal configuration selected:

- 30 features.
- RBF kernel.
- C = 0.1.
- Gamma = auto.

Unlike the baseline model, this variant successfully identified escalation events.

The confusion matrix (Figure 39) shows:

- True positives: 9
- False negatives: 11
- False positives: 135
- True negatives: 241

The model detected 45% of escalation events, representing a substantial improvement over the baseline version.

Performance metrics were:

- Recall = 0.45
- Precision = 0.063
- F1-score = 0.110
- Balanced Accuracy = 0.545
- ROC-AUC = 0.637
- PR-AUC = 0.075
- MCC = 0.041

From a clinical perspective, this behaviour reflects a more useful operating point than the baseline model because escalation events are no longer completely ignored.

However, the improvement in sensitivity came at the cost of a large increase in false positives. More than one-third of normal windows were incorrectly classified as escalation events.

This phenomenon is consistent with the behaviour previously observed in the Logistic Regression SMOTE variant. By artificially balancing the classes, the classifier becomes more willing to predict escalation but sacrifices specificity.

Although the overall discriminative performance remains limited, these results demonstrate that SVMs are capable of extracting part of the escalation signal when provided with a more balanced and reduced feature space.

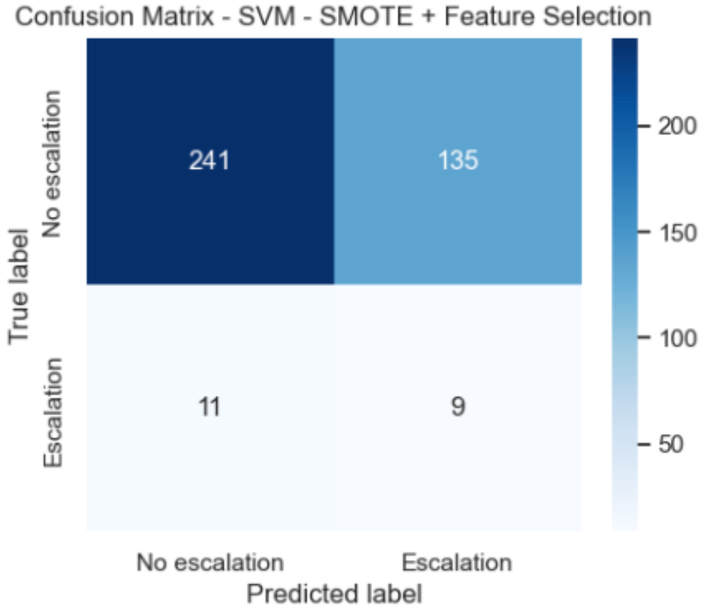


Figure 39. Confusion matrix obtained by the SVM with SMOTE and feature selection.

3.5.3. SVM WITH SHAP-BASED FEATURE SELECTION

Direct SHAP explanations for SVMs require KernelSHAP, which is computationally expensive for datasets of this size. Consequently, the SHAP-selected feature subset obtained from the Logistic Regression explainability analysis was used as a proxy feature selection strategy.

This approach preserved the twenty features previously identified as the most informative predictors while drastically reducing dimensionality.

The optimal configuration selected:

- Linear kernel.
- C = 0.1.
- Balanced class weighting.

The resulting confusion matrix (Figure 40) shows:

- True positives: 5
- False negatives: 15
- False positives: 153
- True negatives: 223

This configuration detected fewer escalation events than the SMOTE-based version.

Performance metrics were:

- Recall = 0.25
- Precision = 0.032
- F1-score = 0.056
- Balanced Accuracy = 0.422
- ROC-AUC = 0.470
- PR-AUC = 0.049
- MCC = -0.070

The reduction in recall and the negative MCC indicate that this feature selection strategy was not beneficial for SVM performance.

Unlike tree-based models such as XGBoost, which benefited from SHAP-driven dimensionality reduction, the SVM appears to require a richer representation of the feature space to construct an effective separating hyperplane.

This result suggests that the interactions captured by SVMs are distributed across a larger set of variables and cannot be fully represented by only twenty features.

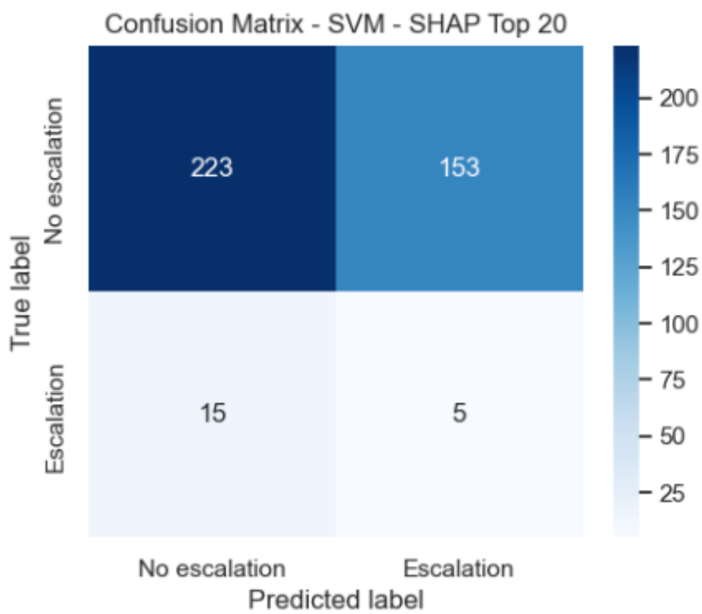


Figure 40. Confusion matrix obtained by the SHAP Top-20 SVM model.

3.5.4. COMPARATIVE ANALYSIS

Table 7 and Figure 41 compare the three SVM variants.

Table 7. Performance comparison of the SVM variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**SVM – baseline**	0.95	0.50	0.00	0.00	0.00	0.53	0.06	0.00

1	**SVM – SMOTE + FS**	0.63	0.55	0.06	0.45	0.11	0.64	0.08	0.04
2	**SVM - SHAP**	0.58	0.42	0.03	0.25	0.06	0.47	0.05	-0.07

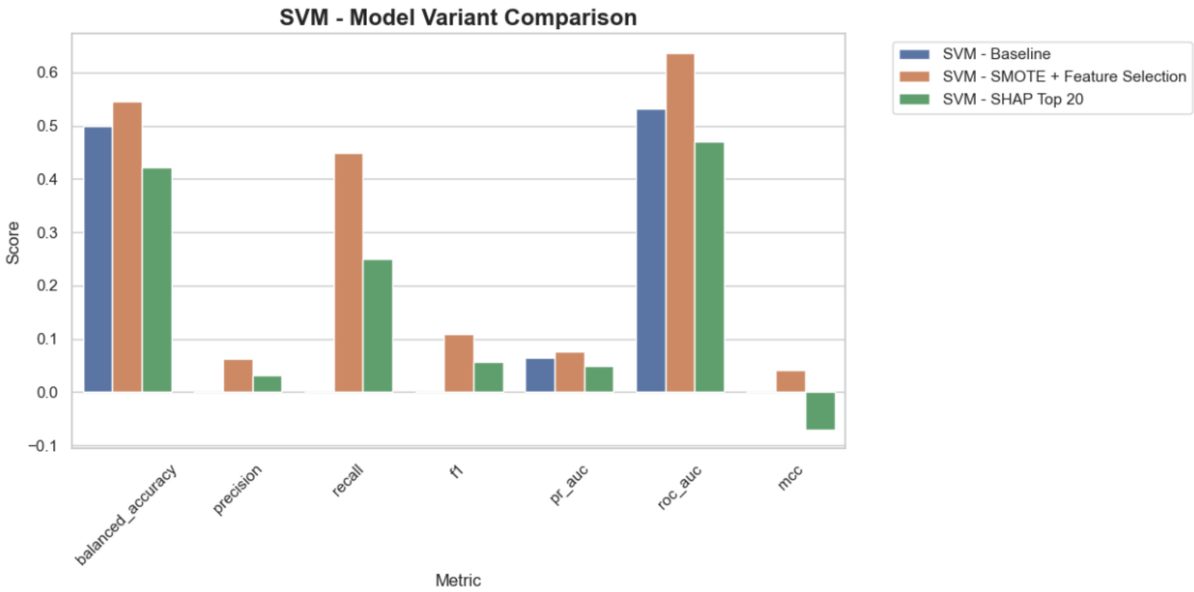


Figure 41. SVM Model Variant Comparison.

Several findings emerge from this comparison.

First, the baseline model completely failed to identify escalation events despite achieving very high overall accuracy. This again demonstrates that accuracy is not an informative metric in highly imbalanced prediction problems.

Second, the combination of SMOTE and feature selection produced the strongest SVM variant. This model achieved the highest recall, balanced accuracy, PR-AUC and MCC within the SVM family.

Third, SHAP-based feature reduction negatively affected performance. While the selected features were physiologically meaningful, the resulting reduction in dimensionality appears to have removed information necessary for the SVM to construct robust decision boundaries.

Finally, even the best SVM variant remained substantially weaker than the best XGBoost configuration. The SVM family showed moderate sensitivity improvements but generated a large number of false alarms, limiting its practical applicability.

3.5.5. INTERPRETATION OF THE SELECTED FEATURES

Although SHAP-based dimensionality reduction did not improve predictive performance, the selected features remain informative from a physiological standpoint.

The feature subset was dominated by:

- Head rotation statistics.
- Head Euler angle dynamics.
- Gaze direction variability.
- Eye openness measures.
- Pupil position descriptors.
- Vergence-related variables.

These features are highly consistent with those identified previously by Logistic Regression, Random Forest, XGBoost and HistGradientBoosting.

The repeated appearance of these variables across independent model families reinforces the hypothesis that cybersickness escalation is associated with subtle alterations in head stability, gaze behaviour and ocular control mechanisms.

Therefore, the poor performance of the SHAP-reduced SVM should not be interpreted as evidence against the relevance of these biomarkers. Instead, it suggests that the SVM learning mechanism was unable to exploit them as effectively as other algorithms.

3.5.6. COMPARISON WITH PREVIOUS MODELS

Compared with the models evaluated previously, the SVM family occupies an intermediate position.

- It clearly outperformed Random Forest and HistGradientBoosting, which failed to detect escalation events entirely.
- It exhibited behaviour very similar to Logistic Regression with SMOTE.
- It remained inferior to XGBoost, which achieved stronger recall, precision, F1-score, PR-AUC and MCC simultaneously.

These findings suggest that while kernel-based methods can partially recover escalation-related patterns, gradient boosting approaches are more effective at exploiting the complex non-linear relationships present in physiological cybersickness data.

3.6. CATBOOST

CatBoost is a gradient boosting algorithm specifically designed to improve the learning process of decision tree ensembles while reducing overfitting and prediction bias. Compared with traditional boosting approaches, CatBoost introduces ordered boosting and advanced regularization mechanisms that often result in stronger generalization performance, particularly when complex interactions exist among features.

Although CatBoost is especially well known for handling categorical variables, it can also perform effectively on high-dimensional numerical datasets such as the physiological and behavioural measurements used in this study. Given its strong performance in many imbalanced classification tasks, CatBoost was evaluated as a potential alternative to XGBoost and HistGradientBoosting.

Three variants were investigated:

1. Baseline CatBoost.
2. CatBoost with SMOTE.
3. CatBoost with SHAP-based feature selection.

3.6.1. BASELINE CATBOOST

The baseline CatBoost model was trained using the complete feature set and optimized through grid search.

The optimal configuration selected:

- Tree depth = 5.
- Learning rate = 0.1.
- 100 boosting iterations.
- L2 regularization = 1.
- Automatic balanced class weighting.

Despite incorporating class balancing internally, the resulting classifier failed to identify any escalation events.

The confusion matrix (Figure 42) shows:

- True positives: 0
- False negatives: 20
- False positives: 0
- True negatives: 376

Performance metrics were:

- Recall = 0.00
- Precision = 0.00
- F1-score = 0.00
- Balanced Accuracy = 0.50
- ROC-AUC = 0.663
- PR-AUC = 0.081
- MCC = 0.00

This behaviour mirrors that observed in the baseline Logistic Regression, Random Forest and SVM models. The classifier converged to a majority-class solution, maximizing overall accuracy (94.95%) while completely ignoring escalation events.

However, the ROC-AUC of 0.663 suggests that the underlying probability estimates retained some discriminatory information. In other words, the model learned patterns associated with escalation but these patterns were not sufficiently strong to cross the default classification threshold.

This observation is important because it indicates that the model contained useful predictive information despite its poor classification behaviour.

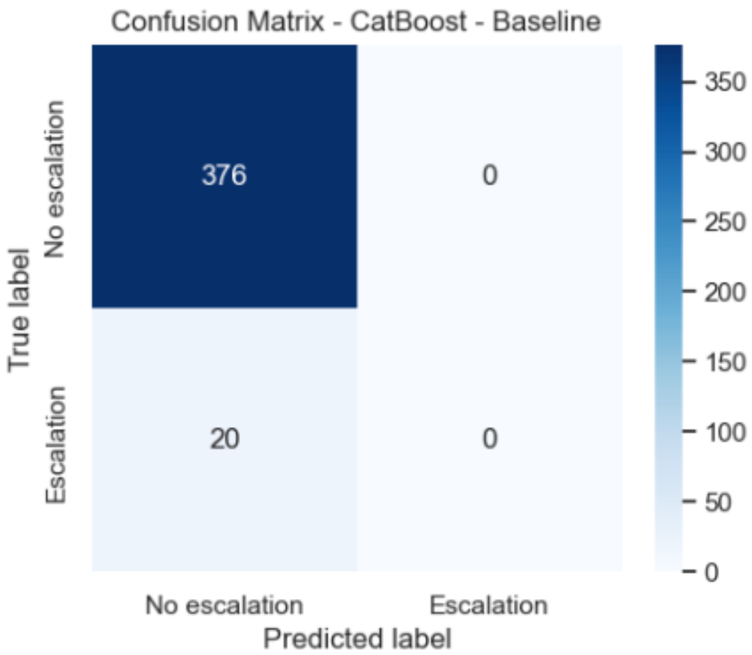


Figure 42. Confusion matrix obtained by the baseline CatBoost model.

3.6.2. CATBOOST WITH SMOTE

To improve sensitivity toward escalation events, SMOTE oversampling was incorporated before training.

The optimal configuration selected:

- Tree depth = 3.
- Learning rate = 0.1.
- 100 boosting iterations.
- L2 regularization = 3.
- SMOTE k-neighbours = 3.

Surprisingly, the introduction of SMOTE did not improve escalation detection.

The confusion matrix (Figure 43) shows:

- True positives: 0
- False negatives: 20
- False positives: 15
- True negatives: 361

Performance metrics were:

- Recall = 0.00
- Precision = 0.00
- F1-score = 0.00

- Balanced Accuracy = 0.480
- ROC-AUC = 0.717
- PR-AUC = 0.106
- MCC = -0.046

Although the model generated some false-positive escalation predictions, it still failed to identify any true escalation event.

This result differs markedly from Logistic Regression and SVM, where SMOTE increased recall substantially.

A possible explanation is that CatBoost already incorporates internal mechanisms for handling class imbalance through automatic class weighting. Consequently, external oversampling may have introduced additional noise without providing genuinely informative minority-class examples.

Interestingly, ROC-AUC and PR-AUC both increased relative to the baseline model, suggesting that ranking performance improved despite the lack of improvement in hard classification decisions.

Therefore, the issue appears to be related more to threshold calibration than to the model's ability to learn informative patterns.

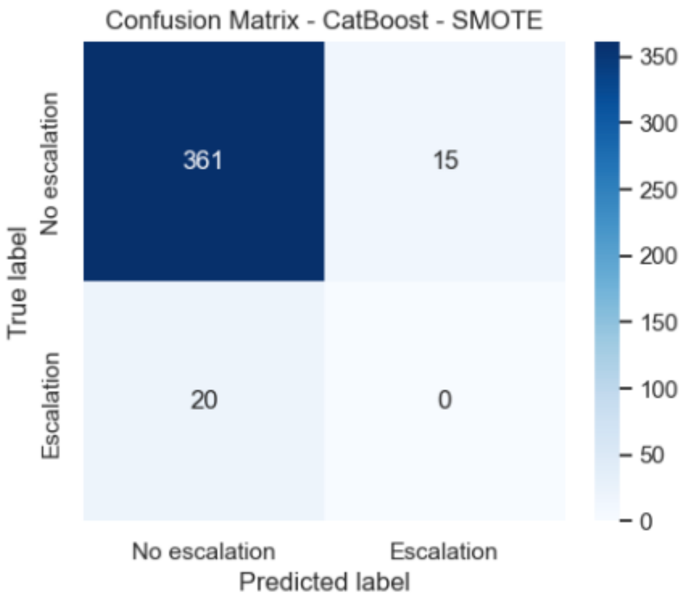


Figure 43. Confusion matrix obtained by the CatBoost-SMOTE model.

3.6.3. CATBOOST WITH SHAP-BASED FEATURE SELECTION

The third configuration employed SHAP values to identify the twenty most influential variables and retrained CatBoost using only this reduced feature set.

The most important predictors included:

- Eye-gaze direction statistics.
- Head rotation dynamics.
- Head Euler angles.
- Eye movement variability.
- Pupil position measurements.
- Eye openness descriptors.

Many of these variables also appeared among the top-ranked features in XGBoost, Random Forest and HistGradientBoosting analyses.

The optimized CatBoost configuration selected:

- Tree depth = 3.
- Learning rate = 0.05.
- 100 boosting iterations.
- L2 regularization = 3.
- Automatic balanced class weighting.

The resulting confusion matrix (Figure 44) shows:

- True positives: 6
- False negatives: 14
- False positives: 70
- True negatives: 306

Performance metrics were:

- Recall = 0.30
- Precision = 0.079
- F1-score = 0.125
- Balanced Accuracy = 0.557
- ROC-AUC = 0.659
- PR-AUC = 0.150
- MCC = 0.063

Unlike the previous CatBoost variants, the SHAP-reduced model successfully detected escalation events.

The model correctly identified 30% of escalation windows while maintaining substantially fewer false positives than the Logistic Regression and SVM SMOTE variants.

Although recall remained modest, the improvement demonstrates that reducing the feature space to the most informative variables helped CatBoost focus on relevant physiological patterns and reduced the impact of noisy or redundant predictors.

Notably, the PR-AUC reached 0.150, representing the highest value among all CatBoost variants and one of the strongest precision-recall performances observed throughout the study.

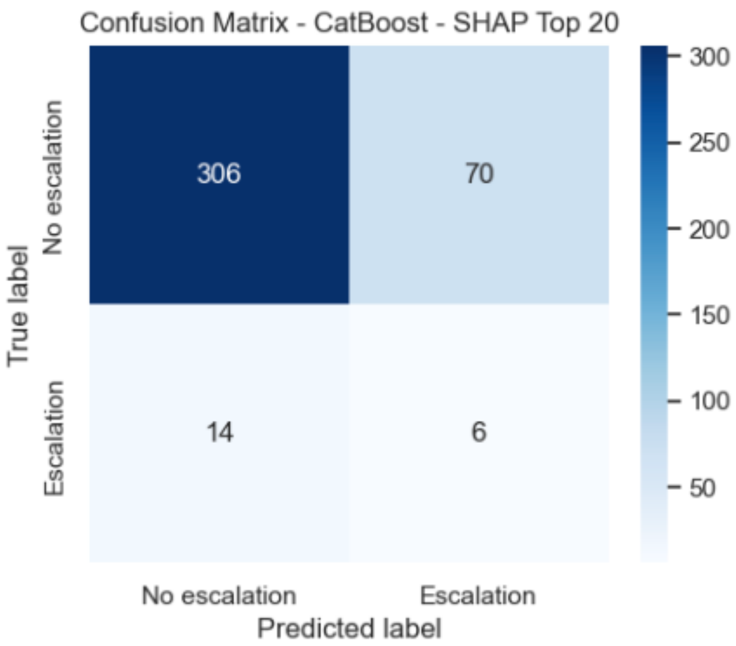


Figure 44. Confusion matrix obtained by the CatBoost SHAP Top-20 model.

3.6.4. COMPARATIVE ANALYSIS

Table 8 and Figure 45 compare the three CatBoost variants.

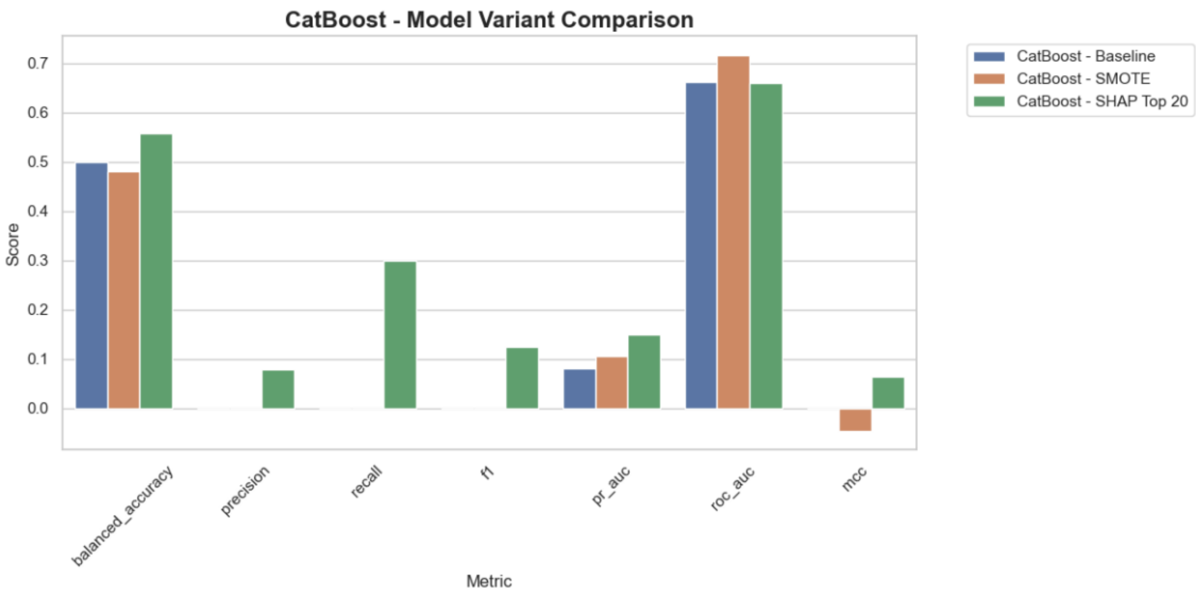


Figure 45. CatBoost Model Variant Comparison.

Several important observations emerge.

First, neither the baseline model nor the SMOTE-enhanced version managed to identify any escalation events. This finding indicates that class balancing alone was insufficient to overcome the severe imbalance and overlap present in the physiological data.

Second, SHAP-based feature selection produced a substantial improvement across nearly all relevant metrics:

Table 8. Performance comparison of the CatBoost variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**CB – baseline**	0.95	0.50	0.00	0.00	0.00	0.66	0.08	0.00
1	**CB – SMOTE**	0.91	0.48	0.00	0.00	0.00	0.72	0.11	-0.05
2	**CB - SHAP**	0.79	0.56	0.08	0.3	0.13	0.66	0.15	0.06

These improvements suggest that feature selection was considerably more beneficial than oversampling for CatBoost.

Third, the results reinforce an observation already noted with XGBoost: gradient boosting models appear particularly sensitive to feature quality rather than feature quantity. Eliminating noisy variables can improve their ability to identify meaningful patterns associated with escalation.

3.6.5. INTERPRETATION OF THE SELECTED FEATURES

The SHAP analysis revealed remarkable consistency with previous model families.

The most influential predictors were dominated by:

- Head rotation dynamics.
- Eye-gaze directional variability.
- Eye movement statistics.
- Head orientation descriptors.
- Eye openness measures.
- Pupil position metrics.

Particularly prominent variables included:

- NrmSRLeftEyeGazeDirX_mean.
- HeadQRotationW_median.
- NrmRightEyeOriginZ_std.
- Head rotation magnitude descriptors.
- Head Euler angle statistics.

These variables describe how users stabilize gaze, coordinate head movements and maintain visual focus while immersed in virtual environments.

The repeated emergence of these features across Logistic Regression, Random Forest, XGBoost, HistGradientBoosting, SVM and CatBoost strongly supports their role as robust physiological indicators of cybersickness escalation.

This consistency increases confidence that the identified biomarkers reflect genuine behavioural phenomena rather than model-specific artefacts.

3.6.6. COMPARISON WITH PREVIOUS MODELS

Within the gradient boosting family, CatBoost produced mixed results.

Compared with HistGradientBoosting:

- CatBoost achieved higher ROC-AUC.
- CatBoost obtained better PR-AUC.
- CatBoost was the only boosting model besides XGBoost that achieved meaningful escalation detection after SHAP feature selection.

Compared with XGBoost:

- CatBoost achieved lower recall (0.30 vs 0.40 in the XGBoost baseline).
- CatBoost achieved lower MCC.
- CatBoost achieved slightly higher PR-AUC in the SHAP configuration (0.150 vs 0.107).
- XGBoost remained the overall strongest performer.

Compared with Logistic Regression and SVM:

- CatBoost SHAP produced fewer false positives.
- CatBoost achieved a more balanced trade-off between sensitivity and specificity.

Therefore, CatBoost can be considered a competitive alternative, particularly when interpretability-driven feature selection is incorporated.

3.7. MULTILAYER PERCEPTRON (MLP)

Artificial Neural Networks have demonstrated strong performance in a wide range of classification tasks due to their ability to model complex non-linear relationships between variables. In this study, a Multilayer Perceptron (MLP) was evaluated to determine whether deep non-linear representations could improve the detection of cybersickness escalation events from physiological and behavioural features.

Given the high dimensionality of the dataset and the severe class imbalance, three variants were explored:

1. Baseline MLP.
2. MLP with SMOTE and feature selection.
3. MLP trained using the SHAP Top-20 features.

3.7.1. BASELINE MLP

The baseline MLP was trained using the complete feature set without oversampling or feature reduction.

The resulting confusion matrix (Figure 46) shows:

- True positives: 0
- False negatives: 20
- False positives: 4
- True negatives: 372

Performance metrics were:

- Accuracy = 0.939
- Balanced Accuracy = 0.495
- Precision = 0.000
- Recall = 0.000
- F1-score = 0.000
- ROC-AUC = 0.638
- PR-AUC = 0.072
- MCC = -0.023

Similarly to several previously evaluated baseline models, the neural network converged to a majority-class solution and failed to identify any escalation events.

Despite achieving a high overall accuracy, the model was clinically useless because all escalation cases were classified as non-escalation. The balanced accuracy close to 0.5 confirms that performance was essentially equivalent to random guessing when both classes are considered equally important.

The ROC-AUC value of 0.638 indicates that the network learned some discriminatory information from the data, but this information was not sufficiently reflected in the final classification decisions.

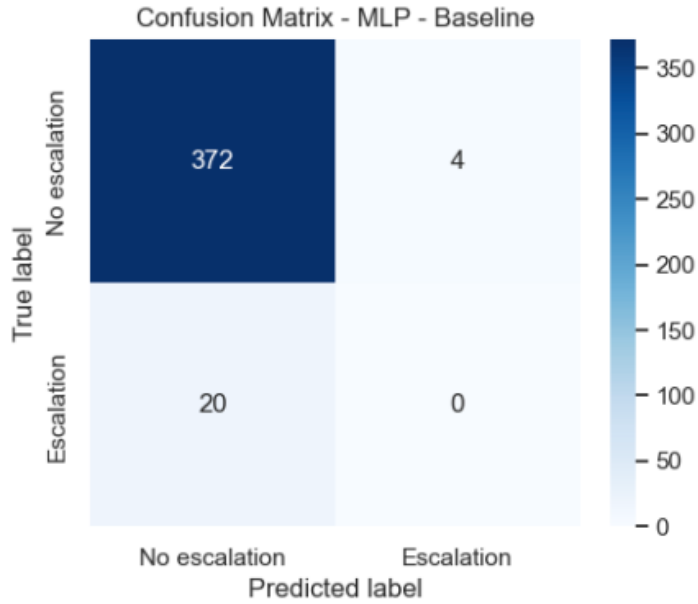


Figure 46. Confusion matrix obtained by the baseline MLP model.

3.7.2. MLP WITH SMOTE AND FEATURE SELECTION

The second configuration combined minority-class oversampling with feature selection before model training.

The confusion matrix (Figure 47) shows:

- True positives: 9
- False negatives: 11
- False positives: 61
- True negatives: 315

Performance metrics were:

- Accuracy = 0.818
- Balanced Accuracy = 0.644
- Precision = 0.129
- Recall = 0.450
- F1-score = 0.200
- ROC-AUC = 0.800
- PR-AUC = 0.121
- MCC = 0.165

This variant produced a substantial improvement over the baseline model.

The network successfully detected 45% of escalation events, representing one of the highest recall values observed among all evaluated models. Although the increase in sensitivity was accompanied by a larger number of false positives, the model achieved the strongest overall balance between sensitivity and discrimination within the MLP family.

Particularly noteworthy is the ROC-AUC of 0.800, which is the highest value obtained among all MLP configurations and one of the strongest discrimination scores observed throughout the entire study.

The MCC value of 0.165 also represents the highest correlation between predictions and ground truth achieved by any MLP variant.

These results suggest that neural networks benefit considerably from balancing strategies and dimensionality reduction when learning from highly imbalanced physiological datasets.

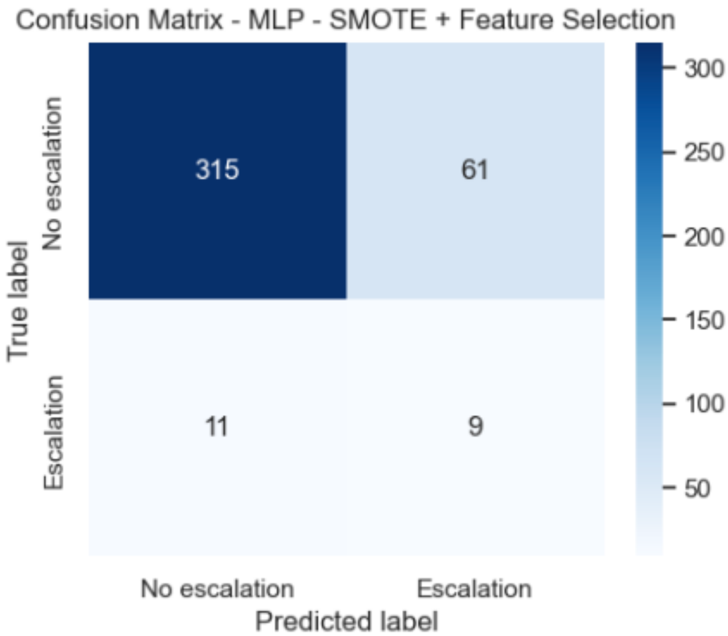


Figure 47. Confusion matrix obtained by the MLP with SMOTE and feature selection.

3.7.3. MLP WITH SHAP TOP-20 FEATURES

The third configuration was trained exclusively using the twenty most relevant features identified through SHAP analysis.

The confusion matrix (Figure 48) shows:

- True positives: 0
- False negatives: 20
- False positives: 7
- True negatives: 369

Performance metrics were:

- Accuracy = 0.932
- Balanced Accuracy = 0.491
- Precision = 0.000

- Recall = 0.000
- F1-score = 0.000
- ROC-AUC = 0.544
- PR-AUC = 0.072
- MCC = -0.031

Contrary to the behaviour observed in XGBoost and CatBoost, restricting the feature space to the SHAP Top-20 variables did not improve MLP performance.

The network again converged to a majority-class solution and failed to identify any escalation events.

One possible explanation is that neural networks typically benefit from richer feature representations and complex interactions among variables. While tree-based models often improve after aggressive feature reduction, neural networks may lose useful complementary information when the feature space becomes excessively constrained.

The reduction from hundreds of variables to only twenty features may therefore have limited the network's capacity to learn the subtle patterns associated with escalation events.

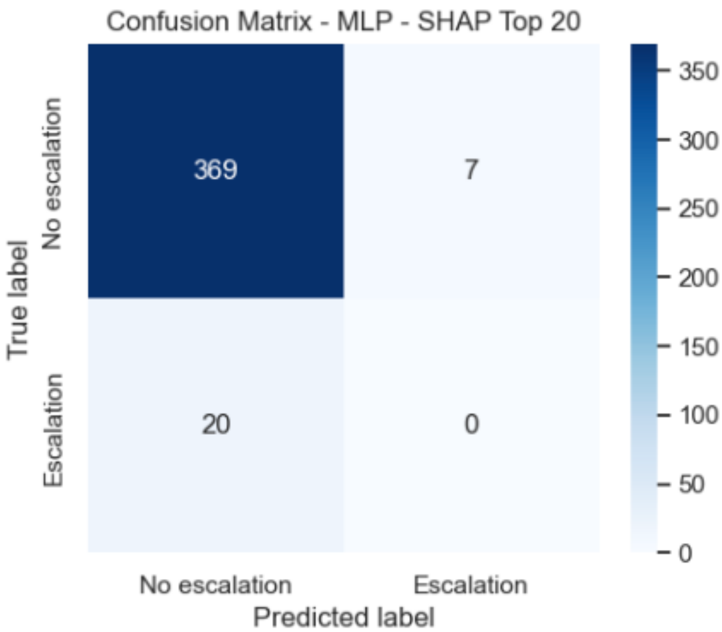


Figure 48. Confusion matrix obtained by the MLP SHAP Top-20 model.

3.7.4. COMPARATIVE ANALYSIS

Table 9 and Figure 49 compare the three MLP variants.

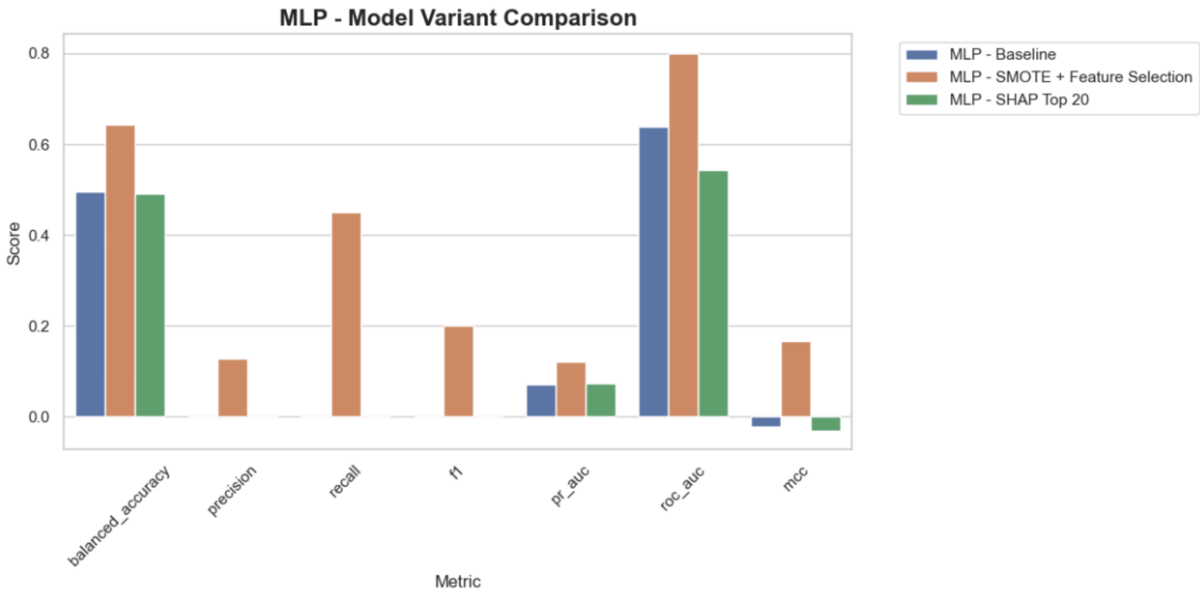


Figure 49. MLP Model Variant Comparison.

Several important findings emerge.

First, both the baseline and SHAP Top-20 models completely failed to detect escalation events. This indicates that feature reduction alone was insufficient to address the severe class imbalance and class overlap present in the dataset.

Second, the combination of SMOTE and feature selection dramatically improved performance:

Table 9. Performance comparison of the three MLP variants.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**MLP – baseline**	0.94	0.49	0.00	0.00	0.00	0.64	0.07	-0.02
1	**MLP – SMOTE + FS**	0.82	0.64	0.13	0.45	0.20	0.80	0.12	0.17
2	**MLP - SHAP**	0.93	0.49	0.00	0.00	0.00	0.54	0.07	-0.03

The SMOTE-based variant clearly dominates the other two configurations across all relevant metrics.

Third, the large increase in ROC-AUC from 0.638 to 0.800 indicates that balancing the minority class enabled the network to learn substantially more discriminative internal representations.

3.7.5. COMPARISON WITH PREVIOUS MODELS

When compared with the other model families, the MLP produced competitive results.

Compared with Logistic Regression:

- Similar recall (0.45 vs 0.50).
- Higher precision.
- Higher ROC-AUC.
- Higher MCC.

Compared with SVM:

- Identical recall (0.45).
- Higher precision.
- Considerably higher ROC-AUC (0.800 vs 0.637).
- Much stronger MCC.

Compared with CatBoost:

- Higher recall (0.45 vs 0.30).
- Higher MCC.
- Similar PR-AUC.

Compared with XGBoost:

- Slightly higher recall (0.45 vs 0.40).
- Higher ROC-AUC.
- Similar precision.
- Slightly higher MCC.

These results place the MLP among the strongest performing models in terms of escalation detection capability.

While XGBoost offered a more balanced trade-off overall, the MLP demonstrated that neural networks can successfully learn relevant escalation patterns when supported by appropriate balancing and feature-selection strategies.

3.7.6. INTERPRETATION AND LIMITATIONS

The results highlight several important characteristics of neural-network-based approaches.

The baseline and SHAP-reduced models demonstrate that MLPs remain highly sensitive to severe class imbalance. Without appropriate balancing mechanisms, the network tends to optimise overall accuracy by ignoring the minority class entirely.

However, once oversampling and feature selection are introduced, the network becomes capable of identifying nearly half of all escalation events.

The main limitation remains the relatively high false-positive rate, reflected by the 61 false escalation predictions. Although this behaviour improves sensitivity, it may reduce practical usability if excessive false alarms are generated.

Future work could explore:

- Threshold optimization.

- Cost-sensitive loss functions.
- Focal loss.
- Ensemble neural architectures.
- Temporal deep-learning models such as LSTMs or Transformers.
- Calibration techniques to improve probability estimates.

3.8. 1D-CONVOLUTIONAL NEURAL NETWORK

In addition to the classical machine learning models trained on engineered statistical features, a one-dimensional Convolutional Neural Network (1D-CNN) was evaluated using the raw temporal structure of the forecasting windows.

Unlike the previous models, which received one vector of aggregated features per window, the 1D-CNN received each window as a multivariate temporal sequence. Therefore, the network was able to learn directly from the temporal evolution of the physiological signals across the three-second observation interval.

The resulting input tensor had the following shape:

$$X_{sequences} = (1980{,}180{,}44)$$

where 1,980 represents the total number of windows, 180 corresponds to the number of temporal samples per window, and 44 represents the number of original signal features.

The positive ratio remained 5.56%, confirming that the sequential dataset preserved the same class imbalance as the engineered-feature dataset.

3.8.1. TRAINING, VALIDATION AND TEST STRUCTURE

The same participant-independent train-test split used for the classical machine learning models was applied to the CNN dataset.

The training set contained:

- 1,584 sequences.
- 180 time steps per sequence.
- 44 signal features.
- 90 positive escalation windows.

The test set contained:

- 396 sequences.
- 180 time steps per sequence.
- 44 signal features.
- 20 positive escalation windows.

Before training, the temporal sequences were imputed and standardized. The sequence tensors were temporarily reshaped into two-dimensional matrices so that median imputation and standard scaling could be applied consistently across all time points and features. After preprocessing, the data were reshaped back to their original three-dimensional sequence format.

A validation split was then created from the training data using StratifiedGroupKFold, again preserving participant-level separation. The CNN training subset contained 1,188 sequences, while the validation subset contained 396 sequences. The training subset included 70 positive windows and the validation subset included 20 positive windows.

This participant-aware validation strategy was particularly important for the CNN because temporal neural networks can easily learn participant-specific movement patterns if windows from the same participant appear in both training and validation data.

3.8.2. HYPERPARAMETER SEARCH

Three CNN configurations were evaluated, shown in Table 10.

The best validation performance was obtained with the following architecture:

- 32 convolutional filters in the first Conv1D layer.
- Kernel size of 7.
- 64 dense units.
- Dropout rate of 0.4.
- Learning rate of 0.0005.

This configuration achieved the highest validation PR-AUC among the tested architectures.

The selection of a larger kernel size suggests that the network benefited from analysing slightly longer temporal patterns within each window. Since each window was sampled at 60 Hz, a kernel size of 7 corresponds to a short but temporally meaningful segment of the physiological signal. This may allow the model to detect brief temporal motifs related to head movement, gaze instability, or eye-tracking fluctuations preceding cybersickness escalation.

Class weights were also applied during training to compensate for class imbalance. The minority escalation class received a substantially higher weight than the non-escalation class. This encouraged the model to penalize missed escalation events more strongly during optimization.

Table 10. 1D-CNN hyperparameter search results.

	filters	kernel_size	dense_units	dropout	learning_rate	val_pr_auc
0	16	3	32	0.3	0.0010	0.048750
1	32	5	32	0.3	0.0010	0.054060
2	32	7	64	0.4	0.0005	0.070258

3.8.3. TEST PERFORMANCE

The final CNN model was evaluated on the independent participant-held-out test set.

The confusion matrix (Figure 50) shows:

- True positives: 13
- False negatives: 7
- False positives: 99
- True negatives: 277

This corresponds to the strongest escalation detection capability observed in the study.

Performance metrics (Table 11) were:

- Accuracy = 0.732
- Balanced Accuracy = 0.693
- Precision = 0.116
- Recall = 0.650
- F1-score = 0.197
- ROC-AUC = 0.778
- PR-AUC = 0.150
- MCC = 0.188

The model correctly identified 65% of the escalation events, outperforming all classical machine learning approaches in terms of recall.

This result is highly relevant because the objective of the system is not merely to classify current states, but to anticipate future worsening of cybersickness. In such an early-warning context, high recall is particularly valuable because it reduces the probability of missing impending discomfort escalation.

However, this improvement came at the cost of a considerable number of false positives. The model incorrectly classified 99 non-escalation windows as escalation windows. This indicates that the CNN adopted a more sensitive detection strategy, prioritizing the identification of risky windows even when this increased the number of false alarms.

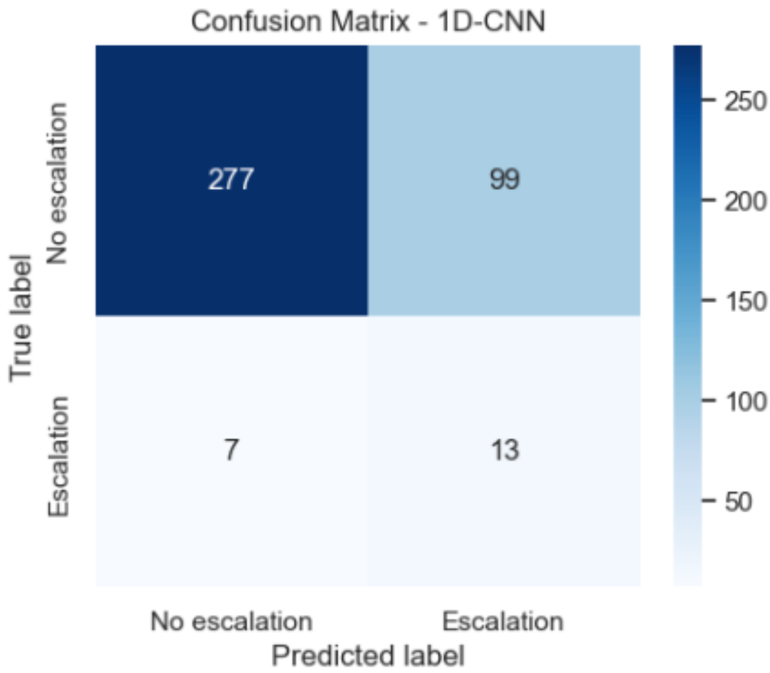


Figure 50. Confusion matrix obtained by the 1D-CNN model.

Table 11. 1D-CNN test performance metrics.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**1D-CNN**	0.73	0.69	0.12	0.65	0.20	0.78	0.15	0.19

3.8.4. NEUROTECHNOLOGICAL INTERPRETATION

The superior recall obtained by the 1D-CNN suggests that relevant information may be contained not only in summary statistics but also in the temporal shape of the physiological signals.

Classical models relied on window-level descriptors such as mean, standard deviation, range and slope. Although these features are informative, they compress the original temporal signal into a fixed set of aggregate values. This compression may remove short-lived fluctuations or transient patterns occurring before escalation events.

In contrast, the 1D-CNN processes the full temporal sequence and can learn local patterns directly from the raw signal trajectories. These patterns may include transient changes in head orientation, gaze direction, eye openness, convergence distance or pupil dynamics.

From a neurotechnology perspective, this is particularly meaningful. Cybersickness is not expected to emerge as a static physiological state, but rather as a dynamic process involving continuous adaptation of visuovestibular, proprioceptive and oculomotor systems. Therefore, temporal variations in head and eye behaviour may carry important information about the progression toward discomfort.

The CNN results support the hypothesis that pre-escalation signatures are partly temporal in nature. In other words, the way physiological signals evolve during the seconds before escalation appears to be more informative than their average value alone.

3.8.5. COMPARISON WITH CLASSICAL MACHINE LEARNING MODELS

Compared with the best classical models, the 1D-CNN achieved the highest recall and balanced accuracy.

The recall of 0.65 indicates that the CNN detected 13 out of 20 escalation events, whereas the best classical models detected fewer positive cases.

The MCC of 0.188 was also among the strongest values obtained, suggesting that the model predictions were more aligned with the true labels than most alternatives.

Nevertheless, precision remained low because of the high number of false positives. This limitation is shared by most models evaluated in the study and reflects the intrinsic difficulty of forecasting rare escalation events.

Compared with XGBoost and MLP, the CNN showed stronger sensitivity but a less conservative decision behaviour. This may be advantageous in early-warning scenarios where missing an escalation is more harmful than issuing occasional false alerts, but it would require careful threshold calibration before deployment in a real-time VR system.

3.8.6. LIMITATIONS

Despite its promising results, the CNN approach presents several limitations.

First, the dataset size was relatively small for training deep learning models. Although the number of windows was high, many windows were derived from the same participants and therefore were not fully independent.

Second, the number of positive escalation sequences remained limited. Only 90 positive examples were available in the training set and 20 in the test set. This restricts the model's ability to learn diverse escalation patterns.

Third, the false-positive rate was high. In a real-time system, excessive false warnings could interrupt immersion unnecessarily or reduce user trust.

Fourth, the CNN model is less directly interpretable than classical models. While its predictive behaviour suggests that temporal patterns are important, additional explainability techniques would be required to determine which signals and time regions contributed most strongly to each prediction.

These limitations motivate the subsequent XAI analysis, where feature importance and physiological interpretability are investigated in more detail.

3.9. LONG SHORT-TERM MEMORY NETWORK (LSTM)

In addition to the one-dimensional Convolutional Neural Network, a Long Short-Term Memory (LSTM) network was evaluated as an alternative deep learning architecture for cybersickness escalation forecasting.

Unlike convolutional networks, which learn local temporal patterns through sliding filters, LSTM networks are specifically designed to model sequential dependencies through recurrent memory mechanisms. This allows the network to retain information from previous time steps and potentially capture longer temporal relationships within physiological and behavioral signals.

As with the 1D-CNN, the LSTM received each forecasting window as a multivariate temporal sequence rather than as a vector of engineered features. Therefore, the model operated directly on the temporal evolution of the original VR signals without requiring explicit statistical feature extraction.

The same participant-independent train-test split, preprocessing strategy, and validation procedure used for the 1D-CNN were maintained to ensure a fair comparison between deep learning approaches.

3.9.1. HYPERPARAMETER SEARCH

Three LSTM configurations were evaluated, shown in Table 12.

The best validation performance was obtained with the following architecture:

- 32 LSTM units.
- 32 dense units.
- Dropout rate of 0.3.
- Learning rate of 0.001.

This configuration achieved the highest validation PR-AUC among the tested recurrent architectures.

The selected architecture corresponds to the simplest of the evaluated configurations. This suggests that increasing model complexity did not improve validation performance, possibly because the forecasting dataset contains a relatively limited number of escalation events. Under such conditions, larger recurrent architectures may be more susceptible to overfitting and may struggle to generalize to unseen participants.

Class weights were applied during training to compensate for the severe class imbalance present in the forecasting dataset. As in the CNN experiments, escalation windows received a substantially higher weight than non-escalation windows, encouraging the model to prioritize the detection of rare escalation events.

Table 12. LSTM hyperparameter search results.

	lstm_units	dense_units	dropout	learning_rate	val_pr_auc	epochs_trained
0	32	32	0.3	0.0010	0.114401	33
1	64	32	0.3	0.0010	0.043926	20
2	64	64	0.4	0.0005	0.033765	20

3.9.2. TEST PERFORMANCE

The final LSTM model was evaluated on the independent participant-held-out test set.

The confusion matrix (Figure 51) shows:

- True positives: 9
- False negatives: 11
- False positives: 49
- True negatives: 327

Performance metrics (Table 13) were:

- Accuracy = 0.848
- Balanced Accuracy = 0.660
- Precision = 0.155
- Recall = 0.450
- F1-score = 0.231
- ROC-AUC = 0.760
- PR-AUC = 0.167
- MCC = 0.198

The model correctly identified 45% of the escalation events while generating substantially fewer false positives than the 1D-CNN. This resulted in the highest precision, F1-score, PR-AUC, and MCC among all evaluated models.

Although the recall was lower than that obtained by the CNN, the recurrent model produced a more conservative decision strategy. Rather than maximizing sensitivity, the LSTM generated fewer escalation predictions overall, which reduced the number of false alarms while maintaining a competitive ability to detect true escalation events.

This behavior may be advantageous in practical deployments where excessive warnings could negatively affect user experience or reduce confidence in the forecasting system.

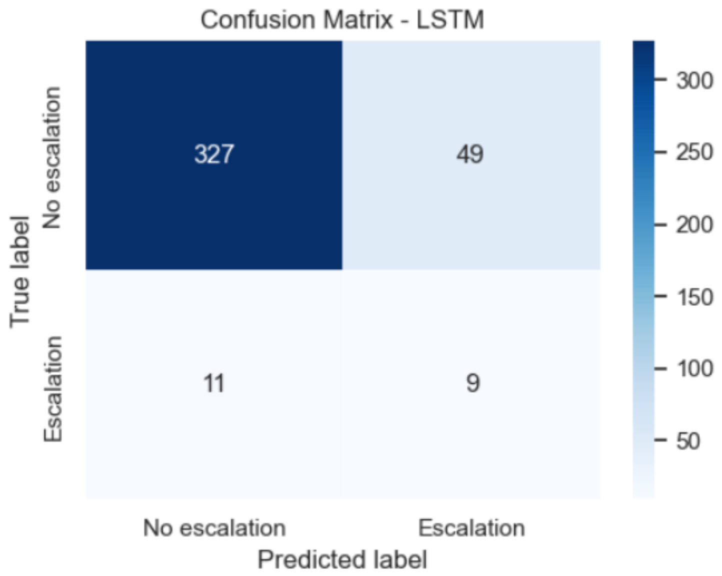


Figure 51. Confusion matrix obtained by the LSTM model.

Table 13. LSTM test performance metrics.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**LSTM**	0.85	0.66	0.16	0.45	0.23	0.76	0.17	0.20

3.9.3. NEUROTECHNOLOGICAL INTERPRETATION

The performance of the LSTM suggests that temporal dependencies within eye-tracking and head-tracking signals contain useful information for anticipating cybersickness escalation. Unlike classical machine learning models that rely on aggregated statistics, the recurrent architecture can directly process the temporal evolution of physiological variables and maintain internal representations of recent signal history.

From a neurotechnological perspective, this capability is particularly relevant because cybersickness is believed to emerge progressively through the accumulation of sensory conflict between visual, vestibular, and proprioceptive systems. Consequently, temporal relationships extending across multiple seconds may carry predictive information that cannot be fully captured through static summary descriptors.

The results indicate that recurrent modeling can successfully identify a subset of escalation-related temporal patterns while maintaining relatively stable prediction behavior. The reduced number of false positives compared with the CNN suggests that the LSTM learned more conservative representations of escalation risk, potentially focusing on more robust temporal signatures before issuing a positive prediction.

3.9.4. COMPARISON WITH THE 1D-CNN

Although both deep learning approaches outperformed most classical machine learning models in several metrics, their predictive behavior differed substantially.

The CNN achieved the highest recall (0.65), detecting 13 of the 20 escalation events present in the test set. In contrast, the LSTM achieved a recall of 0.45, identifying 9 escalation events.

However, the LSTM produced considerably fewer false positives (49 versus 99), resulting in higher precision (0.16 versus 0.12), higher F1-score (0.23 versus 0.20), higher PR-AUC (0.17 versus 0.15), and the highest MCC obtained in the study (0.20).

These results reveal a trade-off between sensitivity and reliability. The CNN adopted a more aggressive detection strategy that maximized escalation identification, whereas the LSTM generated more conservative predictions and achieved better overall balance between positive and negative classifications.

For an early-warning system, the preferred model depends on the operational objective. If minimizing missed escalation events is the primary concern, the CNN may be preferable. Conversely, if reducing false alarms and improving overall prediction reliability is considered more important, the LSTM provides a more balanced solution.

3.9.5. LIMITATIONS

Several limitations should be considered when interpreting the LSTM results.

First, the dataset remains relatively small for recurrent deep learning models. Although nearly two thousand temporal windows were available, many windows originated from the same participants and therefore cannot be considered fully independent observations.

Second, the number of positive escalation windows remained limited. Only 90 positive windows were available during training and 20 during testing, restricting the diversity of escalation patterns that the network could learn.

Third, only a small number of LSTM configurations were evaluated. A more extensive hyperparameter optimization procedure could potentially improve performance, although this would significantly increase computational requirements.

Finally, the internal representations learned by recurrent neural networks are inherently difficult to interpret. While the model achieved promising forecasting performance, additional explainability techniques would be required to determine which temporal segments and physiological variables contributed most strongly to the final predictions.

3.10. GLOBAL MODEL COMPARISON AND FINAL MODEL SELECTION

After evaluating all classical machine learning algorithms and the temporal deep learning approach, a final comparison was performed using the best-performing variant of each model family.

The selected models were:

- Logistic Regression + SMOTE + Feature Selection

- XGBoost Baseline
- MLP + SMOTE + Feature Selection
- 1D-CNN
- LSTM

The comparison focused on metrics that are particularly relevant for rare-event forecasting:

- Balanced Accuracy
- Precision
- Recall
- F1-score
- PR-AUC
- ROC-AUC
- Matthews Correlation Coefficient (MCC)

3.10.1. COMPARATIVE PERFORMANCE

The comparison, as seen in Table 14 and Figure 52 reveals substantial differences in the behavior of the models.

Table 14. Best 5 test performance metrics.

	model	accuracy	bal_acc	precision	recall	F1	roc_auc	pr_auc	mcc
0	**LR – SMOTE + FS**	0.64	0.57	0.07	0.50	0.12	0.69	0.08	0.07
1	**XGB – Baseline**	0.81	0.62	0.11	0.40	0.18	0.70	0.14	0.13
2	**MLP – SMOTE + FS**	0.82	0.64	0.13	0.45	0.20	0.80	0.12	0.17
3	**1D-CNN**	0.73	0.69	0.12	0.65	0.20	0.78	0.15	0.19
4	**LSTM**	0.85	0.66	0.16	0.45	0.23	0.76	0.17	0.20

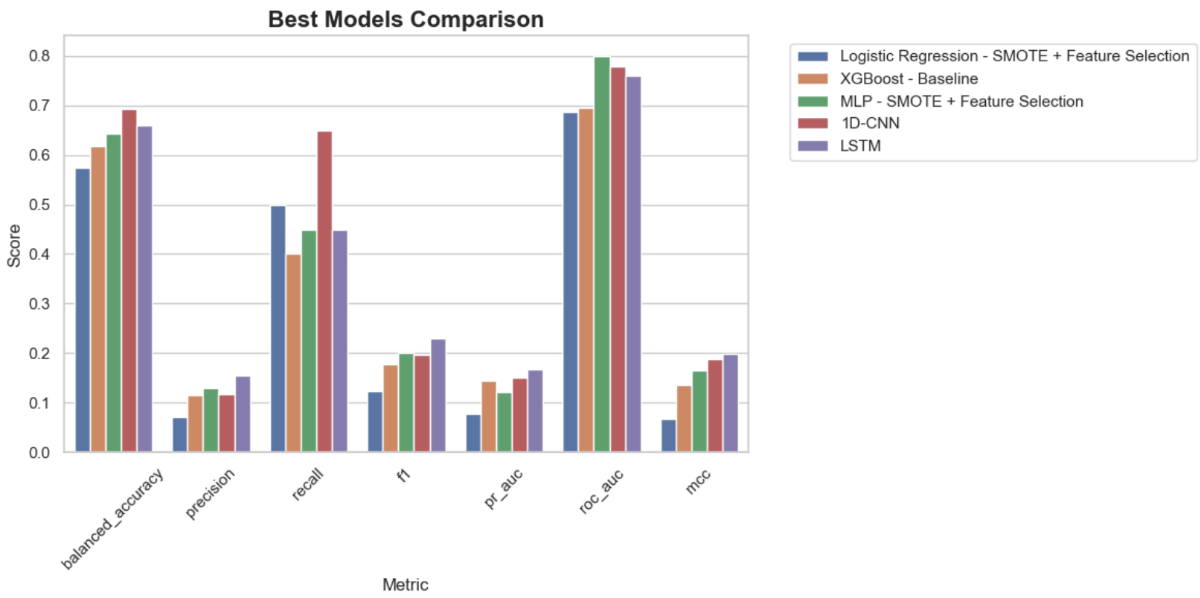


Figure 52. Global Model Variant Comparison.

LOGISTIC REGRESSION

The Logistic Regression model achieved:

- Balanced Accuracy = 0.574
- Recall = 0.50
- PR-AUC = 0.077
- MCC = 0.068

Although its discrimination ability was moderate, the model demonstrated relatively high sensitivity to escalation events. Its main advantage is interpretability, but overall predictive performance remained limited compared with more advanced approaches.

XGBOOST

The XGBoost baseline model achieved:

- Balanced Accuracy = 0.618
- Precision = 0.114
- Recall = 0.40
- PR-AUC = 0.144
- MCC = 0.135

Among the classical machine learning models, XGBoost provided one of the strongest trade-offs between sensitivity and specificity. It achieved the second-highest PR-AUC and produced relatively few false alarms compared with neural approaches.

MLP

The MLP combined with SMOTE and feature selection obtained:

- Balanced Accuracy = 0.644
- Precision = 0.129
- Recall = 0.45
- F1-score = 0.20
- ROC-AUC = 0.800
- PR-AUC = 0.121
- MCC = 0.165

The MLP achieved the highest ROC-AUC among all evaluated models, indicating strong ranking ability between escalation and non-escalation windows. However, its PR-AUC remained below that of XGBoost and the CNN, suggesting that its probability estimates were less reliable under severe class imbalance.

1D-CNN

The temporal CNN achieved:

- Balanced Accuracy = 0.693
- Precision = 0.116
- Recall = 0.65
- F1-score = 0.197
- ROC-AUC = 0.778
- PR-AUC = 0.150
- MCC = 0.188

The CNN produced the highest balanced accuracy and recall among all evaluated models.

Most importantly, it correctly detected:

$13/20 = 65\%$ of escalation events in the independent test set.

LSTM

The LSTM model achieved:

- Balanced Accuracy = 0.660
- Precision = 0.155
- Recall = 0.45
- F1-score = 0.231
- ROC-AUC = 0.760
- PR-AUC = 0.167
- MCC = 0.200

The LSTM provided the most balanced overall performance among the evaluated models. Although its recall was lower than that of the CNN, it achieved the highest F1-score, PR-AUC, and MCC.

Most importantly, it correctly detected:

$9/20 = 45\%$ of escalation events while generating substantially fewer false positives than the CNN.

This result suggests that the recurrent architecture adopted a more conservative prediction strategy, sacrificing some sensitivity in exchange for greater reliability and overall classification quality. The superior PR-AUC and MCC indicate that the LSTM achieved the strongest balance between escalation detection and false-alarm control under severe class imbalance.

The CNN achieved the highest balanced accuracy and recall, whereas the LSTM obtained the highest F1-score, PR-AUC, and MCC. Together, these results indicate a trade-off between maximum escalation detection capability and overall classification reliability.

All results are shown in Table 15.

Table 15. All-models test performance metrics ordered by pr_auc.

model	**accuracy**	**bal_acc**	**precision**	**recall**	**F1**	**roc_auc**	**pr_auc**	**mcc**
LSTM	0.85	0.66	0.16	0.45	0.23	0.76	0.17	0.20
1D-CNN	0.73	0.69	0.12	0.65	0.20	0.78	0.15	0.19
CB - SHAP	0.79	0.56	0.08	0.3	0.13	0.66	0.15	0.06
XGB – Baseline	0.81	0.62	0.11	0.40	0.18	0.70	0.14	0.13
MLP – SMOTE + FS	0.82	0.64	0.13	0.45	0.20	0.80	0.12	0.17
XGB - SHAP	0.92	0.58	0.20	0.20	0.20	0.67	0.10	0.16
CB – SMOTE	0.91	0.48	0.00	0.00	0.00	0.72	0.11	-0.05
LR - baseline	0.95	0.50	0.00	0.00	0.00	0.70	0.09	0.00
XGB – SMOTE	0.84	0.54	0.08	0.20	0.11	0.64	0.09	0.05
CB – baseline	0.95	0.50	0.00	0.00	0.00	0.66	0.08	0.00
LR – SMOTE + FS	0.64	0.57	0.07	0.50	0.12	0.69	0.08	0.07
RF – SMOTE	0.93	0.49	0.00	0.00	0.00	0.64	0.08	-0.03

SVM – SMOTE + FS	0.63	0.55	0.06	0.45	0.11	0.64	0.08	0.04
MLP - SHAP	0.93	0.49	0.00	0.00	0.00	0.54	0.07	-0.03
MLP – baseline	0.94	0.49	0.00	0.00	0.00	0.64	0.07	-0.02
RF - baseline	0.95	0.50	0.00	0.00	0.00	0.64	0.07	0.00
RF - SHAP	0.88	0.46	0.00	0.00	0.00	0.63	0.07	-0.06
HGB – baseline	0.84	0.44	0.00	0.00	0.00	0.58	0.07	-0.08
SVM – baseline	0.95	0.50	0.00	0.00	0.00	0.53	0.06	0.00
HGB - SHAP	0.89	0.47	0.00	0.00	0.00	0.59	0.06	-0.06
LR - SHAP	0.95	0.50	0.00	0.00	0.00	0.51	0.06	0.00
HGB – SMOTE	0.85	0.45	0.00	0.00	0.00	0.52	0.06	-0.08
SVM - SHAP	0.58	0.42	0.03	0.25	0.06	0.47	0.05	-0.07

3.10.2. ROC CURVE ANALYSIS

The ROC curves, shown in Figure 53, further highlight the differences between the models.

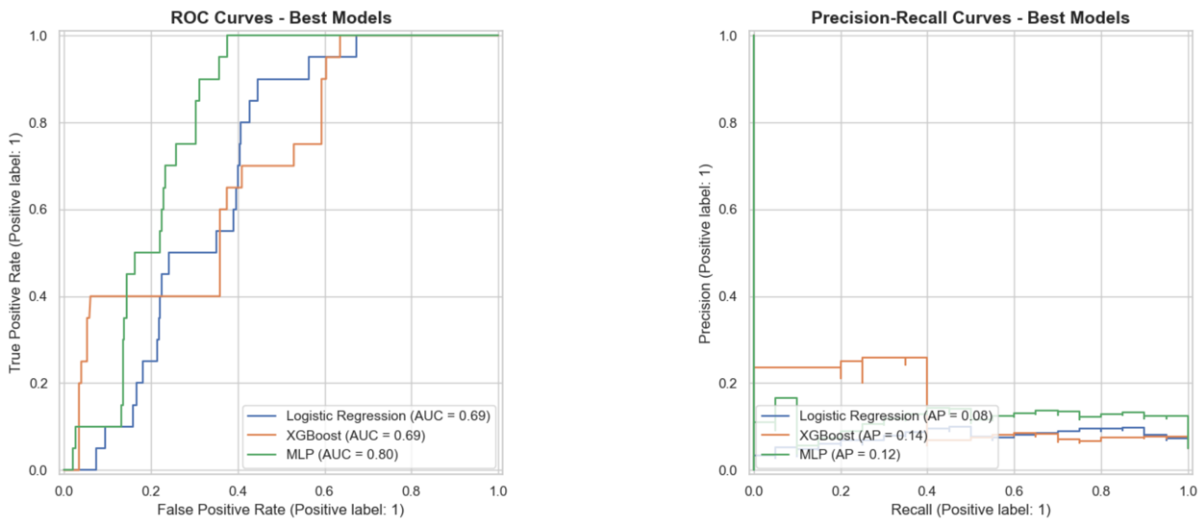


Figure 53. ROC and Prec/Rec curves of best models.

The MLP achieved the highest ROC-AUC:

ROC-AUC = 0.800,

followed by the 1D-CNN:

ROC-AUC = 0.778,

and the LSTM:

ROC-AUC = 0.760.

XGBoost and Logistic Regression obtained lower values, with ROC-AUC scores of approximately 0.69. These results indicate that the neural-network-based approaches learned more informative decision boundaries than the classical linear and tree-based methods.

However, ROC-AUC can be misleading in highly imbalanced datasets because it assigns equal importance to both classes regardless of their prevalence. For this reason, Precision–Recall analysis was considered more informative for model comparison.

3.10.3. PRECISION–RECALL CURVE ANALYSIS

The Precision–Recall curves reveal a different ranking. The highest PR-AUC was achieved by the LSTM:

PR-AUC = 0.167,

followed by the 1D-CNN:

PR-AUC = 0.150,

and XGBoost:

PR-AUC = 0.144.

The MLP obtained a PR-AUC of 0.121, despite achieving the highest ROC-AUC.

Because only 5.56% of the forecasting windows corresponded to escalation events, improvements in PR-AUC directly reflect a better ability to identify the minority escalation class. Consequently, PR-AUC was considered one of the most relevant discrimination metrics for the final comparison of forecasting performance.

The results indicate that although the MLP achieved the strongest global ranking capability, the LSTM and 1D-CNN provided superior performance when focusing specifically on the detection of rare escalation events. This behaviour is particularly relevant for early-warning systems, where identifying impending cybersickness escalation is more important than maximizing overall classification accuracy.

3.10.4. MODEL RANKING

To obtain an overall ranking, a composite score was calculated using:

$Score = 0.4 \cdot PR\text{-}AUC + 0.3 \cdot BalancedAccuracy + 0.3 \cdot MCC$. The weighting scheme emphasizes:

- Minority-class detection (PR-AUC)
- Balanced performance (Balanced Accuracy)
- Overall prediction quality (MCC)

The resulting ranking (Table 16) was:

Table 16. Ranking of the best models.

Rank	model	bal_acc	F1	pr_auc	roc_auc	mcc	rank_score
1	1D-CNN	0.69	0.20	0.15	0.78	0.19	0.32
2	LSTM	0.66	0.23	0.17	0.76	0.20	0.32
3	MLP – SMOTE + FS	0.64	0.20	0.12	0.80	0.17	0.29
4	XGB – Baseline	0.62	0.18	0.14	0.69	0.13	0.28
5	LR – SMOTE + FS	0.57	0.12	0.08	0.69	0.07	0.22
6	CB – Baseline	0.50	0.00	0.08	0.66	0.00	0.18
7	RF – Baseline	0.50	0.00	0.07	0.64	0.00	0.18
8	SVM – Baseline	0.50	0.00	0.06	0.53	0.00	0.18
9	HGB - Baseline	0.44	0.00	0.07	0.58	-0.08	0.14

3.10.5. FINAL MODEL SELECTION

The final model selection was based on the overall objective of the study: forecasting cybersickness escalation events as early as possible.

Although the LSTM achieved the highest F1-score (0.231), PR-AUC (0.167), and MCC (0.200), the 1D-CNN was selected as the final model because it provided the strongest escalation detection capability, which was considered the most important requirement for an early-warning system.

Several observations support this decision:

1. It achieved the highest recall (0.65), correctly detecting 13 of the 20 escalation events present in the independent test set.
2. It achieved the highest balanced accuracy (0.693).

3. It directly learned temporal patterns from the physiological signals without relying on handcrafted statistical descriptors.
4. It demonstrated the strongest sensitivity to impending cybersickness escalation, which is particularly valuable in preventive intervention scenarios.
5. Its performance remained competitive across all other evaluation metrics, including ROC-AUC, PR-AUC, and MCC.

Unlike the classical machine learning approaches, which relied on engineered statistical summaries, both the CNN and the LSTM operated directly on the temporal evolution of head motion, gaze behaviour, eye movements, and pupil dynamics. However, the CNN proved more effective at identifying escalation events before they occurred, whereas the LSTM adopted a more conservative prediction strategy that reduced false positives but also missed a larger proportion of true escalation events.

This finding suggests that cybersickness escalation is characterized not only by static physiological states but also by dynamic temporal patterns that emerge during the seconds preceding discomfort progression.

MAIN FINDING OF THE STUDY

The overall results indicate that temporal deep learning provides a clear advantage for forecasting cybersickness escalation in immersive virtual reality environments.

While several classical machine learning algorithms were able to identify part of the escalation events, the deep learning architectures consistently achieved the strongest overall performance. The CNN provided the highest sensitivity and early-warning capability, whereas the LSTM achieved the strongest balance between escalation detection and overall classification quality.

These results support the hypothesis that short-term temporal dynamics of eye-tracking and head-movement signals contain predictive information that can be leveraged to anticipate cybersickness before users explicitly report discomfort.

From a practical perspective, the CNN represents the most suitable candidate when maximizing escalation detection is the primary objective, whereas the LSTM may be preferable in scenarios where reducing false alarms is equally important. Overall, the results demonstrate the value of temporal deep-learning approaches for the development of future adaptive VR systems capable of proactively mitigating cybersickness.

The CNN was selected as the final model because the objective was early warning, where missing escalation events is more critical than generating occasional false positives. However, the LSTM achieved the best overall balance between precision, PR-AUC, F1-score, and MCC.

3.11. EXPLAINABILITY AND NEUROTECHNOLOGICAL INTERPRETATION

3.11.1. FEATURE SELECTION CONSISTENCY ACROSS PREDICTIVE MODELS

Before applying model-specific explainability techniques, an initial analysis was conducted to investigate which neurophysiological variables were consistently identified as relevant by the two best-performing feature-selection pipelines: Logistic Regression with SMOTE and MLP with SMOTE.

Both models employed a feature-selection stage during training. The objective of this analysis was to determine whether different learning algorithms converged toward the same physiological indicators of cybersickness escalation.

The results revealed a remarkable level of agreement. Both models selected exactly the same set of twenty temporal features, resulting in a complete overlap between the variables retained by Logistic Regression and those retained by the MLP.

This finding is particularly relevant because the two algorithms rely on fundamentally different learning mechanisms. Logistic Regression assumes a linear relationship between predictors and outcome, whereas the MLP is capable of learning complex non-linear patterns. The convergence toward the same feature subset suggests that these variables contain robust predictive information that is independent of the specific modeling approach.

Consequently, the selected features can be considered strong candidate biomarkers of impending cybersickness escalation.

3.11.2. NEUROPHYSIOLOGICAL DOMAIN ANALYSIS

To facilitate interpretation, the selected variables were grouped according to the neurophysiological system they represent. Each feature was assigned to one of several signal domains, including gaze behavior, eye position, pupil dynamics, head motion, vergence-related measures, and locomotion indicators.

Figure 54 summarizes the distribution of selected features across these domains.

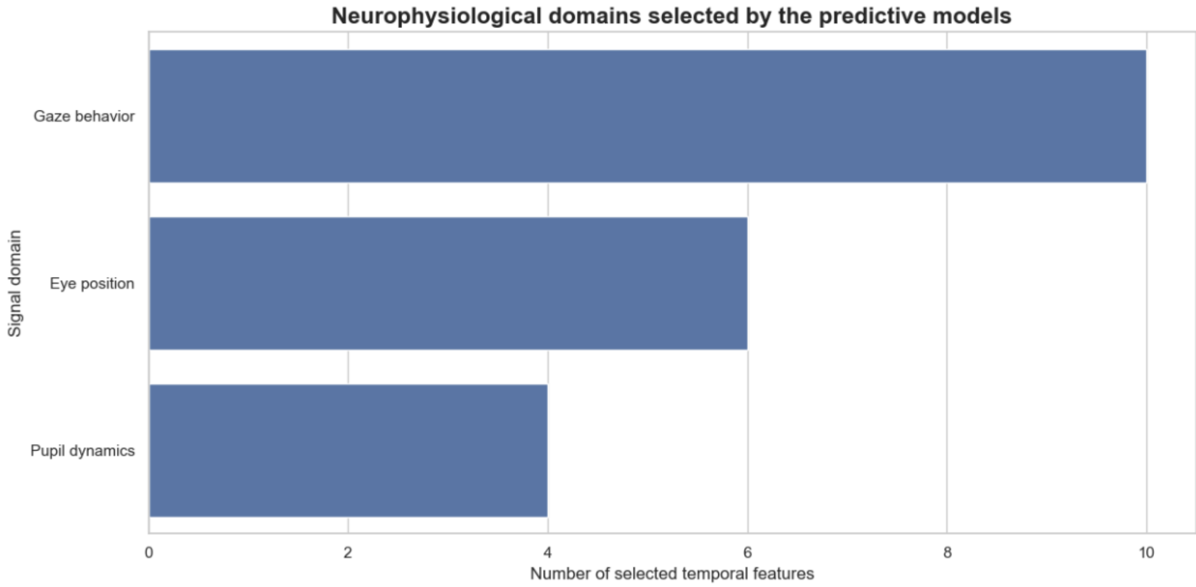


Figure 54. Neurophysiological domains selected by the predictive models

The analysis revealed that gaze-related variables dominated the selected feature set, accounting for ten of the twenty retained predictors. Eye-position descriptors contributed six variables, while pupil-dynamics measures contributed four variables.

Interestingly, no head-motion, locomotion, vergence, or eye-openness features survived the feature-selection process in the final models.

This result suggests that, within the temporal window used for prediction, ocular signals contain substantially more predictive information about future cybersickness escalation than gross motor behavior.

3.11.3. INTERPRETATION FROM A NEUROTECHNOLOGY PERSPECTIVE

GAZE BEHAVIOR FEATURES

The predominance of gaze-related features is consistent with current neurotechnology and cybersickness literature.

Many contemporary theories describe cybersickness as a consequence of sensory conflict between visual, vestibular, and proprioceptive systems. Before users consciously report discomfort, subtle abnormalities often emerge in visual exploration patterns, fixation stability, and gaze-control mechanisms.

Several of the selected variables describe characteristics such as:

- Gaze direction variability.
- Gaze-origin displacement.
- Eye-movement geometry.
- Temporal gaze drift.
- Spatial instability of visual exploration.

These measures may reflect compensatory visual strategies that users unconsciously employ when attempting to maintain perceptual stability in immersive virtual environments.

From a neurophysiological perspective, such behaviors may indicate the early emergence of sensory conflict before overt symptoms become apparent.

EYE POSITION FEATURES

The second most represented group corresponds to eye-position descriptors.

These variables characterize the spatial configuration of the eyes within the headset coordinate system and include measures related to ocular alignment and eye-movement trajectories.

Eye-position abnormalities have previously been associated with:

- Increased oculomotor effort.
- Visual fatigue.
- Difficulties maintaining binocular coordination.
- Adaptation processes during prolonged VR exposure.

The presence of six eye-position variables among the most predictive features suggests that cybersickness escalation may be preceded by measurable alterations in ocular control mechanisms.

This observation aligns with previous findings indicating that oculomotor instability frequently appears before the onset of severe cybersickness symptoms.

PUPIL DYNAMICS FEATURES

Four selected variables were related to pupil dynamics.

Pupil responses are strongly influenced by the autonomic nervous system and are frequently used as indirect indicators of:

- Cognitive workload.
- Physiological arousal.
- Stress responses.
- Mental effort.

In immersive virtual environments, pupil dilation has also been linked to discomfort, sensory overload, and increasing task demands.

The fact that pupil-based descriptors were consistently selected by both predictive models suggests that autonomic responses may emerge during the early stages of cybersickness escalation, potentially reflecting physiological adaptation processes occurring before subjective symptom reporting.

3.11.4. IMPLICATIONS FOR CYBERSICKNESS PREDICTION

One of the most important findings of this analysis is that all retained predictors originated from eye-tracking-derived measurements.

No physiological sensors, questionnaires, or manually engineered cybersickness indicators were required for the final predictive models.

This result supports the central hypothesis of the study: that eye-tracking signals contain sufficient information to anticipate cybersickness escalation before severe symptoms become observable.

From a practical perspective, these findings are highly relevant for future adaptive virtual reality systems. Since modern VR headsets increasingly incorporate integrated eye-tracking hardware, the identified biomarkers could potentially be monitored in real time without additional sensors, enabling proactive interventions aimed at preventing cybersickness before user performance or immersion are significantly affected.

3.11.5. TEMPORAL EVOLUTION OF PREDICTED CYBERSICKNESS RISK BEFORE ESCALATION

To investigate whether the predictive model could identify increasing cybersickness risk before symptom escalation occurred, the predicted probabilities generated by the Logistic Regression model were analyzed as a function of time relative to each escalation event.

For every escalation event observed in the dataset, all preceding temporal windows were aligned relative to the escalation onset. The predicted escalation probabilities were then averaged across participants to obtain a mean temporal risk trajectory.

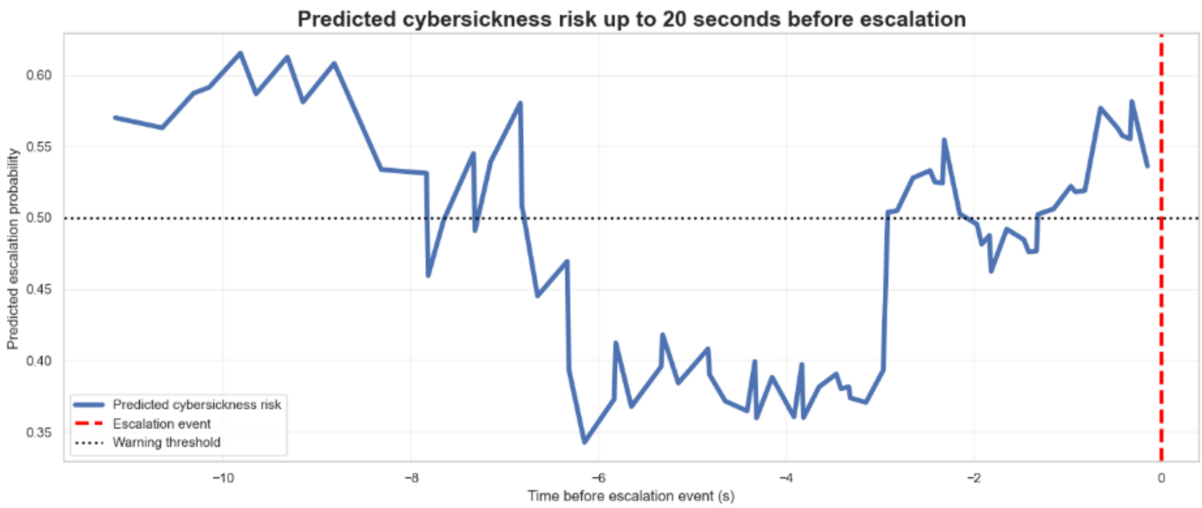


Figure 55. Predicted cybersickness risk up to 20 seconds before escalation.

The resulting trajectory presented in Figure 55 suggests that the model detects elevated cybersickness risk several seconds before escalation occurs. Although the probability evolution is not strictly monotonic, the average risk remains close to or above the decision threshold during the final seconds preceding escalation.

From a neurotechnological perspective, this finding is consistent with the hypothesis that subtle ocular biomarkers emerge before participants consciously report severe discomfort. Changes in gaze stability, eye-position dynamics and pupil-related measures may therefore provide early indicators of impending cybersickness episodes.

These results support the feasibility of real-time warning systems capable of triggering adaptive interventions before symptom escalation becomes severe.

3.11.6. MODEL GENERALIZATION ANALYSIS

To assess model robustness and generalization, the final Logistic Regression and MLP models were evaluated on the training set, a validation subset derived from the training data, and the independent test set. This comparison was used to identify potential overfitting and to verify that model performance remained stable across unseen participants, as presented in Figure 56.

Figure 56. Train / Validation / Test performance of final trained models.

The Logistic Regression model exhibited relatively stable performance across all data partitions. Although some degradation was observed when moving from training to testing data, the overall behavior remained consistent, suggesting limited overfitting.

In contrast, the MLP achieved substantially higher performance on the training partition, particularly in terms of recall and ROC-AUC. However, a noticeable reduction was observed on the independent test set, indicating a higher degree of overfitting.

Despite this reduction, the MLP maintained the highest overall predictive performance among the evaluated models, achieving the best balance between sensitivity, discriminative capacity and early-event detection.

These findings suggest that the neural-network approach captures more complex cybersickness-related patterns than linear models, although at the cost of reduced generalization stability.

3.11.7. IDENTIFICATION OF NEUROPHYSIOLOGICAL SIGNALS ASSOCIATED WITH CYBERSICKNESS ESCALATION

The temporal features jointly selected by the Logistic Regression and MLP models were mapped back to their original eye-tracking signals in order to identify the underlying neurophysiological variables associated with cybersickness escalation. This approach enabled the interpretation of the predictive models beyond their statistical performance and provided insight into the physiological mechanisms preceding symptom worsening.

A total of 20 predictive temporal features were consistently selected by both models, corresponding to approximately 15 unique raw eye-tracking signals. These signals were primarily associated with three neurophysiological domains: gaze behavior, eye position, and pupil dynamics.

The convergence of feature selection across two independent machine learning architectures suggests that these signals contain robust and stable information related to cybersickness progression. In particular, gaze-related variables represented the largest proportion of the selected features, indicating that oculomotor behavior may play a central role in the development of cybersickness symptoms.

These findings are consistent with previous research suggesting that alterations in visual exploration patterns, gaze stability, and eye movement control emerge before the subjective manifestation of severe discomfort in immersive virtual environments.

PRE-ESCALATION NEUROPHYSIOLOGICAL SIGNATURE

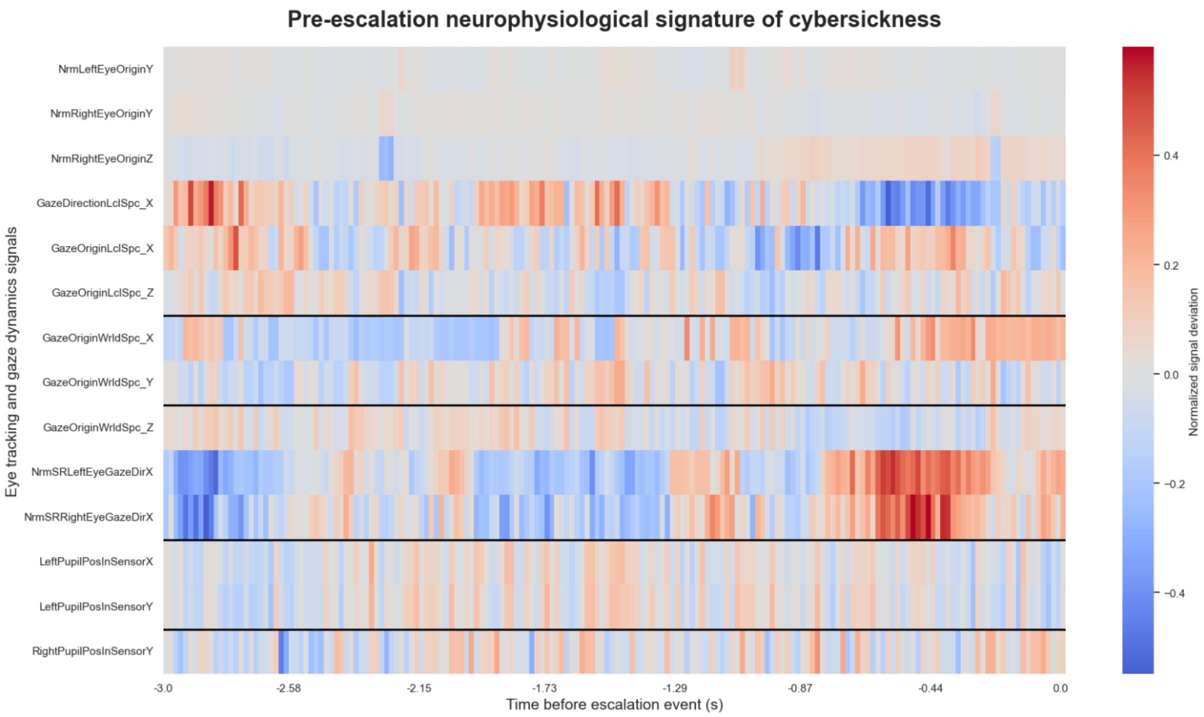


Figure 57. Pre-escalation neurophysiological signature of cybersickness.

To further investigate the physiological processes preceding symptom escalation, the temporal evolution of the most relevant eye-tracking signals was analyzed during the three seconds immediately before each escalation event. Signal values were normalized and averaged across participants to generate a unified neurophysiological representation of the pre-escalation period.

The heatmap (Figure 57) reveals that several gaze-related variables exhibited systematic deviations before escalation onset. In particular, horizontal gaze-direction signals showed noticeable temporal transitions during the final second preceding escalation, suggesting increasing visual instability and oculomotor adaptation.

The strongest patterns were observed in the normalized gaze-direction signals of both eyes, which displayed marked shifts shortly before the escalation event. From a neurotechnological perspective, these changes may reflect compensatory eye movements associated with increasing sensory conflict between visual and vestibular information. Such behavior is consistent with the Sensory Conflict Theory of motion sickness [3], which proposes that cybersickness emerges when incoming sensory cues become progressively incongruent.

Conversely, pupil-related signals exhibited higher variability and less consistent temporal patterns across participants. Although these variables were repeatedly selected by the predictive models, their temporal evolution suggests a more heterogeneous physiological response that may depend on individual differences in autonomic nervous system activation.

Overall, the observed results indicate that cybersickness escalation is preceded by measurable alterations in gaze-control mechanisms and oculomotor dynamics. These findings support the hypothesis that eye-tracking biomarkers can provide early objective indicators of symptom worsening and may therefore be useful for the development of adaptive real-time cybersickness monitoring systems.

3.11.8. PRE-EVENT NEUROPHYSIOLOGICAL INSTABILITY ANALYSIS

To complement the previous heatmap-based analysis, a neurophysiological instability index was computed from the selected eye-tracking signals. Instead of analyzing the direction of each signal change, this index quantified the magnitude of signal deviation during the seconds preceding cybersickness escalation.

For each selected signal, values were first normalized. Then, the absolute deviation from the normalized baseline was calculated. This transformation generated a measure of instability independently of whether the signal increased or decreased:

$Instability = |z|$, where z represents the normalized value of each signal.

The instability values were then averaged within each neurophysiological domain: gaze behavior, eye position, and pupil dynamics.

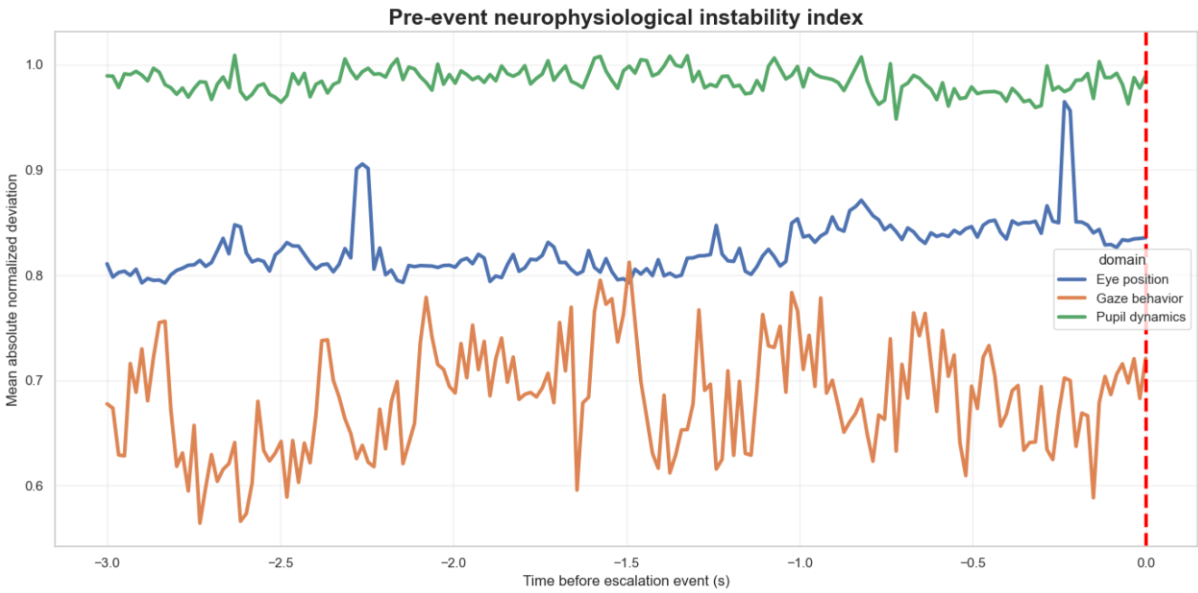


Figure 58. Pre-event neurophysiological instability index. The figure shows the mean absolute normalized deviation of the main eye-tracking domains during the three seconds preceding cybersickness escalation.

Figure 58 shows that pupil-related variables exhibited the highest overall instability throughout the pre-escalation interval. This suggests that pupil-position dynamics may be particularly sensitive to physiological variability before cybersickness worsening. However, this pattern should be interpreted cautiously, since pupil-related signals may also be affected by tracking noise, blinks, partial occlusions, and individual differences in eye-tracking quality.

Eye-position variables showed a more stable profile, with relatively small fluctuations across the pre-event period. This indicates that ocular alignment descriptors remained comparatively consistent before escalation.

Gaze-behavior variables presented an intermediate pattern, with visible fluctuations and a slight increase near the final part of the pre-escalation interval. From a neurotechnological perspective, this may reflect increasing instability in visual exploration and gaze-control mechanisms as the participant approaches symptom escalation.

Overall, the instability analysis supports the idea that cybersickness escalation is not preceded by a single abrupt physiological change, but by distributed alterations across several oculomotor domains.

3.11.9. PARTICIPANT-LEVEL PRE-EVENT OCULOMOTOR INSTABILITY

A participant-level instability heatmap was also generated to assess whether the pre-escalation instability pattern was homogeneous across subjects or strongly participant-dependent.

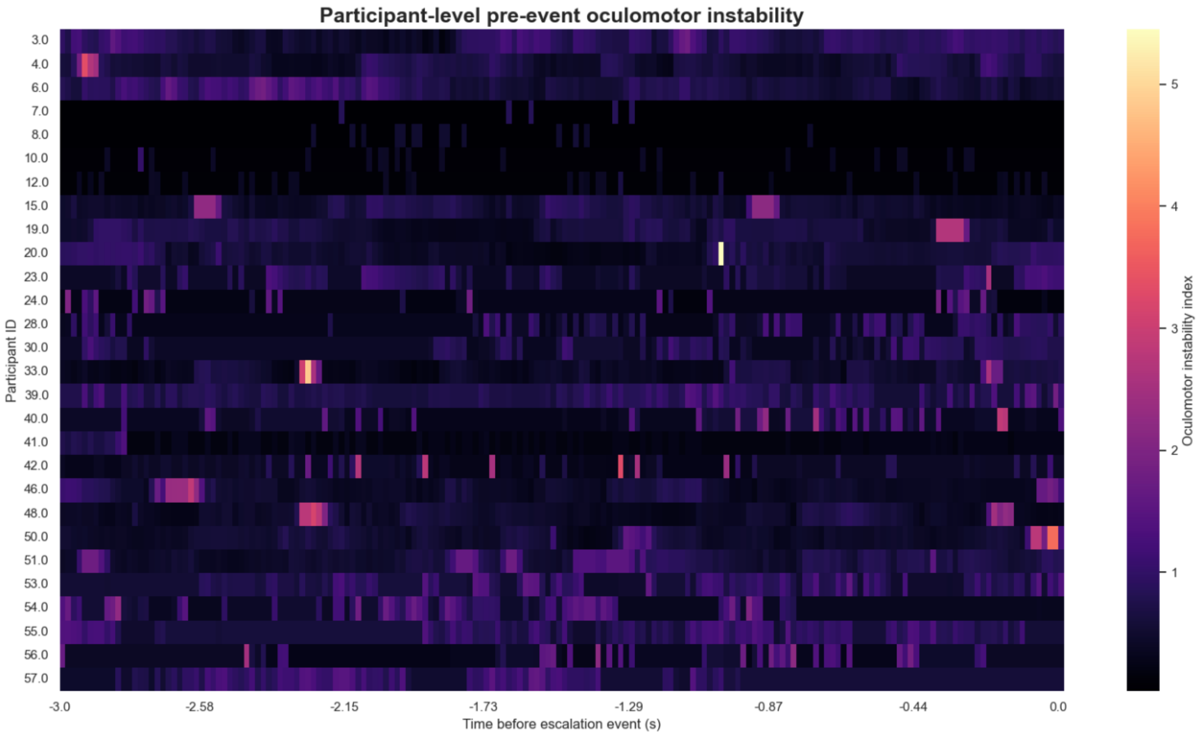


Figure 59. Participant-level pre-event oculomotor instability. Each row represents one participant with at least one valid escalation event, and each column represents time before escalation. Brighter colors indicate higher oculomotor instability.

The heatmap (Figure 59) reveals substantial inter-participant variability in the temporal expression of oculomotor instability. Some participants showed clear increases in instability during specific moments before escalation, whereas others exhibited relatively stable patterns.

This variability is highly relevant from a neurotechnology perspective. Cybersickness susceptibility is known to differ considerably between individuals, and the physiological mechanisms preceding symptom escalation may not follow a single universal trajectory. Instead, different participants may express discomfort through different combinations of gaze instability, pupil dynamics, and ocular-position changes.

Therefore, the participant-level heatmap reinforces the need for adaptive and personalized cybersickness prediction systems. A model intended for real-time VR applications should not only detect generic population-level patterns, but also account for individual differences in oculomotor response.

Together, the instability index and the participant-level heatmap suggest that cybersickness escalation is associated with measurable but heterogeneous oculomotor alterations before the subjective increase in FMS is reported.

3.11.10. NEUROPHYSIOLOGICAL STATE-SPACE TRANSITION BEFORE CYBERSICKNESS ESCALATION

To further investigate the temporal organization of the pre-escalation response, gaze-instability and pupil-instability indices were jointly represented in a two-dimensional neurophysiological state space. Each point represents the average physiological state at a given moment during the three seconds preceding escalation, while the color gradient indicates temporal proximity to the escalation event.

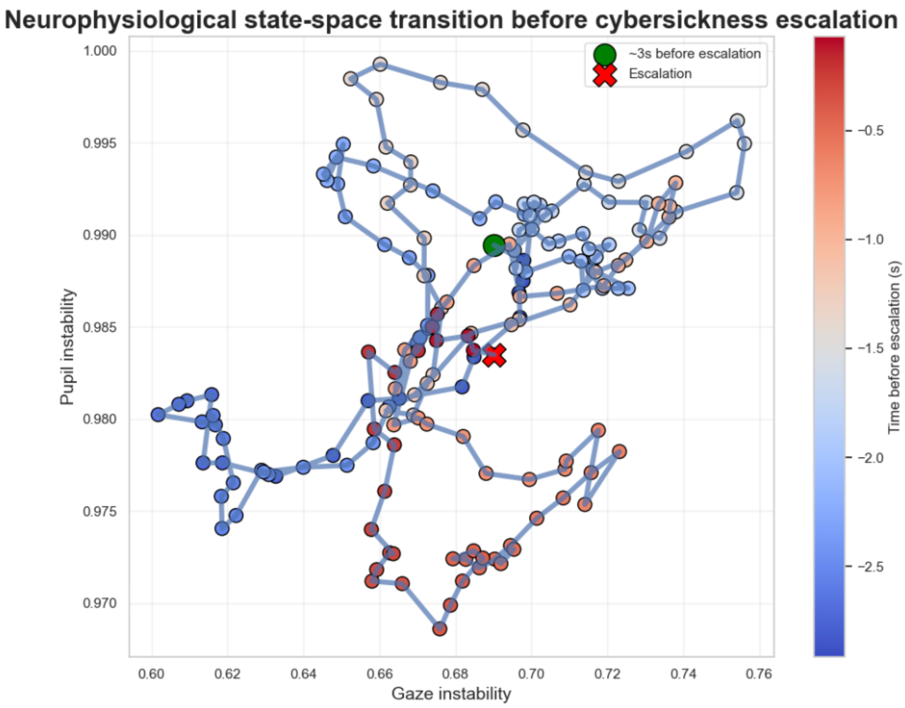


Figure 60. Neurophysiological state-space transition before cybersickness escalation.

The trajectory shown in Figure 60 reveals a gradual transition of the neurophysiological state during the pre-escalation period. Early time points, located approximately three seconds before escalation, occupy a relatively compact region of the state space, suggesting a more stable oculomotor condition.

As the escalation event approaches, the trajectory progressively moves through different regions of the state space and exhibits increased dispersion. This behavior suggests growing variability in the interaction between gaze-control mechanisms and pupil-related dynamics.

From a neurotechnology perspective, this pattern is consistent with the progressive accumulation of sensory conflict proposed by cybersickness theories. Rather than appearing abruptly, cybersickness escalation seems to emerge through a continuous transition involving multiple oculomotor and autonomic processes.

Importantly, the final pre-escalation states do not converge to a single point. Instead, several nearby configurations are observed before escalation, indicating that different physiological trajectories may lead to a similar subjective cybersickness outcome. This observation is consistent with the substantial inter-individual variability observed throughout the study.

3.11.11. INDIVIDUAL EARLY-WARNING CASE STUDY

To illustrate the practical behavior of the proposed cybersickness prediction framework, an individual participant presenting a clear escalation event was analyzed in detail.

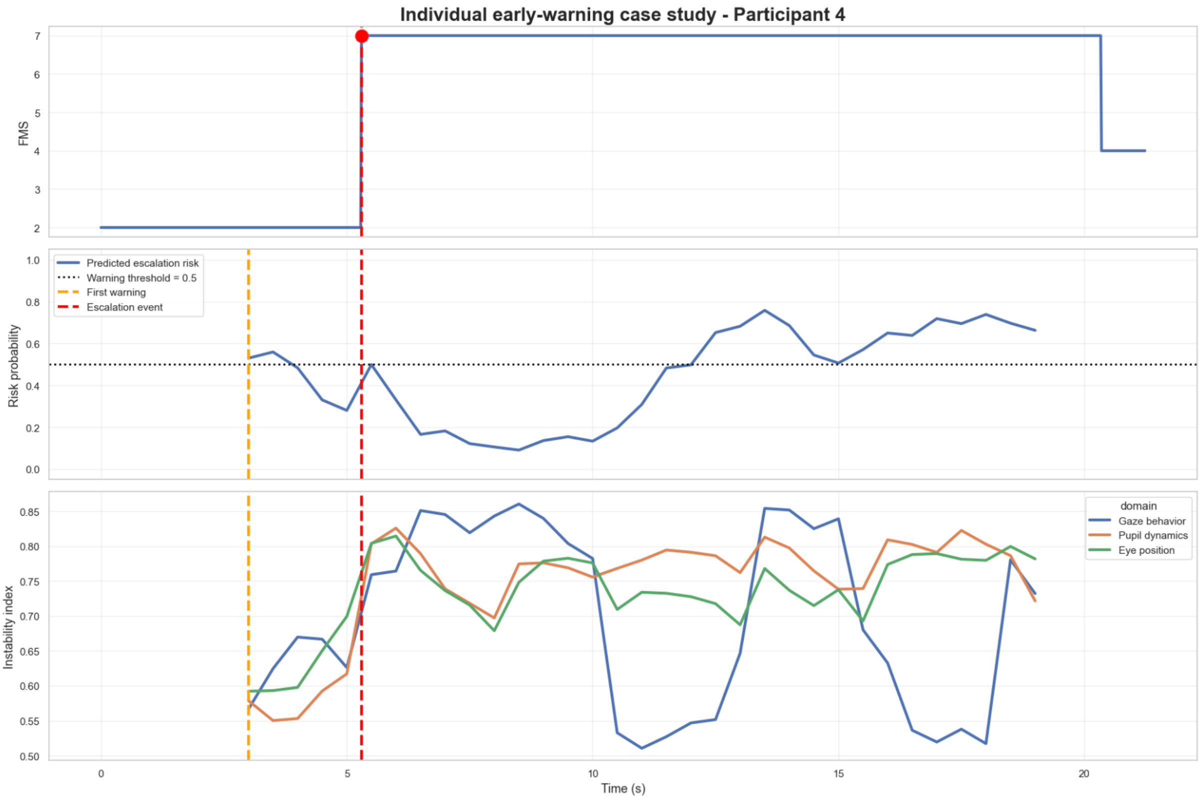


Figure 61. Individual early-warning case study (p.4).

Figure 61 combines three complementary sources of information:

1. the participant's reported Fast Motion Sickness (FMS) score,
2. the escalation risk estimated by the predictive model,
3. the evolution of the neurophysiological instability indices extracted from the eye-tracking signals.

The participant experienced a sudden increase in FMS from 2 to 7 at approximately 5.3 seconds, indicating a significant cybersickness escalation event.

Importantly, the predictive model generated its first warning at approximately 3.0 seconds, corresponding to an anticipation interval of roughly 2.3 seconds before the subjective escalation was reported.

This result demonstrates that the proposed framework is capable of identifying elevated cybersickness risk before the participant explicitly reports severe symptoms.

The lower panel further shows concurrent alterations in the neurophysiological instability indices. Variations in gaze behavior, pupil dynamics, and eye-position signals emerged around the warning period, suggesting that the model captures meaningful physiological changes associated with the progression of cybersickness.

From a neurotechnology perspective, this case study illustrates the potential of integrating eye-tracking biomarkers with machine-learning prediction to enable real-time early-warning systems in immersive virtual reality environments.

3.11.12. FUNCTIONAL INTERACTION BETWEEN NEUROPHYSIOLOGICAL DOMAINS

To further investigate the relationships between the neurophysiological domains identified by the predictive models, the temporal activity profiles of eye position, gaze behavior, and pupil dynamics were correlated during the three-second period preceding cybersickness escalation, as shown in Figure 62.

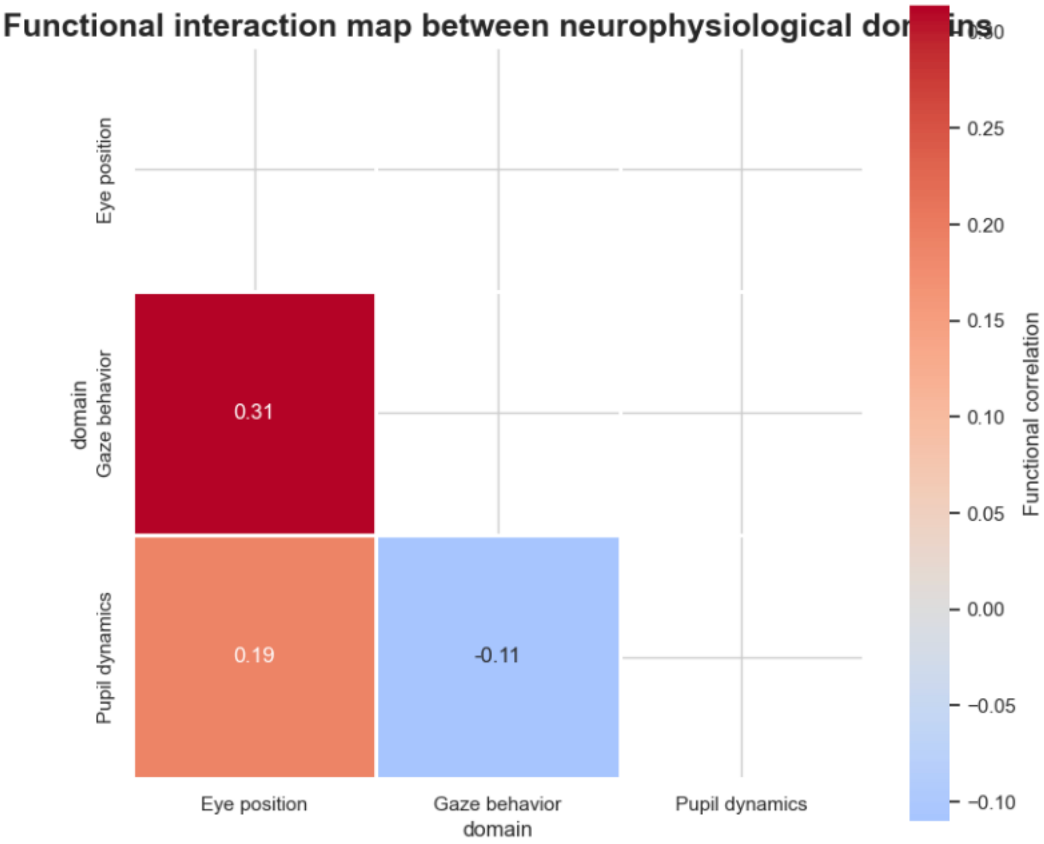


Figure 62. Functional interaction map between neurophysiological domains.

The strongest association was observed between eye position and gaze behavior (r = 0.31), indicating a moderate positive coupling between ocular alignment dynamics and gaze-control processes. This finding is neurophysiologically plausible, as both domains are closely involved in visual exploration, oculomotor control, and compensation mechanisms that emerge during visuovestibular conflict.

From a neurotechnology perspective, this result supports the hypothesis that cybersickness escalation is primarily associated with alterations in visuomotor processing. Disturbances in gaze stability and eye-position control are expected consequences of increasing sensory mismatch between visual and vestibular information, one of the main mechanisms proposed in cybersickness theory.

A weaker positive association was observed between eye position and pupil dynamics (r = 0.19), suggesting limited interaction between oculomotor behavior and autonomic responses during the pre-escalation period.

Interestingly, gaze behavior and pupil dynamics exhibited a small negative correlation ($r = -0.11$). Although the magnitude of this relationship is low, it may indicate a partial dissociation between visuomotor alterations and autonomic activation. In other words, pupil-related responses do not appear to evolve in complete synchrony with gaze-control abnormalities during cybersickness progression.

Overall, these results suggest that the neurophysiological signature preceding cybersickness escalation is dominated by visuomotor alterations, whereas autonomic responses may contribute more heterogeneously across participants.

3.11.13. EARLY-WARNING CAPABILITY OF THE PREDICTIVE FRAMEWORK

A clinically relevant objective of the proposed framework is not only the detection of cybersickness escalation events, but also their anticipation before symptom severity increases.

To evaluate this capability, a warning was generated whenever the predicted escalation probability exceeded a threshold of 0.5 within the three-second period preceding an escalation event.

Across all escalation episodes, the model successfully generated an early warning in 81.25% of cases. The average anticipation time was 2.36 seconds, with a median anticipation time of 2.58 seconds.

These results indicate that the predictive system was generally capable of identifying the transition towards cybersickness escalation before the actual increase in symptom severity occurred.

From a neurotechnology perspective, even short anticipation intervals may be operationally valuable, as they provide sufficient time for adaptive VR interventions such as visual scene modification, motion attenuation, exposure reduction, or personalized warning systems.

Overall, the results support the feasibility of using eye-tracking-based neurophysiological markers for real-time cybersickness forecasting.

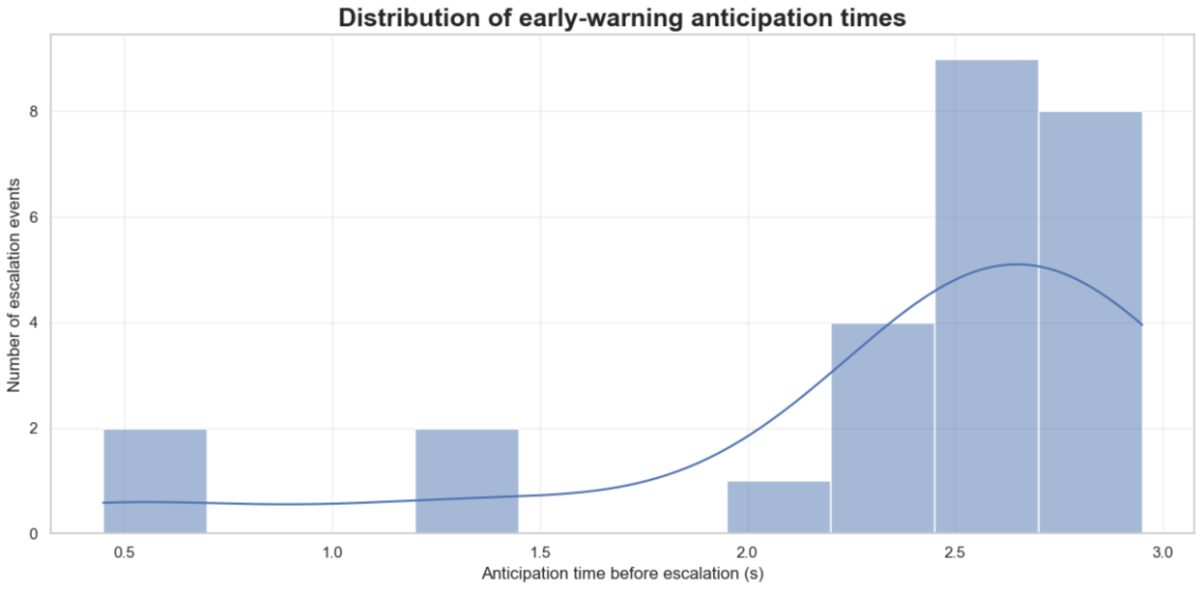


Figure 63. Distribution of anticipation times before escalation.

Most escalation events were detected between approximately 2 and 3 seconds before symptom escalation, indicating a relatively consistent early-warning window across participants, as presented in Figure 63.

Only a small number of events exhibited shorter anticipation times, suggesting that rapid escalation episodes may be inherently more difficult to predict.

The concentration of warnings around the 2–3 second interval reinforces the practical feasibility of implementing proactive cybersickness mitigation strategies.

3.11.14. INDIVIDUAL EARLY-WARNING CASE STUDY

Figure 64 presents a representative example of the proposed early-warning framework.

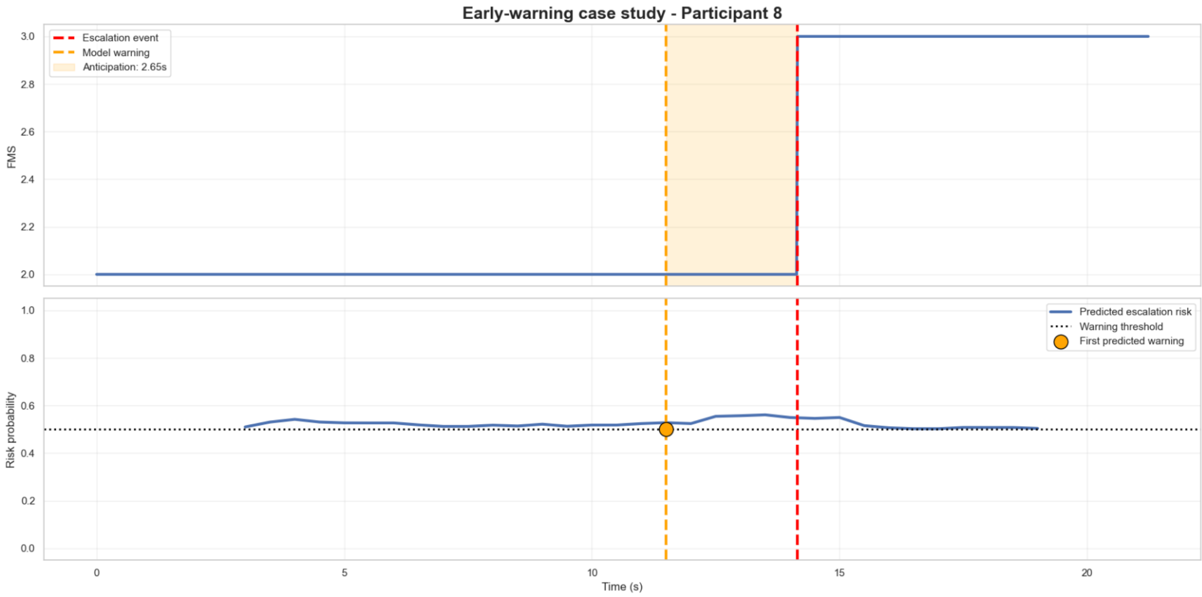


Figure 64. Individual early-warning case study (p.8)

Table 17. Important metrics extracted from the study.

Participant	8.00
Escalation time	14.15
First warning time	11.50
Anticipation time	2.65

Table 17 summarizes the key metrics extracted at this point. For this participant, the model-generated warning occurred approximately 2.65 seconds before the escalation event. During this period, the predicted escalation probability exceeded the predefined warning threshold while the subjective symptom severity remained stable.

This example illustrates how neurophysiological alterations captured through eye-tracking measurements may emerge before overt symptom escalation becomes observable.

The result highlights the potential of predictive neurotechnology systems to support real-time adaptive interventions aimed at preventing or mitigating cybersickness progression.

Although individual trajectories may vary across participants, the observed anticipation interval is consistent with the global early-warning analysis and provides further evidence that cybersickness escalation can be forecast using neurophysiological markers extracted from eye-tracking signals.

4. CONCLUSIONS AND FUTURE WORKS

4.1. CONCLUSIONS

This Master's Thesis addressed the problem of forecasting cybersickness escalation in immersive virtual reality environments using eye-tracking and head-related neurophysiological signals. The main objective was not only to classify whether a participant was experiencing cybersickness, but to anticipate future increases in discomfort before they were explicitly reported through the Fast Motion Sickness (FMS) score.

To achieve this objective, a complete machine learning framework was developed, including data exploration, temporal preprocessing, window-based feature extraction, model development, model comparison, and explainability analysis. The original continuous signals were sampled at 60 Hz, while cybersickness severity was provided as discrete FMS values. Therefore, a forecasting strategy based on temporal windows was implemented. Each window summarized the physiological state of the participant during a short observation period, while the target label indicated whether an escalation event occurred in the following forecasting horizon.

This formulation transformed the problem into an early-warning task. Instead of detecting cybersickness once it had already increased, the system attempted to identify physiological patterns preceding the escalation. This approach is especially relevant in neurotechnology, where the goal is to develop adaptive systems capable of responding to changes in the user's internal state in real time.

Several artificial intelligence models were evaluated, including Logistic Regression, Random Forest, XGBoost, HistGradientBoosting, Support Vector Machine, CatBoost, Multilayer Perceptron, a 1D Convolutional Neural Network and a LSTM. Different strategies were also tested to address the strong class imbalance observed in the dataset, including class weighting, SMOTE oversampling, feature selection, and SHAP-based feature reduction.

The results showed that the prediction of cybersickness escalation is a difficult task due to the low number of positive escalation windows and the high similarity between pre-escalation and normal physiological states. Several baseline models achieved high accuracy but failed to detect escalation events, demonstrating that accuracy alone was not an appropriate metric for this problem. For this reason, balanced accuracy, recall, PR-AUC, ROC-AUC, F1-score, and Matthews Correlation Coefficient were considered more informative.

Among the classical machine learning models, XGBoost and the MLP with SMOTE and feature selection showed the most relevant predictive behavior. XGBoost provided a balanced compromise between false positives and true positive detection, while the MLP achieved strong discrimination ability but showed signs of reduced generalization when moving from training to test data.

The strongest results were obtained by the temporal deep-learning models. Unlike the classical approaches, which were trained on aggregated statistical descriptors, both the 1D-CNN and the LSTM processed the full temporal structure of the physiological windows. This allowed the models to learn temporal patterns directly from the eye-tracking and movement signals.

The 1D-CNN achieved the highest recall and balanced accuracy, correctly identifying 65% of escalation events in the independent test set. This indicates a strong early-warning capability and suggests that short-term temporal dynamics contain predictive information that may be partially lost when the data are reduced to handcrafted summary features. In contrast, the LSTM achieved the highest F1-score, PR-AUC, and Matthews Correlation Coefficient, indicating a better overall balance between escalation detection and false-positive control.

Taken together, these findings support the hypothesis that cybersickness escalation is characterized by meaningful temporal patterns that can be exploited through deep-learning approaches. While the CNN proved more effective for maximizing escalation detection, the LSTM provided the most balanced overall classification performance, highlighting a trade-off between sensitivity and reliability in early-warning cybersickness forecasting.

The explainability analysis provided additional insight into the neurophysiological mechanisms associated with cybersickness escalation. The selected features were mainly related to gaze behavior, eye position, and pupil dynamics. In particular, gaze-related variables represented the largest proportion of the selected predictors, suggesting that oculomotor control and visual exploration patterns play an important role in the progression of cybersickness.

This result is coherent with the theoretical basis of cybersickness, especially the sensory conflict hypothesis. In immersive virtual reality, inconsistencies between visual, vestibular, and proprioceptive information may lead to compensatory changes in gaze control, eye alignment, and pupil-related responses. The models appeared to capture these subtle changes before the subjective FMS score increased.

The neurophysiological signature analysis showed that several eye-tracking signals changed during the seconds preceding escalation. These alterations were not limited to a single variable, but appeared distributed across multiple oculomotor domains. This supports the idea that cybersickness escalation is not caused by one isolated biomarker, but by a combination of visual, oculomotor, and autonomic processes.

The early-warning analysis further demonstrated the practical value of the proposed framework. Using a probability threshold of 0.5, the system generated an early warning in 81.25% of escalation events, with an average anticipation time of 2.36 seconds and a median anticipation time of 2.58 seconds. Although this anticipation interval is short, it may be sufficient for real-time adaptive interventions in VR, such as reducing visual motion, modifying scene dynamics, adjusting exposure intensity, or warning the user before discomfort becomes severe.

Overall, the findings of this thesis support the feasibility of using eye-tracking-based neurophysiological markers to forecast cybersickness escalation. The results indicate that cybersickness progression can be partially anticipated from short-term temporal changes in gaze behavior, eye-position dynamics, and pupil-related signals. From a neurotechnology perspective, this work contributes to the development of adaptive VR systems capable of monitoring the user’s physiological state and responding proactively to prevent or reduce discomfort.

However, the results must be interpreted with caution. The dataset contained a limited number of escalation events, and the positive class was strongly underrepresented. In addition, the models were trained and evaluated on windowed samples derived from a relatively small number of participants. Therefore, although participant-level separation was applied during evaluation, further validation with larger and more diverse datasets is required before the system can be considered robust for real-world deployment.

In conclusion, this thesis demonstrates that artificial intelligence and eye-tracking signals can be combined to develop predictive neurotechnology systems for cybersickness monitoring. The best-performing approach was the 1D-CNN, which confirmed the importance of temporal signal dynamics. The explainability analysis further showed that oculomotor and gaze-related features are key components of the predictive process. These findings provide a promising foundation for future real-time cybersickness prevention systems in immersive virtual reality.

4.2. FUTURE WORKS

Several future research directions can be proposed based on the results and limitations of this thesis.

First, the dataset should be expanded with a larger number of participants and escalation events. The current study was limited by the low prevalence of positive escalation windows, which made the classification task highly imbalanced. Increasing the number of participants would improve the diversity of physiological responses and allow the models to learn more robust escalation patterns. In particular, future datasets should include participants with different levels of cybersickness susceptibility, previous VR experience, age ranges, and visual characteristics.

Second, future work should validate the proposed framework using external datasets. In this thesis, the models were evaluated using participant-level separation, which reduced the risk of data leakage. However, external validation on an independent dataset would provide stronger evidence of generalization. This is especially important because eye-tracking signals can be influenced by headset type, calibration quality, display characteristics, task design, and individual physiology.

Third, the forecasting strategy could be improved by exploring different window sizes, step sizes, and prediction horizons. In this work, a fixed temporal window and forecasting horizon were selected. However, cybersickness escalation may occur at different temporal scales across participants. Future studies could compare shorter and longer windows to determine whether early symptoms are better captured by rapid transient changes or slower physiological trends.

Fourth, additional deep learning architectures should be investigated. The 1D-CNN showed the best overall performance, suggesting that temporal information is highly relevant. Future work could therefore evaluate recurrent neural networks, Gated Recurrent Units, Temporal Convolutional Networks, and Transformer-based architectures. These models may be able to capture longer temporal dependencies and more complex progression patterns.

Fifth, future work should explore threshold optimization and probability calibration. In this thesis, a fixed threshold of 0.5 was used to generate warnings. However, in real-world applications, the optimal threshold may depend on the desired balance between missed escalation events and false alarms. A lower threshold may increase sensitivity, while a higher threshold may reduce unnecessary interventions. Calibration methods could also improve the reliability of predicted probabilities.

Sixth, personalized models should be developed. The participant-level analyses showed considerable variability in oculomotor instability before escalation. This suggests that cybersickness does not evolve identically across users. Future systems could combine a general population-level model with user-specific adaptation, allowing the system to learn each participant’s baseline physiology and detect deviations from their own normal state.

Seventh, multimodal physiological data should be incorporated. Although this thesis focused mainly on eye-tracking and head-related signals, cybersickness also involves autonomic and postural responses. Future studies could include heart rate variability, electrodermal activity, respiration, EEG, postural sway, and vestibular-related measurements. Combining eye-tracking with these additional modalities may improve prediction performance and provide a more complete neurophysiological representation of cybersickness.

Eighth, the explainability analysis could be extended using more advanced XAI techniques. SHAP and model contribution analyses provided useful information about relevant features, but future work could apply temporal SHAP, integrated gradients, saliency maps, attention visualization, counterfactual explanations, and participant-specific interpretability methods. These techniques would help clarify not only which variables are important, but also when and how they contribute to the predicted escalation risk.

Ninth, future work should evaluate the system in real-time VR conditions. The present study was conducted offline using recorded data. A real-time implementation would require continuous signal acquisition, online preprocessing, window generation, probability estimation, and adaptive intervention triggering. Testing the system in an online environment would allow the evaluation of latency, computational efficiency, false warning burden, and user acceptance.

Tenth, adaptive intervention strategies should be designed and tested. Once a warning is generated, the VR system could respond by reducing motion intensity, narrowing the field of view, stabilizing the visual scene, modifying navigation speed, introducing rest periods, or personalizing exposure. Future studies should determine which intervention strategies are most effective for preventing escalation without disrupting immersion.

Finally, future work should investigate the clinical and usability implications of early-warning systems. Although cybersickness is not usually a clinical disease, it can strongly affect comfort, performance, accessibility, and user safety in immersive environments. Therefore, future research should assess whether predictive systems improve user experience, reduce dropout rates, increase VR tolerance, and support vulnerable users.

In summary, future work should focus on larger datasets, external validation, personalized modeling, temporal deep learning, multimodal physiological integration, advanced explainability, and real-time adaptive VR deployment. These directions would strengthen the current findings and move the proposed framework closer to practical neurotechnology applications for cybersickness prevention.

5. ETHICAL, ECONOMIC, SOCIETAL AND ENVIRONMENTAL ASPECTS

5.1. INTRODUCTION

Virtual Reality (VR) technologies are increasingly being adopted in entertainment, education, healthcare, industrial training, and scientific research. Despite their growing popularity, cybersickness remains one of the main limitations affecting user comfort, performance, and long-term adoption of immersive systems.

This Master's Thesis addresses this challenge through the development of an artificial intelligence framework capable of predicting cybersickness escalation events using eye-tracking and neurophysiological information. The proposed system aims to provide early warnings before symptom severity increases, enabling the implementation of adaptive interventions that may reduce user discomfort and improve the overall VR experience.

The development of predictive neurotechnology systems involves several ethical, societal, economic, and environmental considerations. From an ethical perspective, the use of physiological and behavioral data raises questions related to privacy, informed consent, transparency, and responsible use of artificial intelligence. From a societal perspective, successful cybersickness prediction could improve accessibility and safety for a wider population of VR users. Economically, reducing cybersickness may contribute to increased adoption of VR technologies and decrease costs associated with interrupted training sessions or reduced user engagement. Finally, although the environmental impact of this project is relatively limited, the computational resources required for machine learning model training also represent a relevant sustainability consideration.

For these reasons, a sustainability-oriented analysis was conducted to evaluate the broader implications of the proposed framework beyond its technical performance.

5.2. DESCRIPTION OF RELEVANT IMPACTS RELATED TO THE PROJECT

Several potential impacts associated with the project were identified and evaluated according to their relevance and expected influence on stakeholders.

5.2.1. ETHICAL IMPACTS

The most significant ethical consideration concerns the collection and processing of eye-tracking data. Eye movements, gaze patterns, and pupil responses can be considered biometric and behavioral information that may reveal aspects of a user's physiological and cognitive state. Consequently, appropriate data protection mechanisms, anonymization procedures, and informed consent practices are essential.

Another important ethical issue relates to the use of artificial intelligence for decision-making. Although the proposed framework generates predictions regarding cybersickness risk, these predictions are probabilistic and should not be interpreted as deterministic assessments of a user's condition. Therefore, transparency regarding model limitations and prediction uncertainty is necessary.

5.2.2. SOCIETAL IMPACTS

The project has the potential to generate positive societal benefits by improving user safety and comfort during VR exposure. Cybersickness is a major factor limiting accessibility to immersive technologies, particularly among sensitive individuals. By enabling proactive interventions, predictive systems may reduce discomfort and increase inclusiveness.

The proposed framework may also contribute to safer deployment of VR technologies in educational, clinical, and professional environments where prolonged exposure is required.

5.2.3. ECONOMIC IMPACTS

Cybersickness represents a significant barrier to the adoption of virtual reality systems in industry and healthcare. Training interruptions, reduced productivity, and shortened exposure times may decrease the effectiveness of VR-based applications.

A reliable early-warning system could increase user retention, improve training efficiency, reduce session interruptions, and enhance the overall economic viability of VR solutions. Consequently, predictive cybersickness monitoring may provide value for developers, healthcare institutions, training organizations, and technology companies.

5.2.4. ENVIRONMENTAL IMPACTS

The environmental impact of the project is relatively low compared with physical engineering systems. The main environmental cost is associated with computational resources used during model training and experimentation.

Although machine learning training requires energy consumption, the dataset size and model complexity used in this project remain moderate. Therefore, the environmental footprint is limited. Furthermore, successful deployment of VR technologies may indirectly reduce travel requirements for training and education activities, potentially generating positive environmental effects through reduced transportation-related emissions.

5.2.5. STAKEHOLDERS

The main stakeholders identified in this project are:

- VR users and participants.
- Researchers in virtual reality and neurotechnology.
- Healthcare professionals using immersive technologies.
- VR software developers.
- Educational and industrial training institutions.
- Companies deploying immersive applications.
- Regulatory bodies responsible for data protection and AI governance.

Considering all identified impacts, ethical and societal dimensions were determined to be the most relevant and therefore selected for deeper analysis.

5.3. DETAILED ANALYSIS OF ONE OF THE PRINCIPAL IMPACTS

5.3.1. ETHICAL IMPLICATIONS OF AI-BASED NEUROPHYSIOLOGICAL MONITORING

The most relevant impact associated with this project is the ethical management of physiological and behavioral data used for cybersickness prediction.

The proposed framework relies on eye-tracking measurements, including gaze direction, eye position, and pupil-related variables. While these signals are collected primarily to estimate cybersickness risk, they may also contain information about attention, fatigue, emotional responses, cognitive workload, or other aspects of user behavior. Consequently, the responsible use of such data becomes essential.

A first ethical requirement is informed consent. Participants should clearly understand what information is collected, how it is processed, and the purpose of the predictive system. Data collection must remain limited to variables strictly necessary for the intended application.

A second consideration involves privacy protection. Physiological data should be anonymized whenever possible, and storage procedures must comply with applicable data protection regulations such as the General Data Protection Regulation (GDPR) within the European Union. Access to raw data should be restricted to authorized personnel.

A third issue concerns algorithmic transparency. Artificial intelligence systems may generate false positives or false negatives. In the context of cybersickness prediction, a false warning may unnecessarily interrupt a VR session, whereas a missed warning may allow discomfort to continue increasing. Therefore, predictions should be presented as decision-support information rather than definitive judgments.

Another important aspect is user autonomy. Adaptive interventions triggered by the predictive system should not remove the user’s ability to make decisions regarding their VR experience. Instead, warnings should assist users by providing additional information while preserving individual control.

Finally, fairness and inclusiveness should be considered. Physiological responses may vary substantially across individuals due to age, experience, susceptibility to motion sickness, or other personal factors. Future implementations should therefore ensure that predictive performance remains acceptable across diverse user populations and does not disproportionately disadvantage specific groups.

Overall, the ethical deployment of AI-driven neurophysiological monitoring systems requires balancing technological benefits with privacy protection, transparency, user autonomy, and responsible data governance.

5.4 CONCLUSIONS

The sustainability analysis performed in this thesis demonstrates that the proposed cybersickness prediction framework presents predominantly positive societal and economic impacts while maintaining a relatively limited environmental footprint.

From a societal perspective, the system may improve accessibility, safety, and comfort during immersive virtual reality experiences by enabling proactive mitigation strategies before symptom escalation occurs. Such capabilities may facilitate broader adoption of VR technologies across educational, healthcare, industrial, and entertainment applications.

From an economic perspective, predictive cybersickness monitoring may reduce interruptions, improve user retention, increase training efficiency, and enhance the practical viability of immersive systems.

The principal challenges identified are ethical rather than technical. Because the framework relies on physiological and behavioral information, particular attention must be paid to privacy protection, informed consent, transparency, fairness, and responsible AI deployment. These considerations are essential to ensure that technological innovation remains aligned with user rights and societal expectations.

Regarding environmental sustainability, the computational requirements associated with model training are moderate and therefore generate a relatively limited impact. Moreover, future VR applications supported by predictive monitoring systems may indirectly contribute to reducing travel-related emissions through wider adoption of remote immersive experiences.

In conclusion, incorporating sustainability criteria into the development of this project has provided significant added value. The analysis demonstrates that the proposed framework is not only technically feasible but also compatible with responsible, ethical, and socially beneficial deployment of artificial intelligence within immersive virtual reality environments.

6. FINANCIAL BUDGET

PERSONNEL COSTS (direct costs)	Hours	Hourly cost	Total
	360	15 €	**5.400 €**

MATERIAL RESROURCES COSTS (direct costs, DC)	Purchase price	Months of use	Depreciation (in years)	Total
Personal computer, including software	1500,00 €	6	5	150,00 €
VR headset and eye-tracking system*	3000,00 €	6	5	300,00 €
Other equipment				

TOTAL COSTS OF MATERIAL RESOURCES	**450,00 €**

GENERAL COSTS (indirect costs, IC)	15%	on DC	**877,50 €**
INDUSTRIAL BENEFIT	6%	on DC+IC	**403,65 €**

CONSUMABLES	
Printouts	**30,00 €**
Binding	**20,00 €**

SUBTOTAL BUDGET		**7.181,15 €**
VAT	21%	**1.508,04 €**

TOTAL BUDGET	**8.689,19 €**

*Equipment not bougth, but needed in case of data acquisition.

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8. ANNEX — DECLARATION OF RESPONSIBLE USE OF ARTIFICIAL INTELLIGENCE

8.1. AI TOOLS USED

Tool 1 (name and version/date): ChatGPT (OpenAI, GPT-5.5, June 2026)

8.2. TYPES OF USE (PLEASE TICK AS APPROPRIATE)

[ ] Draft text generation
[yes] Rewriting / style editing
[yes] Translation / linguistic correction
[ ] Structure or outline suggestions
[ ] Methodological support (experimental design, metrics, etc.)
[ ] Data analysis / exploration
[yes] Code generation
[yes] Image / figure generation or editing
[ ] Other (please specify): ________________________________

8.3. BRIEF DESCRIPTION OF USE AND VALIDATION

Artificial Intelligence was used exclusively as a supporting tool during the preparation of this Master's Thesis. ChatGPT assisted with improving the academic writing style, translating and refining text, generating illustrative figures and providing programming support for implementing and documenting machine learning and deep learning models. AI was also used to generate schematic figures summarizing the forecasting pipeline and the validation strategy.

All AI-generated outputs were critically reviewed, manually verified, and adapted before being incorporated into the thesis. Experimental results, statistical analyses, figures, code execution, and scientific conclusions were independently validated using the original dataset and the implemented software pipeline. The author assumes full responsibility for all final content.

8.4. EXAMPLE PROMPT(S) USED

Prompt 1

"Based on the forecasting pipeline implemented in this thesis (EDA, preprocessing, sliding windows, feature engineering, machine learning, deep learning and SHAP explainability), generate a horizontal figure summarizing the complete workflow suitable for inclusion in the Development chapter."

Response / use

A schematic pipeline figure was generated and later reviewed, adapted, and incorporated as the overview of the proposed methodology.

Prompt 2

"Improve the academic writing of this section while preserving the scientific meaning and adapting it to the style of an English Master's Thesis."

Response / use

The generated text was manually revised, corrected where necessary, and integrated only after verification of its scientific accuracy and consistency with the rest of the thesis.

Prompt 3

"Using the existing implementation of the 1D-CNN forecasting model and a previously implemented LSTM example as a reference, adapt the code to build an LSTM model that follows the same preprocessing pipeline, participant-aware train/validation/test split, feature dimensions, evaluation metrics, and output format used throughout the project."

Response / use

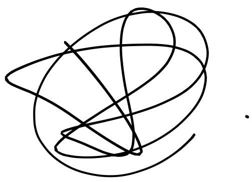
The generated code was used as a starting point for adapting the existing 1D-CNN implementation to an LSTM architecture while preserving the same experimental pipeline. The proposed implementation was manually reviewed, modified to fit the project structure, executed locally, and validated using the MazeSick dataset before being incorporated into the final experiments.

8.5. ETHICAL AND LEGAL CHECKLIST

[yes] I have not entered personal or sensitive data, nor confidential material, without a legal basis or prior anonymization.
[yes] I have respected copyright and licensing requirements and have cited AI-generated content where applicable.
[yes] I have verified that the terms of use of the tools allow the academic purposes described.
[yes] I have critically reviewed and validated the AI-generated outputs and assume full responsibility for the final content.

8.6. AUTHORSHIP DECLARATION

I, Daniel Llopis Conejo, declare that the content of this Master's Thesis is my own work. The AI tools listed above were used solely as support; I have critically verified their outputs, cited them where appropriate, and complied with the course guidelines and applicable regulations.

Signature:  Date: 26 / 06 / 2026